
A Causal Decomposition Approach for Fair Contextual Multi-Armed Bandits

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Abstract

Counterfactual reasoning is one of the fundamental facets of human cognition, involved in various tasks such as explanation, credit assignment, blame, and responsibility. It describes the queries what would have happened had some intervention been performed given that something else, corresponding to Layer 3 of the *Pearl Causal Hierarchy*. In this project, we examine specific types of counterfactual quantities, called counterfactual direct (Ctf-DE), indirect (Ctf-IE), and spurious (Ctf-SE) effects for quantifying fairness in a sequential decision-making framework. Building on these measures, we formulate an online causally-fair learning problem with multiple long-term constraints and study it in both non-parametric contextual bandits and parametric logistic bandits settings. We achieve sublinear regret and violations bounds for both bandits settings with round-wise counterfactual fairness constraints (that are a priori unknown) without Slater’s condition. For logistic bandits, our method achieves a regret with leading term matches that of the unconstrained setting.

1. Introduction

Explainable artificial intelligence (XAI) (Lipton, 2018; Miller, 2019; Longo et al., 2024) seeks to identify underlying mechanisms that govern a system’s behavior through observations, interventions, and imaginations, which induce the three layers in *Pearl Causal Hierarchy* (a.k.a. the Ladder of Causation) (Pearl & Mackenzie, 2018; Bareinboim et al., 2022). In particular, imaginations, namely counterfactuals, entail thinking about alternative ways the world could be, including ways that conflict with how the world currently is. For instance, would the employer have taken the same action

had the employee been of a different religion (age, national origin, etc.) and everything else had been the same? For answering this question, a wider variety of counterfactual quantities including conditional average treatment effect (CATE), direct effect (DE), indirect effect (IE), and path-specific effect (PSE) have been used for measuring fairness in the machine learning community (Kusner et al., 2017; Zhang et al., 2017; Zhang & Bareinboim, 2018; Nabi & Shpitser, 2018; Chiappa, 2019; Wu et al., 2019a;b; Mhasawade & Chunara, 2021; Plečko & Bareinboim, 2024).

More recently, there has been growing interest in counterfactual, causally grounded notions of fairness for sequential decision-making under uncertainty, including in settings such as multi-armed bandits (MABs) (Huang et al., 2022b) and Markov decision processes (MDPs) (Hu & Zhang, 2022); (Hu et al., 2023). However, neither the conditional average treatment effect (CATE) nor path-specific effects (PSE), as commonly employed above, yields a quantitative explanation of the total observed disparity into contributions of underlying discriminatory mechanisms (Zhang & Bareinboim, 2018; Plečko & Bareinboim, 2024). CATE conflates all causal paths and might average out meaningful variations, e.g., a positive effect along one path could be canceled by a negative effect in another path (MacKinnon et al., 2000). PSE, by construction, excludes the spurious associations through the non-causal (backdoor/confounding) pathways.

In this paper, we use fine-grained counterfactual measures (Ctf-DE, Ctf-IE, and Ctf-SE) (Plečko & Bareinboim, 2024) for attributing the total observed variations into structural counterfactual mechanisms along direct, indirect, and spurious paths. In practice, this allows one to explain the effect of a certain discriminatory mechanism in terms of the observed disparity found in the data. We focus on the challenge of learning the best policy in contextual MABs subject to a set of counterfactual fairness constraints. The contexts, decisions, and outcomes are generated by a structural causal model (SCM) consistent with a causal graph. More specifically, our contributions are as follows.

1. We formulate an online causally-fair learning problem where the fairness is measured by the underlying structural counterfactual mechanisms (Ctf-DE, Ctf-IE, Ctf-SE) that are treated as multiple long-term constraints. We study learning in both non-

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Table 1. Comparison of frequentist regret and constraint violations upper bounds for stochastic non-parametric contextual bandits and parametric logistic bandits. Notations: dimension n ; horizon T ; ρ is a parameter related with Slater’s constant; m is the number of constraints; ω and T_0 are the user-selected exploration parameters; γ_1 and γ_2 are the context coverage and covariate diversity parameters; κ_* and κ are the problem dependent constants where $\sqrt{1/\kappa_*} \leq 1 \ll \kappa$. † Requires prior knowledge of a safe action or policy (per-round zero constraint violations with high probability). ‡ Considers a generalized linear model. * Considers a (discrete) non-parametric bandit setting with adversarially generated rewards and stochastic constraints, in the absence of contextual information. § Considers the counterfactual constraints involving multiple contexts (factual and counterfactual contexts). N/A (not applicable) denotes the logistic bandits setting without constraints. ¶ Considers one aggregate constraint (CATE) and assumes the cumulative factual and counterfactual norms have the same order.

Setting	Constraint Feedback	Algorithm	Regret	Violation	Cost Per-Round
Non-Parametric	No §	F-UCB (Huang et al., 2022b)	$\mathcal{O}(\sqrt{T})$	0 †	$\mathcal{O}(1)$
	Yes	Alg.2 (Bernasconi et al., 2024) *	$\mathcal{O}(\sqrt{T})$	$\mathcal{O}(\sqrt{T \log(mT^2)})$	$\mathcal{O}(1)$
	No §	Algorithm 1 (Theorem 1)	$\tilde{\mathcal{O}}(T^{\max\{1/2, 1-\omega\}})$	$\tilde{\mathcal{O}}(\frac{T^{\max\{\frac{1+\omega}{2}, 1-\omega\}}}{\sqrt{\gamma_1}})$	$\mathcal{O}(1)$
Logistic	No	Safe UCB-GLM (Amani et al., 2020) ‡	$\tilde{\mathcal{O}}(\kappa n \sqrt{T})$	0 †	$\mathcal{O}(t)$
	N/A	OFU-MLogB (Fauray et al., 2022)	$\tilde{\mathcal{O}}(n \sqrt{T/\kappa_*})$	N/A	$\mathcal{O}(\log(t))$
	N/A	Cor. 1 (Zhang & Sugiyama, 2023)	$\tilde{\mathcal{O}}(n \sqrt{T/\kappa_*})$	N/A	$\mathcal{O}(1)$
	N/A	GLB-OMD (Zhang et al., 2025) ‡	$\tilde{\mathcal{O}}(n \sqrt{T/\kappa_*})$	N/A	$\mathcal{O}(1)$
	No §	CCLB (Chen et al., 2025) ¶	$\tilde{\mathcal{O}}(n \rho \sqrt{T})$	$\tilde{\mathcal{O}}(n \sqrt{T} + n^2 \kappa)$	$\mathcal{O}(t)$
	No §	Algorithm 2 (Theorem 2)	$\tilde{\mathcal{O}}(n \sqrt{T} + T_0)$	$\tilde{\mathcal{O}}(T_0 + T \sqrt{\frac{n \kappa}{\gamma_2 T_0}})$	$\mathcal{O}(1)$

parametric contextual bandits and parametric logistic bandits settings and establish sublinear regret and violations bounds for both bandits settings with (unknown) roundwise counterfactual fairness constraints.

- For handling constraints, we optimistically estimate both factual and counterfactual mean rewards and safety violations to select actions. Our method achieves sublinear regret and violations bounds without Slater’s condition and are independent of the number of constraints. For logistic bandits, the leading term of our regret, $\tilde{\mathcal{O}}(n \sqrt{T})$, matching that of the unconstrained problem with $\mathcal{O}(1)$ per round time complexity.

1.1. Related Works

For non-parametric contextual bandits, the technical challenge is to ensure the learner have increasingly good estimates of decisions’ empirical means under both the factual (observed) and counterfactual contexts (Remark 1). (Huang et al., 2022b) were the first to study multi-armed bandits with counterfactual fairness constraints. They considered the non-parametric setting with fairness defined using CATE. They obtained sublinear regret with high probability guarantees of no constraint violations. However, their approach involved conservatively estimating the feasible set which can be challenging to ensure is non-empty when there is a lack prior knowledge of the mean rewards.

For (parametric) logistic bandits, the main technical challenge is to (1) construct an optimistic and convex estimated feasible set without strong assumptions (e.g., a known strictly feasible policy or Slater’s condition); and (2) ensure that the selected feature vectors

$f(Z_{X_1, d_1}), f(Z_{X_2, d_2}), \dots, f(Z_{X_t, d_t})$ collectively span $\mathbb{R}^n$ well enough over time as accurate estimates of counterfactual rewards are needed to estimate fairness, and counterfactual feature vectors may be orthogonal to the putatively best arms’ feature vectors.

The closest prior work is (Chen et al., 2025). The authors studied counterfactual fairness in MAB problems with logistic rewards. Counterfactual fairness was captured through a single, long-term constraint on the CATE. Our work improves it on three aspects. (1) We use fine-grained counterfactual fairness metrics (Ctf-DE, Ctf-IE, Ctf-SE) that decompose the total variations. By contrast, (Chen et al., 2025) operationalize fairness via CATE which is pathway-agnostic and susceptible to mediation inconsistency for both linear and non-linear causal mechanisms (VanderWeele, 2015). (2) We generalize the setting from a single constraint to multiple constraints. In our problem, each constraint must account for a number of counterfactual contexts on the order of the context space, while (Chen et al., 2025) only needs to handle a single counterfactual context per round. (3) We achieve a constant per-round time complexity.

Another work, Lear & Zhang (2025), were the first to apply causal fairness decomposition in an online MDP setting. They study long-term fairness and decompose total interventional disparity into three terms: the direct policy effect, the indirect policy effect (mediated by environment dynamics), and the spurious policy effect (disparity attributable only to the environment). They report encouraging empirical results but do not provide any theoretical analysis (convergence or regret guarantees). See Appendix C for more works.

Preliminaries. We provide the basic notations used throughout the paper. A structural causal model (SCM) (Pearl, 2009, Ch. 7) M is a tuple $\langle U, V, \mathcal{F}, \mathbb{P}(u) \rangle$, where U is a set of exogenous (unobserved or latent) variables, V is a set of endogenous (observed) variables, $\mathcal{F}$ is a set of structural functions, and $\mathbb{P}(u)$ is a distribution over the latent variables. For the set of structural functions $\mathcal{F}$, $f_i \in \mathcal{F}$ decides values of an endogenous variable $i \in V$ taking as argument a combination of other variables. An intervention, denoted by $do(i = c)$ ($do(c)$, for short) (Pearl, 2009), represents a model manipulation where the values of a set of variables $i \in V$ are set fixed to c regardless of how their values are ordinarily determined by f_i . For random variables such as the outcome Y , we use the counterfactual distribution $P(Y_{i=c} = y)$ to denote the causal effect of the intervention $do(i = c)$ on the outcome Y , where the counterfactual variable $Y_{i=c}$ (Y_c , for short) denotes the potential response of Y to intervention $do(i = c)$. $\mathcal{F}_{t-1}$ denotes the history available before round t , formally represented as a σ -algebra.

2. Problem Formulation

In this section, we formulate an online causal fair decision-making problem, where the fairness is measured using counterfactual structural mechanisms through direct, indirect, and spurious pathways, which will be treated as (unknown) constraints over the course of sequential learning.

2.1. Online Causal Fairness Decomposition

At each round $t = 1, \dots, T$, the learner observes the contextual feature vector X_t that is drawn stochastically from an underlying fixed distribution and then is presented a set of decisions $\mathcal{D}$. The learner chooses a decision (an intervention) $d_t \in \mathcal{D}$ and receives an outcome reward y_t . The decision maker will seek to maximize (cumulative) rewards while limiting (cumulative) violations of fairness constraints. The decision-making process is characterized by the extended Standard Fairness Model (SFM) with a causal diagram shown in Figure 1 (Zhang & Bareinboim, 2018; Plečko & Bareinboim, 2023; Plečko & Bareinboim, 2024; Chen et al., 2025). The variable A_t represents the (binary) protected attribute, W_t is a set of features (confounders), M_t is a set of intermediate features, D_t represents the decision, and Y_t represents the outcome reward. For simplicity, we write $X_t = \{A_t, W_t, M_t\}$. We denote by a_t and a'_t two realizations (factual and counterfactual) of the sensitive attribute A_t (e.g. two demographic groups), which we will use to measure the changes of the outcome. We motivate our online causal fairness analysis by the following example:

Example 1 (Online Recommendation System (Adapted from (Li et al., 2010))). Consider an online job platform that recommends a job opportunity D_t to each arriving individual using contextual information (A_t, W_t, M_t). The

customer’s religious belief is the protected attribute A_t (1 for religious, 0 for non-religious); M_t and W_t represent user’s intermediate features (e.g., application/job history) and confounded features (e.g., demographics, zip code). Once a job recommendation (D_t) is made, the platform observes user-click feedback Y_t (1 click, 0 no click). The platform seeks to maximize expected user-click rewards from its recommendations, subject to fairness constraints. Fairness concerns arise since the recommendation determines which users are exposed to valuable job opportunities; thus, discriminatory recommendation rules may create unequal opportunities across sensitive groups. See Appendix D for more details.

For quantifying fairness, one popular criterion is the demographic parity (Dwork et al., 2012), measured by the total variation (Ashurst et al., 2022) (also called parity gap). For our online setting, we write it as (adapted from (Zhang & Bareinboim, 2018; Plečko & Bareinboim, 2024)):

Definition 1 (Total Variation (TV)). Under an intervention $do(D_t = d_t)$, the total variation of the event $A_t = a'_t$ on Y_t (with factual value a_t) is defined as:

$$TV_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{d_t}|A_t = a'_t] - \mathbb{E}[Y_{d_t}|A_t = a_t]. \quad (1)$$

Holding the recommendation fixed at d_t , $TV_{a_t, a'_t}(d_t)$ measures the difference between the conditional distributions of Y_t when (passively) observing A_t changing from a_t to a'_t over the entire population. Natural questions that arise are (1) what are the structural mechanisms of the observed disparity in the data and (2) what empirical measures could identify and potentially control such mechanisms? In this paper, we use a set of structural counterfactual measures (adapted from (Zhang & Bareinboim, 2018; Plečko & Bareinboim, 2024)) that will allow for a more fine-grained explanation of the observed disparity.

Definition 2 (Online Counterfactual Measures). Given a SCM $\mathcal{M}^*$, under an intervention $do(D_t = d_t)$, the structural counterfactual direct effect (Ctf-DE), indirect effect (Ctf-IE), and the spurious effect (Ctf-SE)¹ of event $A_t = a'_t$ (with factual value a_t) on Y_t are defined as:

$$\text{Ctf-DE}_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{a'_t, d_t, M_{a_t}} - Y_{a_t, d_t, M_{a_t}} | a_t]$$

$$\text{Ctf-IE}_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{a_t, d_t, M_{a'_t}} - Y_{a_t, d_t, M_{a_t}} | a_t]$$

$$\text{Ctf-SE}_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{a_t, d_t} | a'_t] - \mathbb{E}[Y_{d_t} | a_t].$$

The nested quantity above, e.g. $\mathbb{E}[Y_{a'_t, d_t, M_{a_t}} | a_t]$ ², is a more involved counterfactual quantity and is read as “the expected outcome at time t under intervention

¹The spurious effect (backdoor path) measures the average difference in outcome Y_t when observing that $A_t = a_t$, compared to setting $A_t = a_t$ by intervention but conditioned on a'_t .

²The expectation averages over the mediator, confounder, and outcome randomness.

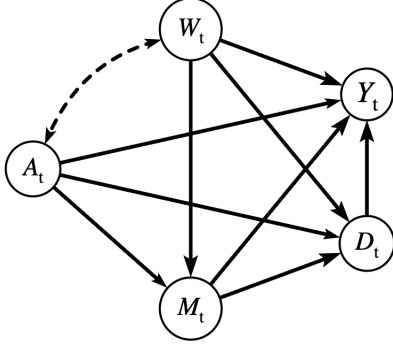


Figure 1. Per-round extended Standard Fairness Model (SFM).

$do(D_t = d_t)$ had the attribute A_t been set to a'_t , while M_t is kept at the same value that it would have attained had A_t been a_t , given that A_t was actually equal to a_t .” Despite the apparently non-trivial reading of such nested counterfactuals, $\text{Ctf-DE}_{a_t, a'_t}(d_t)$, $\text{Ctf-IE}_{a_t, a'_t}(d_t)$, and $\text{Ctf-SE}_{a_t, a'_t}(d_t)$ simply capture, mathematically, the existence of disparate impact (Barocas & Selbst, 2016) along the direct, indirect, and spurious pathways between the protected attribute A and outcome reward Y in Figure 1. Additionally, these counterfactual measures provide a quantitative explanation for the disparities (or lack thereof) observed in TV as follows (Plečko & Bareinboim, 2024):

Proposition 1. *Under an intervention $do(D_t = d_t)$, the total variation measure can be decomposed as³*

$$\text{TV}_{a_t, a'_t}(d_t) = \text{Ctf-DE}_{a_t, a'_t}(d_t) - \text{Ctf-IE}_{a_t, a'_t}(d_t) - \text{Ctf-SE}_{a_t, a'_t}(d_t). \quad (2)$$

TV could be large primarily due to spurious effects, which in some settings might not lead to fairness concerns. TV could also be small due to cancellations along pathways, such as between direct and spurious effect, essentially masking fairness violations when only TV is measured.

2.2. Causally-Fair Constrained Learning

In the online recommendation system (Example 1), the objective for the decision-maker is to learn a policy maximizing the cumulative rewards over a horizon T while achieving user-side fairness. Nominally, this can be formulated as a constrained optimization problem:

$$\max_{\pi_{X_1}, \dots, \pi_{X_T}} \sum_{t=1}^T \mathbb{E}_{d_t \sim \pi_{X_t}} [Y_{d_t}(X_t)], \quad \text{subject to} \quad (3)$$

$$\mathbb{E}_{d_t \sim \pi_{X_t}} |\Delta_{d_t}^{(i)}(X_t)| \leq c_i, \quad i \in \{1, 2, 3\}, t \in [T]. \quad (4)$$

where π_{X_t} denotes a (possibly randomized) policy mapping contexts to a distribution over decisions and c_1, c_2 ,

³The transition runs from a_t to a'_t in TV and Ctf-DE, and in the reverse direction in Ctf-IE and Ctf-SE.

and c_3 are user-defined fairness thresholds. For simplicity, here we denote the counterfactual measures at time t as: $\Delta_{d_t}^{(1)}(X_t) = \text{Ctf-DE}_{a_t, a'_t}(d_t)$, $\Delta_{d_t}^{(2)}(X_t) = \text{Ctf-IE}_{a_t, a'_t}(d_t)$, and $\Delta_{d_t}^{(3)}(X_t) = \text{Ctf-SE}_{a_t, a'_t}(d_t)$.

Technically, the counterfactual fairness constraints in (4) restrict the discrimination along the direct, indirect, and spurious pathways. As for our problem, a disparity in average rewards between two sensitive groups (e.g., believer or non-believer of religion) may arise from: (i) direct discrimination (Ctf-DE) based on individual’s religious belief, (ii) indirect discrimination (Ctf-IE) through intermediate user features (e.g., browsing history) which may be partly influenced by the user’s beliefs, or (iii) spurious discrimination (Ctf-SE) due to background factors (e.g., nationality) that are correlated with user’s beliefs. In practice, this distinction aligns with legal frameworks such as disparate impact (Barocas & Selbst, 2016; Plečko & Bareinboim, 2024) recognized in U.S. and EU anti-discrimination law.

To further analyze these fairness constraints in practice, we need to express the nested counterfactuals as functions of the observational data. Leveraging the assumptions implied by the standard model (Pearl, 2009; Hernán & Robins, 2020), we can identify the structural counterfactual measures as:

Proposition 2 ((Plečko & Bareinboim, 2024)). *Under the extended Standard Fairness Model (SFM) in Figure 1, $\text{Ctf-DE}_{a_t, a'_t}(d_t)$, $\text{Ctf-IE}_{a_t, a'_t}(d_t)$, and $\text{Ctf-SE}_{a_t, a'_t}(d_t)$ can be estimated, respectively, as*

$$\begin{aligned} & \sum_{m, w} (\mathbb{E}[Y_t | d_t, a'_t, w, m] - \mathbb{E}[Y_t | d_t, a_t, w, m]) \mathbb{P}(m | a_t, w) \mathbb{P}(w | a_t), \\ & \sum_{m, w} \mathbb{E}[Y_t | d_t, a_t, w, m] (\mathbb{P}(m | a'_t, w) - \mathbb{P}(m | a_t, w)) \mathbb{P}(w | a_t), \\ & \sum_{m, w} \mathbb{E}[Y_t | d_t, a_t, w, m] \mathbb{P}(m | a_t, w) (\mathbb{P}(w | a'_t) - \mathbb{P}(w | a_t)). \end{aligned}$$

However, in the online setting, it is unclear whether it is possible to certify policies as fair (feasible) before collecting data to estimate the structural counterfactual measures so that we could guarantee that all decisions picked by the learner satisfy the hard constraints in (4) with high probability. In this paper, we do not assume a priori knowledge of fair policies. Instead, we allow for instantaneous violations but seek to bound the cumulative constraints violations. The goals are then to maximize the cumulative rewards and to minimize the maximum cumulative violations of the counterfactual fairness constraints throughout the learning process. We study this problem in Section 3 for non-parametric distributions and in Section 4 for logistic rewards.

3. Non-parametric Contextual Bandits with Counterfactual Fairness Constraints

In this section, we introduce the methods and analysis for solving the causal fair constrained optimization problem

(see Section 2.2) in the non-parametric contextual bandits setting. We consider that the context set $\mathcal{X}$ is finite similar to (Huang et al., 2022b; Varatharajah & Berry, 2022; Wang & Tiwari, 2025). At every round t , the learner observes contextual information $X_t = \{a_t, m_t, w_t\}$, selects an action $d_t \in \mathcal{D}$, and receives a bandit reward feedback $y_t \in [0, 1]$ with mean $\mathbb{E}[Y_{d_t}(X_t)] \in [0, 1]$. Let $\Pi \subset \mathbb{R}^{|\mathcal{D}|}$ be the set of discrete probability distributions (simplex) over the set $\mathcal{D}$, then given a randomized strategy $\pi_t \in \Pi$, the expected reward for policy π_t is $\sum_{d \in \mathcal{D}} \mathbb{E}[Y_d(X_t)] \pi_t(d) = \langle \mathbb{E}[Y_d(X_t)], \pi_t \rangle$. Similarly, $\langle |\Delta_d^{(i)}(X_t)|, \pi_t \rangle = \sum_{d \in \mathcal{D}} |\Delta_d^{(i)}(X_t)| \pi_t(d)$ denotes the expected counterfactual fairness effect for constraint i . Then, we define the pseudo cumulative regret and expected cumulative constraint violations as:

$$R_1(T) = \sum_{t=1}^T \left[\max_{\pi \in \Pi_{X_t}^f} \langle \pi, \mathbb{E}[Y_d(X_t)] \rangle - \mathbb{E}[Y_{d_t}(X_t)] \right],$$

$$V_1(T) = \max_{i \in \{1, 2, 3\}} \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{d \sim \tilde{\pi}_{X_t}} |\Delta_d^{(i)}(X_t)| - c_i \right],$$

where $\Pi_{X_t}^f = \bigcap_{i=1}^3 \{\pi \in \Pi : \mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(X_t)| \leq c_i\}$ is the true feasible set of policies for context X_t and $\tilde{\pi}_{X_t}$ is the learner's chosen policy for round t .

Assumption 1. We assume the existence of feasible policies, that is, $\Pi_{X_t}^f \neq \emptyset$, $\forall t \in [T]$.

Assumption 1 is a weaker assumption than the Slater's condition assumed by primal-dual methods in which $\langle |\Delta_d^{(i)}(X_t)|, \pi_{X_t}^o \rangle \leq c_i - c_0$, $\forall i \in \{1, 2, 3\}$, $\forall t \in [T]$, where c_0 is a strictly positive constant (Boyd & Vandenberghe, 2004; Efroni et al., 2020; Zhou & Ji, 2022).

Assumption 2 (Context Coverage). There exists a $\gamma_1 > 0$ such that for every $(a, w, m) \in (\mathcal{A}, \mathcal{W}, \mathcal{M})$,

$$\mathbb{P}[(A_t, W_t, M_t) = (a, w, m) | \mathcal{F}_{t-1}] \geq \gamma_1, t \in [T]. \quad (5)$$

As for Assumption 2, it is natural to consider the stochastic environment has certain properties, that is, every context (including the counterfactual features) occur with non-vanishing probability, i.e., uniformly over time. It is necessary since the learner needs to collect observations for both the factual contexts and their counterfactual counterparts to estimate the counterfactual effects as we make no other assumptions about how their rewards are related. Our guarantees in Proposition 4 and Theorem 1 are stated for horizons T that are sufficiently large relative to γ_1 (e.g., $\gamma_1 T \gtrsim 1$), so that for each context certain statistics (that arise in the constraints) can be estimated sufficiently well.

Assumption 3 (Context Statistics). We assume the conditional probabilities $\mathbb{P}(m | a, w)$ and $\mathbb{P}(w | a)$ are known to the learner for all $a \in \mathcal{A}$, $m \in \mathcal{M}$, $w \in \mathcal{W}$.

This is a standard assumption in causal bandits with background knowledge (Lattimore et al., 2016; Lu et al., 2020; Huang et al., 2022b), where these conditional distributions can be obtained from domain expertise or offline data collected from the environment. In contrast, we do not assume the reward model is known and learn it online.

3.1. Online Learning Algorithm

In this section, we present the main components of our algorithm. See Algorithm 1 for pseudocode. Let $N_{d,a,m,w}(t) := \sum_{\tau=1}^t \mathbf{1}\{d_\tau = d, a_\tau = a, m_\tau = m, w_\tau = w\}$ denote the number of times that decision d has been played up to round t under context $X = (a, m, w)$ and set $n_{d,a,m,w}(t) := \max\{1, N_{d,a,m,w}(t)\}$. Let $s := \max(\{u < t : X_u = X_t\} \cup \{0\})$ be the previous round with the same context.

The algorithm proceeds in two phases. (i) *Optimistic feasibility estimation.* We form plug-in estimators $\hat{\Delta}_d^{(i)}(X_t)$ of the counterfactual fairness effects, and, together with an optimism bonus $b_t^{(j)}(\pi)$, $j \in \{1, 2, 3\}$ we will later propose in Proposition 3, we construct an estimated feasible set $\hat{\Pi}_{X_t}^f = \bigcap_{i=1}^3 \{\pi \in \Pi : \mathbb{E}_{d \sim \pi} |\hat{\Delta}_d^{(i)}(X_t)| - \sum_{j \in \{1, 2, 3\}} b_t^{(j)}(\pi) \leq c_i\}$, where $\hat{\Delta}_d^{(i)}(X_t)$, $i = \{1, 2, 3\}$ are:

$$\sum_{m,w} (\hat{y}_{a'_t, d_t, m, w} - \hat{y}_{a_t, d_t, m, w}) \mathbb{P}(m|a_t, w) \mathbb{P}(w|a_t) \quad (6)$$

$$\sum_{m,w} \hat{y}_{a'_t, d_t, m, w} (\mathbb{P}(m|a_t, w) - \mathbb{P}(m|a'_t, w)) \mathbb{P}(w|a_t) \quad (7)$$

$$\sum_{m,w} \hat{y}_{a'_t, d_t, m, w} \mathbb{P}(m|a'_t, w) (\mathbb{P}(w|a_t) - \mathbb{P}(w|a'_t)). \quad (8)$$

Here $\hat{y}_{a_t, d_t, m, w}$ and $\hat{y}_{a'_t, d_t, m, w}$ denote the empirical mean rewards for decision d_t evaluated at round t for contexts $\{a_t, m, w\}$ and $\{a'_t, m, w\}$ where m, w are the possible values in the support of M and W . (ii) *Regret minimization over $\hat{\Pi}_{X_t}^f$.* We perform one online mirror-descent step over the simplex with step size β to obtain the unconstrained OMD iterate $\hat{\pi}_{X_t}$. We compute the Bregman minimization onto the convex set $\hat{\Pi}_{X_t}^f$ and select the resulting projected feasible policy $\tilde{\pi}_{X_t}$ with an epsilon greedy exploration.

3.2. Regret and Constraint Violations Bounds

One of the main challenges for the analysis of Algorithm 1 is to design the optimistic bonus such that $\Pi_{X_t}^f \subseteq \hat{\Pi}_{X_t}^f$ holds for all $t \in [T]$ with high probability while remaining sufficiently tight to control constraint violations.

Proposition 3. Given a $\delta > 0$, define the probability kernels $\mathbb{P}_{a_t}^{(1)}(m, w) = \mathbb{P}(m|a_t, w) \mathbb{P}(w|a_t)$, $\mathbb{P}_{a_t}^{(2)}(m, w) = \mathbb{P}(m|a'_t, w) \mathbb{P}(w|a_t)$, $\mathbb{P}_{a_t}^{(3)}(m, w) = \mathbb{P}(m|a'_t, w) \mathbb{P}(w|a'_t)$ on $\mathcal{M} \times \mathcal{W}$ w.r.t. the pair of factual and counterfactual attribute realizations a_t and a'_t , respectively, at

Algorithm 1 Epsilon-Greedy OMD for Contextual Bandits with Counterfactual Fairness Constraints

Require: Decision set $\mathcal{D}$, context set $\mathcal{X}$, horizon T , learning rate $\beta = \sqrt{|\mathcal{X}| \log(2|\mathcal{D}|)/T}$, fairness thresholds c_1, c_2, c_3 , exploration rate ϵ_t , context statistics, failure probability δ , uniform distribution π_0 .

- 1: **for** $t = 1, \dots, T$ **do**
- 2: Observe the context information $X_t = (a_t, m_t, w_t)$.
- 3: Let $s := \max(\{u < t : X_u = X_t\} \cup \{0\})$ be the previous round with the context X_t with the convention $\pi_{X_0} := \pi_0$ and $\hat{\Pi}_{X_0}^f := \Pi$.
- 4: Update the OMD iterate and select the policy:

$$\tilde{y}_{s,d}(X_t) = \begin{cases} 1, & \text{if } N_{a_t,d,m_t,w_t}(s) = 0, \\ \min\left\{1, \hat{y}_{s,d}(X_t) + \sqrt{\frac{\log(2/\delta)}{2N_{a_t,d,m_t,w_t}(s)}}\right\}, & \text{o/w} \end{cases} \quad \forall d \in \mathcal{D}$$

$$\hat{\pi}_{X_t}(d) \leftarrow \pi_{X_s}(d) e^{\beta(\tilde{y}_{s,d}(X_t) - 1)}, \quad \forall d \in \mathcal{D} \quad (9)$$

$$\pi_{X_t} \leftarrow \arg \min_{\pi \in \hat{\Pi}_{X_t}^f} B_{\Psi}(\pi \| \hat{\pi}_{X_t}) \quad (10)$$

$$\tilde{\pi}_{X_t} \leftarrow (1 - \epsilon_t)\pi_{X_t} + \epsilon_t\pi_0. \quad (11)$$

- 5: Sample $d_t \sim \tilde{\pi}_{X_t}$, receive y_t , update empir. means.
- 6: Estimate feasible policy set: $\hat{\Pi}_{X_t}^f = \bigcap_{i=1}^3 \{\pi \in \Pi : \mathbb{E}_{d \sim \pi} |\hat{\Delta}_d^{(i)}(X_t)| - \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) \leq c_i\}$.
- 7: **end for**

round t . Let $b_t^{(i)}(\pi) = \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(i)}, d \sim \pi} \left(\sqrt{\frac{\log(2/\delta)}{2n_{a_t',d,m,w}(t)}} + \sqrt{\frac{\log(2/\delta)}{2n_{a_t,d,m,w}(t)}} \right)$, then for all $t \in [T], i \in \{1, 2, 3\}, \pi \in \Pi$, with probability at least $1 - T|\mathcal{X}||\mathcal{D}|\delta$,

$$\mathbb{E}_{d \sim \pi} [|\Delta_d^{(i)}(X_t)| - |\hat{\Delta}_d^{(i)}(X_t)|] \leq \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi).$$

Then $\Pi_{X_t}^f \subseteq \hat{\Pi}_{X_t}^f$ w.p. at least $1 - T|\mathcal{X}||\mathcal{D}|\delta, \forall t \in [T]$.

See Appendix E.2 for the proof. In contrast to existing constrained (contextual) MABs, at every round, our problem setting only receives bandit feedback for the realized contexts, while the constraints depend on *both* the realized (factual) contexts and the alternative (counterfactual) contexts. From Proposition 2, the fairness of one action d_t in one context X_t depends on mean rewards across many other contexts. In some of those other contexts, that decision d_t may have low rewards and even be unfair (in expectation and/or empirically). Nonetheless, we will need to have played that action d_t in those other contexts to estimate its fairness (sufficiently well) in the current context X_t .

For Algorithm 1, in the unconstrained setting where only the OMD update (9) is applied, the minimal probability for any specific decision could decay exponentially in time. When structural counterfactual fairness constraints are imposed, under the combined OMD and Bregman projection updates

(Equations (9) and (10)), the minimal probability for the output policy $\pi_{X_t}(d)$ depends on the geometry of the constrained set $\hat{\Pi}_{X_t}$ on the face of the simplex, and could be nearly 0. We incorporate an epsilon-greedy step (11).

Proposition 4. Given $\gamma_1 > 0$ and $\epsilon_t = t^{-\omega}$. Let $\mathcal{D}$ be finite domain of decisions. Consider the decisions are draw from policy $\tilde{\pi}_{X_t}(X_t = (a_t, w_t, m_t))$ output from Algorithm 1. Under Assumptions 1, 2, and 3, for any $a \in \mathcal{A}, d \in \mathcal{D}, w \in \mathcal{W}, m \in \mathcal{M}$, the following results hold for Algorithm 1:

$$\mathbb{E} \left[\sum_{t=1}^T \sqrt{1/n_{a,d,w,m}(t)} \right] \leq 2\sqrt{|\mathcal{D}|/\gamma_1} T^{\frac{1+\omega}{2}}.$$

Then with Proposition 3 and 4, we obtain the main theoretical guarantees for Algorithm 1.

Theorem 1. Given $\delta > 0, \gamma_1 > 0, \epsilon_t = t^{-\omega}$. Let $\mathcal{X}$ be a finite context set where $\mathcal{X} = \mathcal{A} \times \mathcal{M} \times \mathcal{W}$, and $\beta = \sqrt{|\mathcal{X}| \log(2|\mathcal{D}|)/T}$. Under Assumptions 1, 2, and 3, Algorithm 1 has the following guarantees:

$$R_1(T) = \tilde{O}(\sqrt{T|\mathcal{X}||\mathcal{D}| \log(1/\delta) \log(2|\mathcal{D}|/\delta)} + T^{1-\omega}) \quad (12)$$

with probability at least $1 - 3T|\mathcal{X}||\mathcal{D}|\delta$, and

$$V_1(T) = \tilde{O}(|\mathcal{D}| \sqrt{|\mathcal{X}|/\gamma_1} T^{\frac{1+\omega}{2}} + T^{1-\omega}). \quad (13)$$

Remark 1. Our proposed Algorithm 1 achieves regret and cumulative constraint violation bounds of order $T^{\max\{1/2, 1-\omega\}}$ and $T^{\max\{\frac{1+\omega}{2}, 1-\omega\}}$, respectively, where ω is the user-selected exploration parameter. To ensure sub-linear regret and constraint violations, over time the learner must have increasingly good estimates of decisions' expected rewards under both the factual (observed) and counterfactual contexts at each time t . Algorithm 1 has $O(1)$ computational complexity per round (with respect to t).

4. Logistic Bandits with Counterfactual Fairness Constraints

The upper bounds in Theorem 1 (see Equations (12) and (13)) exhibit a $\sqrt{|\mathcal{X}|}$ dependence in the nonparametric contextual bandits setting, which can be prohibitive when the context space is large (e.g., exponential in the feature dimension). In this section, we study problem settings with additional structure, where rewards are linked to low-dimensional context features to replace the $\sqrt{|\mathcal{X}|}$ factor with dependence on the feature dimension appeared in the regret and constraint violations upper bounds.

Consider at round t , the learner observes a context $X_t = \{(a_t, w_t, m_t)\} \in \mathcal{X}$, selects an action $d_t \in \mathcal{D}$, and then observes a binary reward $y_t \in \{0, 1\}$. Denote $\mathcal{Z} := \mathcal{X} \times \mathcal{D}$, and let $f : \mathcal{Z} \rightarrow \mathbb{R}^n$ be a known feature map (Lattimore & Szepesvári, 2020; Chen et al., 2025) and $\sigma(x) = (1 +$

$e^{-x})^{-1}$ be the logistic function. Let $f(Z_t) \in \mathbb{R}^n$ denote the feature vector constructed from X_t and d_t . In this section, we assume that the outcome variable Y has a generalized linear relationship (Filippi et al., 2010; Fauray et al., 2020) with its feature vector, specifically,

$$\mathbb{E}[Y|d_t, a_t, w_t, m_t] = \sigma(f(Z_t)^\top \theta_*),$$

where $\theta_* \in \mathbb{R}^n$ with $\|\theta_*\|_2 \leq S$ is an unknown reward parameter in $\mathbb{R}^n$ and S is known to the learner (Lee et al., 2024; Zhang et al., 2025). We denote the counterfactual value of the specified attribute a_t as a'_t . Thus, the expected reward for the counterfactual feature a'_t is represented as

$$\mathbb{E}[Y|a'_t, d_t, w_t, m_t] = \sigma(f(Z'_t)^\top \theta_*),$$

where $f(Z'_t) \in \mathbb{R}^n$ is the counterfactual feature vector constructed from $\{a'_t, d_t, w_t, m_t\}$. We then define the cumulative regret and cumulative constraint violations as:

$$R_2(T) = \sum_{t=1}^T \left[\max_{\pi \in \Pi_t^f} \langle \pi, \sigma(f(Z_d)^\top \theta_*) \rangle - \sigma(f(Z_t)^\top \theta_*) \right]$$

$$V_2(T) = \max_{i \in \{1,2,3\}} \left\{ \sum_{t=1}^T \mathbb{E}_{d \sim \pi_t} |\Delta_d^{(i)}(\theta_*)| - c_i \right\},$$

where π_t (14) is the selected policy at step t from Algorithm 2 and $\Pi_t^f = \bigcap_{i=1}^3 \{\pi \in \Pi : \mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(\theta_*)| \leq c_i\}$ is the (unknown) true feasible set at round t . The structure counterfactual fairness measures $\Delta_d^{(i)}(\theta_*)$, $i \in \{1, 2, 3\}$ for the logistic bandits setting are simplified to:

$$\begin{aligned} & \sum_{m,w} (\sigma(f(Z'_{a'_t, d_t, w, m})^\top \theta_*) - \sigma(f(Z_{a_t, d_t, w, m})^\top \theta_*)) \mathbb{P}(m, w|a_t) \\ & \sum_{m,w} \sigma(f(Z'_{a'_t, d_t, w, m})^\top \theta_*) (\mathbb{P}(m|a_t, w) - \mathbb{P}(m|a'_t, w)) \mathbb{P}(w|a_t) \\ & \sum_{m,w} \sigma(f(Z'_{a'_t, d_t, w, m})^\top \theta_*) \mathbb{P}(m|a'_t, w) (\mathbb{P}(w|a_t) - \mathbb{P}(w|a'_t)). \end{aligned}$$

Here, $f(Z_{a_t, d_t, w, m}) \in \mathbb{R}^n$ and $f(Z'_{a'_t, d_t, w, m}) \in \mathbb{R}^n$ are feature vectors that are known to the learner constructed from $\{a_t, d_t, w, m\}$ and $\{a'_t, d_t, w, m\}$, respectively.

Assumption 4 (Covariate diversity (Li et al., 2017)). There exists a constant $\gamma_2 > 0$, such that

$$\lambda_{\min} \left(\mathbb{E}_{X \sim \mathbb{P}_X} \left[\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} f(Z_{X,d}) f(Z_{X,d})^\top \right] \right) \geq \gamma_2.$$

This condition is analogous to the context coverage condition (Assumption 2) in the non-parametric setting that every context has a positive support.

Without loss of generality, we make the following standard assumption (Slivkins et al., 2019):

Assumption 5. The feature mapping function $f : \mathcal{Z} \mapsto \mathbb{R}^n$ is bounded, i.e., $\|f(Z)\|_2 \leq 1, \forall Z \in \mathcal{Z}$.

Algorithm 2 A Doubly Optimistic Algorithm for Fair Causal Logistic Bandits

Require: Confidence level α , learning rate η , fairness threshold c_1, c_2, c_3 , regularization parameter λ , horizon T , context statistics, parameter norm bound S .

- 1: Initialize $H_1 = \lambda I_n$ and $\hat{\theta}_1 = \mathbf{0}_n \in \mathbb{R}^n$.
 - 2: **for** $t = 1, \dots, T$ **do**
 - 3: Observe the context $X_t = (a_t, m_t, w_t)$.
 - 4: **if** $t \leq T_0$ **then** randomly sample $d_t \sim \mathcal{D}$, receive y_t .
 - 5: **if** $t > T_0$ **then**
 - 6: Construct the confidence region: $\mathcal{C}_t(\alpha) = \{\theta \in \Theta : \|\hat{\theta}_t - \theta\|_{H_t} \leq \gamma_t(\alpha)\}$.
 - 7: Estimate the feasible policy set: $\hat{\Pi}_t^f = \{\pi \in \Pi : \mathbb{E}_{d \sim \pi} [|\Delta_d^{(i)}(X_t; \hat{\theta}_t)| - \xi_i(d)] \leq c_i, \forall i \in [3]\}$.
 - 8: Compute the optimistic reward and select a policy from the estimated feasible set:

$$\tilde{y}_t(d) \leftarrow \max_{\theta \in \mathcal{C}_t(\alpha)} \sigma(f(Z_{a_t, d, m_t, w_t})^\top \theta), \quad \forall d \in \mathcal{D}$$

$$\pi_t \leftarrow \arg \max_{\pi \in \hat{\Pi}_t^f} \langle \pi, \tilde{y}_t \rangle, \quad (14)$$
 - 9: Sample decision $d_t \sim \pi_t$ and receive reward y_t .
 - 10: **end if**
 - 11: Estimate the reward parameter using OMD:

$$\hat{\theta}_{t+1} \leftarrow \arg \min_{\theta \in \Theta} \left[\langle \nabla \mathcal{L}_t(\hat{\theta}_t), \theta - \hat{\theta}_t \rangle + \frac{1}{2\eta} \|\theta - \hat{\theta}_t\|_{H_t}^2 + \frac{1}{2} \|\theta - \hat{\theta}_t\|_{\nabla^2 \mathcal{L}_t(\hat{\theta}_t)}^2 \right]. \quad (15)$$
 - 12: Update the cumulative Hessian matrix: $H_{t+1} = H_t + \dot{\sigma}(f(Z_t)^\top \hat{\theta}_{t+1}) f(Z_t) f(Z_t)^\top$.
 - 13: **end for**
-

We next define a quantity that impacts learnability.

Definition 3 (Problem Dependent Constant).

$$\kappa = \max_{Z \in \mathcal{Z}} \max_{\theta \in \Theta} 1/\dot{\sigma}(f(Z)^\top \theta),$$

where $\Theta = \{\theta \in \mathbb{R}^n : \|\theta\|_2 \leq S\}$. Briefly, κ measures the level of non-linearity of the reward distribution, typically around $\mathcal{O}(e^S)$, growing exponentially with the parameter-norm bound S . We recall other problem dependent constant discussed in Table 1: $\kappa_* = 1/\dot{\sigma}(f(Z_*)^\top \theta_*)$ where $f(Z_*) = \arg \max_{Z \in \mathcal{Z}} \sigma(f(Z)^\top \theta_*)$ and $\sqrt{1/\kappa_*} < 1 \ll \kappa$.

4.1. Efficient Online Estimation

In this section, we describe the estimation method for parametric logistic bandits setting. Many prior works on logistic bandits (Fauray et al., 2020; Abeille et al., 2021; Chen et al., 2025) rely on maximum-likelihood estimation (MLE) to estimate the unknown reward parameter θ_* and construct a confidence set accordingly. However, a significant limitation

of MLE estimation is $\mathcal{O}(t)$ per round complexity. Inspired by recent works on logistic bandits (Zhang & Sugiyama, 2023; Zhang et al., 2025), we adopt online mirror descent (OMD) to estimate the reward parameter θ_* (see (15)) and construct a tight confidence set as follows:

Proposition 5 ((Zhang et al., 2025) Theorem 1). *Let the learning rate $\eta = 1 + S$ and the regularization parameter $\lambda = 2 \max\{7n\eta, 3\eta S, 1\}$. For each $t \in [T]$ and any $\alpha \in (0, 1]$, define the confidence set as*

$$\mathcal{C}_t(\alpha) = \left\{ \theta \in \Theta : \|\hat{\theta}_t - \theta\|_{H_t} \leq \gamma_t(\alpha) \right\}, \quad (16)$$

where $H_t = \lambda I_n + \sum_{s=1}^{t-1} \dot{\sigma}(f(Z_s)^\top \hat{\theta}_{s+1}) f(Z_s) f(Z_s)^\top$ is the cumulative Hessian and $\gamma_t(\alpha) = \sqrt{4\lambda S^2 + 2\eta \ln(\frac{1}{\alpha}) + 6n\eta^2 \ln(2 + \frac{2t}{\lambda})}$ is the confidence radius. Then, we have $\mathbb{P}(\forall t \geq 1, \theta_* \in \mathcal{C}_t(\alpha)) \geq 1 - \alpha$.

The confidence set $\mathcal{C}_t(\alpha)$ is a convex ellipsoid centered at $\hat{\theta}_t$. This construction is both statistically and computationally efficient compared with MLE estimator: (1) The OMD estimator (15) involves a quadratic optimization over the Euclidean ball. It can be solved with a computation cost of $\mathcal{O}(n^3)$ in the worst case, independent of time t ; in contrast to refitting an MLE, whose per-round cost typically increases with t due to repeated passes over past data. (2) The confidence radius (16) satisfies $\gamma_t(\alpha) = \mathcal{O}(\sqrt{n \ln t})$ matching the state of the art radius for the MLE estimator (Lee et al., 2024) with respect to the dimension n and t .

4.2. Regret and Constraint Violations Bounds

In this section, we present theoretical analysis for constrained logistic bandits. Unlike Section 3, this (parametric) logistic setting requires constructing the optimistic estimated feasible set $\hat{\Pi}_t^f$ with a bonus that both guarantees $\Pi_t^f \subseteq \hat{\Pi}_t^f$, $\forall t \in [T]$ with high probability and ensures $\hat{\Pi}_t^f$ is convex in π . We define the bonus term as

$$\begin{aligned} \xi_t(d) = & \left(\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \|f(Z'_{a_t,d,m,w})\|_{H_t^{-1}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \|f(Z'_{a_t,d,m,w})\|_{H_t^{-1}} \right. \\ & \left. + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \|f(Z_{a_t,d,m,w})\|_{H_t^{-1}} \right) \cdot \frac{\gamma_t(\alpha)}{4}. \end{aligned}$$

This construction follows the optimistic feasible set principle from constrained bandits, adapted to the counterfactual constraints. The natural optimistic set $\bigcap_{i=1}^3 \{\pi \in \Pi : \exists \theta \in \mathcal{C}_t(\alpha), \langle \pi, \Delta_d^{(i)}(X_t; \theta) \rangle \leq c\}$ is non-convex in π ; we therefore use a fixed, policy-independent bonus $\xi_t(d)$, which makes the relaxed constraint linear in π while still ensuring $\Pi_t^f \subseteq \hat{\Pi}_t^f$ w.h.p. And we have,

Proposition 6. *In the logistic bandits setting, for any $\alpha \in (0, 1]$, with probability at least $1 - \alpha$, it holds that $\Pi_t^f \subseteq \hat{\Pi}_t^f$ for all $t \in [T]$.*

The main challenge in the parametric logistic bandit setting is to characterize the richness of the feature span available for identifying the shared reward parameter θ_* , in order to accurately estimate fairness (which per Proposition 2 depends on many contexts). Covariate diversity (Assumption 4) guarantees feature diversity under uniformly random decisions, but not necessarily under a context-dependent policy (such as playing OFU while ignoring constraints).

To address this issue, analogous to Section 3, we incorporate an exploration phase and obtain the following result:

Proposition 7. *Let $\delta_2 > 0$, with probability at least $1 - \delta_2$, for every $a \in \mathcal{A}$, $d \in \mathcal{D}$, $m \in \mathcal{M}$, and $w \in \mathcal{W}$,*

$$\sum_{t=1}^T \|f(Z_{a,d,m,w})\|_{H_t^{-1}} = \mathcal{O}\left(\sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}}\right)\right).$$

Then with Proposition 6 and 7, we obtain the main theoretical guarantees for Algorithm 2.

Theorem 2. *Let $\eta = 1 + S$, and $\lambda = 2 \max\{7n\eta, 3\eta S, 1\}$. Under Assumptions 1, 3, 4 and 5, for any $\alpha \in (0, 1]$, $\delta_1 \in (0, 1]$, and $\delta_2 \in (0, 1]$, Algorithm 2 achieves the following bounds with probability at least $1 - \alpha - \delta_1 - \delta_2$,⁴*

$$\begin{aligned} R_2(T) &= \tilde{\mathcal{O}}\left(nS\sqrt{T - T_0} + \kappa n^2 S^2 + T_0\right) \\ V_2(T) &= \tilde{\mathcal{O}}\left(S\sqrt{\frac{n\kappa}{\gamma_2}} \frac{T}{\sqrt{T_0}} + T_0\right). \end{aligned}$$

For unconstrained logistic bandits, κ_* is typical in the leading term. Because of the constraints in our setting, we have different dependence on problem-dependent constants. This manifests from our problem in bounding different terms than standard (unconstrained) logistic bandits.

Remark 2. (i) The leading term of our regret $\tilde{\mathcal{O}}(n\sqrt{T})$, matches the best-known rates reported by (Faury et al., 2022; Lee et al., 2024; Zhang et al., 2025) in the logarithmic dependence on n and T . (ii) Our regret bound does not depend on the Slater's margin (the strict feasibility gap) which appears in (Chen et al., 2025). Consequently, when imposing multiple counterfactual constraints (i.e., Ctf-DE, Ctf-IE, and Ctf-SE), the effective margin may become very small, leading to looser guarantees for (Chen et al., 2025). (iii) Our method achieves $\mathcal{O}(1)$ time per round (with respect to t), matching standard implementations for unconstrained logistic bandits (Zhang & Sugiyama, 2023; Zhang et al., 2025). By contrast, the constrained logistic bandits algorithm of (Chen et al., 2025) incurs $\mathcal{O}(t)$ time complexity per round.

⁴ $R_2(T)$ hides dependencies on $\log \frac{1}{\alpha}$ and $\log(\frac{1}{\delta_1})$. $V_2(T)$ hides dependencies on $\log \frac{1}{\alpha}$ and $\log \frac{1}{\delta_2}$.

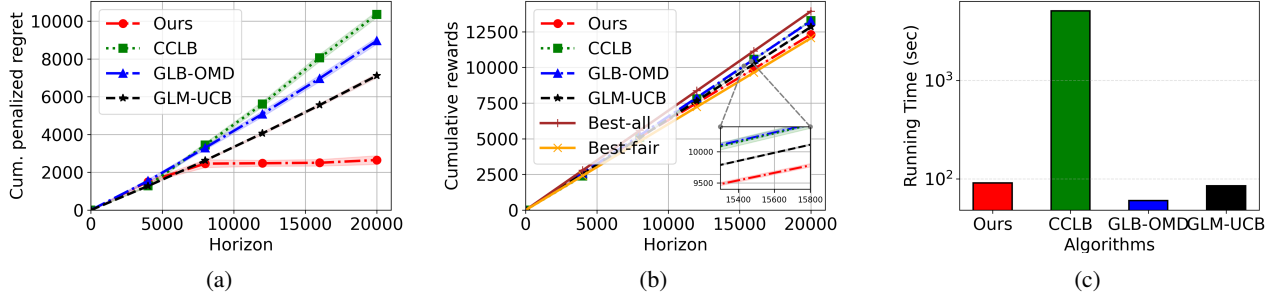


Figure 2. Plots for different algorithms: Algorithm 2 (ours), GLB-OMD, CCLB, GLM-UCB on (a) cumulative penalized regret; (b) cumulative rewards; (c) running time (for $T = 10,000$ steps).

5. Numerical Experiments

We next empirically evaluate our proposed method for the logistic setting on a synthetic data set. See Appendix I for experimental details.

Data set description. We generated the synthetic dataset from a structural causal model, i.e.,

$$\mathcal{F}^* : \begin{cases} A_t \sim \text{Uniform}(\{-1, 1\}), \\ W_t | A_t \sim \text{Bernoulli}(\sigma(1.2A_t)), \\ M_t | A_t, W_t \sim \text{Bernoulli}(\sigma(A_t + 2W_t - 1)), \\ \mathcal{D}_t \leftarrow \{D_{t,1}, D_{t,2}, D_{t,3}, \dots, D_{t,10}\}, \\ Y_t \leftarrow \text{Bernoulli}(\sigma(f(Z_{A_t, W_t, M_t, D_t})^\top \theta_*)), \forall D_t \in \mathcal{D}_t. \end{cases}$$

At every round, we generate a set of 10 feature vectors $\{[A_t, W_t, M_t, D_{t,i}]\}_{i=1}^{10}$ along with their corresponding counterfactual feature vectors. We use rejection sampling over the sets to make sure that at least one of the feature vectors is feasible.

Algorithms. We evaluate four different algorithms: GLM-UCB (Filippi et al., 2010) (unconstrained generalized linear bandits), GLB-OMD (Zhang et al., 2025) (unconstrained logistic bandits), CCLB (Chen et al., 2025) (causal logistic bandits with CATE), and DO-LogB (our method; causal logistic bandits with Ctf-DE, -IE, -SE; Algorithm 2)⁵.

Metrics. We evaluated the algorithms using a penalized form of cumulative regret for different horizons along with cumulative rewards, and running time. For the penalized cumulative regret, when the action picked by the learner violates the counterfactual fairness constraint, the learner still observes the reward value (i.e. the learner can improve the reward parameter estimate $\hat{\theta}$), but we count the reward earned as being 0.

Results. Beginning with penalized cumulative regret, where rewards are only received for fair actions, there is a large gap

between our method and the methods of GLM-UCB, GLB-OMD, and CCLB, with the gap growing larger for longer horizons (Figure 2a). This is expected since GLM-UCB and GLB-OMD do not account for constraints. We observe that the CCLB baseline exhibits even larger penalized cumulative regret than both GLM-UCB and GLB-OMD, despite CCLB being designed for a fairness constrained setting (though for a different fairness metric). In Figure 2b, our method attains cumulative rewards on the same order as other algorithms and consistently dominates the Best-fair oracle. For running time, GLB-OMD performs slightly better than our method (which seeks to satisfy the constraint) (Figure 2c). CCLB has the highest computational cost, taking substantially longer than the other methods.

6. Conclusion

We study the problem of achieving fairness in a sequential decision-making framework, where fairness is measured using structural counterfactual mechanisms along the direct, indirect and spurious pathways that are treated as multiple long-term constraints for the learning process. A notable challenge is that the fairness of one action in the current context depends on expected rewards of that action across many other contexts; in some of those other contexts, the action may have low reward and/or even be unfair. One interesting direction to explore for parametric rewards is whether in some application settings common classes of feature maps exhibit properties that could obviate the need for the exploration phase. Another interesting direction is to extend our framework to MDPs.

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⁵The source code is available at <https://github.com/jchen-research/DO-LogB.git>

Impact Statement

This paper presents work whose goal is to advance the field of machine learning. The results in this paper may have implications bringing together strong practical results in causal decision-making. Such integration is anticipated to contribute to the development of more intelligent and reliable AI systems. While a comprehensive exploration of all implications is beyond the scope of this work, an enhanced understanding of causality and decision-making is expected to increase accountability in AI models.

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A. Table of Notation

We list the notations used in this paper in Table 2, for the convenience of reference.

Table 2. Notations used in the paper.

Symbol	Description
A_t	Sensitive attribute (protected feature).
W_t	Confounding features (non-sensitive covariates).
M_t	Intermediate (mediator) features.
D_t	Decision variable.
Y_t	Reward (outcome) variable.
X_t	Context at round t , including (A_t, W_t, M_t) .
$TV_{a_t, a'_t}(d_t)$	Total variation.
$\text{Ctf-DE}_{a_t, a'_t}(d_t)$	Structural counterfactual direct effect.
$\text{Ctf-IE}_{a_t, a'_t}(d_t)$	Structural counterfactual indirect effect.
$\text{Ctf-SE}_{a_t, a'_t}(d_t)$	Structural counterfactual spurious effect.
β	Learning rate for non-parametric setting.
c_i	User-defined fairness threshold for constraint i .
ω	User-selected exploration parameter.
T	Time horizon.
$\tilde{\pi}_{X_t}$	Output policy in the non-parametric contextual bandit setting at context X_t .
$\Delta_{d_t}^{(i)}(X_t)$	i -th structural counterfactual effect for decision d_t under context X_t .
$\hat{\Delta}_{d_t}^{(i)}(X_t)$	Estimate of the i -th structural counterfactual effect.
$\Pi_{X_t}^f$	Set of true counterfactually fair policies of context X_t .
$\hat{\Pi}_{X_t}^f$	Estimated set of counterfactually fair policies at context X_t .
γ_1	Context-coverage probability.
$n_{d,a,m,w}(t)$	Count of tuples (d, a, m, w) up to round t .
δ	Failure probability level for the non-parametric bandit analysis.
$f(Z_{d_t})$	Feature vector constructed from (X_t, d_t) .
$f(Z'_{d_t})$	Counterfactual feature vector constructed from (X'_t, d_t) .
n	Dimension of the reward parameter.
S	Known upper bound on $\ \theta\ _2$.
$\sigma(\cdot)$	Logistic (sigmoid) function.
θ_*	Unknown true reward parameter.
$\hat{\theta}_t$	Estimated reward parameter.
κ	Problem-dependent constant.
α	Failure probability level for parametric logistic bandits.
$\mathcal{L}_s(\theta)$	Logistic loss function at round s .
$\mathcal{C}_t(\alpha)$	Confidence set at round t with confidence level $1 - \alpha$.
$\gamma_t(\alpha)$	Confidence radius at round t with confidence level $1 - \alpha$.
H_t	Cumulative Hessian matrix of the (regularized) logistic loss up to round t .
λ	Regularization parameter.
$b_t^{(i)}(\pi), i \in \{1, 2, 3\}$	Fairness bonus term for non-parametric setting.
$\xi_t(d)$	Fairness bonus term for parametric setting.

B. Preliminaries

We provide additional notations and definitions used throughout the paper. Each SCM is associated with a directed acyclic graph (DAG) $\mathcal{G}$, called the causal diagram, where nodes correspond to endogenous variables $i \in V$, solid arrows represent arguments of each function f_i . A bi-directed arrow between nodes i and j indicates an unobserved confounder affecting both i and j , i.e., $U_i \cap U_j \neq \emptyset$. We will use the graph-theoretic family abbreviations, e.g., $Pa(V)_{\mathcal{G}}$ stand for the set of parents of V in $\mathcal{G}$. We omit the subscript $\mathcal{G}$ when it is obvious. A path from a node X to a node Y in $\mathcal{G}$ is a sequence of edges which does not include a particular node more than once. For a twice-differentiable function σ , the notation $\dot{\sigma}$ and $\ddot{\sigma}$ denote the first and second derivative of function σ respectively. For a random variable Z , let $\mathcal{Z}$ represent the domain of Z and for discrete-valued Z let $|\mathcal{Z}|$ denote its cardinality. For two real-valued symmetric matrices A and B , the notation $A \geq B$ indicates that $A - B$ is positive semi-definite, and when A is positive definite, we denote A -norm for a vector z as $\|z\|_A = \sqrt{z^\top A z}$. Finally, for two univariate real-valued functions f and g , we denote $f = \tilde{\mathcal{O}}(g)$ to indicate that g dominates f up to logarithmic factors; and for an event $E \in \Omega$, we write $\mathbb{1}\{E\}$ the indicator function of E .

Definition 4 (Conditional Average Treatment Effect (CATE)). Given a SCM $\mathcal{M}^*$, under an intervention $do(D = d_t)$, the conditional average treatment effect of event $A = a'_t$ (with factual value a_t) on Y conditioned on m_t and w_t is defined as:

$$\text{CATE}_{a_t, a'_t}(d_t | m_t, w_t) = \mathbb{E}[Y_{d_t} | do(a'_t), w_t, m_t] - \mathbb{E}[Y_{d_t} | do(a_t), w_t, m_t]. \quad (17)$$

Definition 5 (Path-specific Effect (PSE)). Given a SCM $\mathcal{M}^*$, suppose $\mathcal{C}_0$ describes a causal pathway from A to Y , under an intervention $do(D = d_t)$, the path-specific effect of event $A = a'_t$ (with factual value a_t) on Y along path $\mathcal{C}_0$ is defined as:

$$\text{PSE}_{a_t, a'_t}^{\mathcal{C}_0}(Y_{d_t}) = \mathbb{E}[Y_{d_t, (a'_t)^{\mathcal{C}_0}, (a_t)^{\overline{\mathcal{C}_0}}}] - \mathbb{E}[Y_{d_t, a_t}].$$

Where $Y_{d_t, (a'_t)^{\mathcal{C}_0}, (a_t)^{\overline{\mathcal{C}_0}}}$ denotes the path-specific potential outcome of Y under the edge intervention of $do(A = a'_t)$ along edges in $\mathcal{C}_0$ and $A = a_t$ along the remaining outgoing edges from A to Y (those not in $\mathcal{C}_0$).

Definition 6 (Probability Coupling). Using the standard formalism of probability theory, let X_1 and X_2 be random variables defined on probability spaces $(\Omega_1, \mathcal{F}_1, \mathbb{P}_1)$ and $(\Omega_2, \mathcal{F}_2, \mathbb{P}_2)$, respectively. A *coupling* of X_1 and X_2 is a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ together with random variables Y_1, Y_2 on $(\Omega, \mathcal{F}, \mathbb{P})$ such that

$$Y_1 \stackrel{d}{=} X_1 \quad \text{and} \quad Y_2 \stackrel{d}{=} X_2.$$

Definition 7 (First-Order Stochastic Dominance). Let X and Y be real-valued random variables with cumulative distribution functions $F_X(x) = \mathbb{P}(X \leq x)$ and $F_Y(x) = \mathbb{P}(Y \leq x)$. We say that X *first-order stochastically dominates* Y if

$$F_X(x) \leq F_Y(x) \quad \text{for all } x \in \mathbb{R}.$$

C. Additional Related Works

Logistic bandits. The logistic bandit model represents a sequential decision-making framework in which the expected reward is a logistic function of a low-dimensional context and can be viewed as a special case of generalized linear contextual bandits. Early work on parametric and generalized linear bandits established $\tilde{O}(\sqrt{dT})$ -type regret bounds for UCB-style algorithms based on confidence sets around the maximum-likelihood estimator, and already covered logistic rewards as a key binary-feedback instance (Li et al., 2010; Filippi et al., 2010; Li et al., 2017; Dong et al., 2019; Liu et al., 2024). However, these analyses typically rely on a global linearization of the link function and lead to regret bounds that depend on a non-linearity constant κ , which can be prohibitively large in adversarially chosen instances.

Focusing explicitly on logistic bandits, (Faury et al., 2020) proposed an optimistic algorithm (OFULog) based on a refined self-concordant analysis of the logistic loss, showing that one can significantly improve the dependence on κ by exploiting the curvature of the model instead of linearizing it. Building on this perspective, (Abeille et al., 2021) derived instance-wise minimax lower bounds of order $\Omega(n\sqrt{T/\kappa_*(T)})$ together with matching algorithms, demonstrating that the non-linearity of the logistic link can actually help exploration in the long run. More recently, (Faury et al., 2022) resolved the tension between statistical and computational efficiency by introducing ECOLog-type algorithms and a new convex confidence set while still attaining the instance-optimal regret rate of (Abeille et al., 2021). Additionally, another line of research studies the multinomial logit (MNL) contextual bandit problem (Agrawal et al., 2017; Oh & Iyengar, 2019; Zhang & Sugiyama, 2023; Lee et al., 2024; Hwang et al., 2025), which generalizes the binary logistic bandit by allowing the learner to offer a subset of items and observe a multiclass choice.

Causal Fairness. The body of research in fair machine learning is expanding and encompasses a variety of contexts. Within this field, three distinct tasks can be identified: (1) the detection and quantification of biases in currently deployed policies; (2) the development of fair predictive models for outcomes; and (3) the formulation of fair decision-making policies. Most early work on algorithmic fairness addressed tasks (1) and (2) using purely statistical criteria such as demographic parity, equalized odds, or calibration, which do not explicitly model the causal mechanisms that generate sensitive attributes, mediators, and outcomes.

Our work falls under the setting of online outcome control (task (3)) that explores fairness through a causal lens (Zhang et al., 2018; Huang et al., 2022a;b; Nilforoshan et al., 2022; Plecko & Bareinboim, 2023; 2024; Plečko & Bareinboim, 2025; Chen et al., 2025). Unlike us, Plecko & Bareinboim (2023); Plečko & Bareinboim (2024) explored fairness through path-specific counterfactual effects in an offline setting along with budget constraints, focusing on identifying policies that satisfy fairness constraints given a fixed batch of data. As for the online setting, Huang et al. (2022b); Chen et al. (2025) and Hu & Zhang (2022) studied achieving causal fairness on Multi-armed Bandits (MABs) and Markov Decision Process (MDP) frameworks.

Constrained MABs. There is a large body of work on bandits with different types of constraints, including knapsack bandits (Wu et al., 2015; Agrawal & Devanur, 2016), submodular maximization (Krause & Guestrin, 2007; Niazadeh et al., 2023; Nie et al., 2023), bandits with hard safety constraints (Amani et al., 2019; Pacchiano et al., 2021), and bandits with cumulative long term constraints (Liu et al., 2021; Zhou & Ji, 2022; Bernasconi et al., 2024). These formulations capture a variety of practically motivated restrictions: knapsack and budgeted bandits model global resource or capacity limits over the horizon, submodular bandits address combinatorial decision sets with diminishing returns, and safe bandits impose almost-sure or high-probability feasibility requirements at every round, often via confidence sets over an unknown safe region. As for handling long-term constraints, Zhou & Ji (2022) considered a general unknown reward function and a general unknown constraint function with constraint feedback. More broadly, this type of constrained problem has also been studied in the reinforcement learning (RL) setting (Efroni et al., 2020; Ding et al., 2021), where constrained Markov decision processes are handled by optimistic planning or primal–dual policy optimization, and constraints are again encoded as long-term costs and managed through convex optimization methods. In contrast to these works, our constraints (without direct feedback) arise from structural counterfactual fairness requirements, which couple the learner’s policy with a causal model of the environment and lead to a different geometric and statistical structure.

D. Proof for Section 2

In this section, we provide proofs and additional details for Section 2 for completeness.

Example 1 (Details). We compute different fairness measures for the following structural causal model. Here D_t is modeled as a scalar-valued intervention level. The baseline $D_t = 0$ corresponds to the null recommendation (“no item”), and the term $0.1D_t$ specifies a linear effect of intervention intensity on Y_t . W.l.o.g., at round t , let $a'_t = 1$, $a_t = 0$. The following SCM $\mathcal{M}^*$ describes the data-generating mechanisms (unknown to the decision-maker):

$$\mathcal{F}, \mathbb{P}(U) : \begin{cases} A_t & \leftarrow \mathbb{1}\{U_{A_t} < 0.5\}, \\ W_t & \leftarrow \mathbb{1}\{U_{W_t} < 0.1 + 0.7A_t\}, \\ M_t & \leftarrow \mathbb{1}\{U_{M_t} < 0.2 + 0.7A_t + 0.1W_t\}, \\ D_t & \leftarrow f(A_t, W_t, M_t, U_{D_t}), \\ Y_t & \leftarrow \mathbb{1}\{U_{Y_t} < 0.45 + 0.05A_t - 0.4W_t + 0.4M_t + 0.1D_t\}. \end{cases} \quad (18)$$

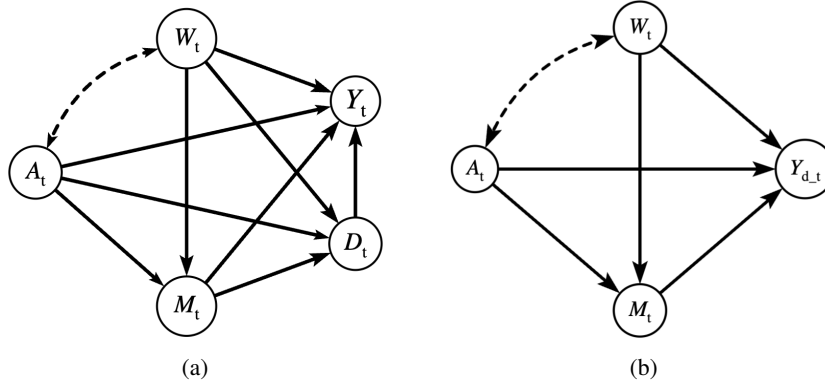


Figure 3. (a). Per-round Extended standard fairness model; (b). Interventional causal diagram $\mathcal{G}_{\bar{D}}$.

For the potential response Y_{d_t} (intervention $do(D_t = d_t)$) (see Figure 3b), the structural equation become

$$Y_{d_t} | (A_t, W_t, M_t) \sim \text{Bernoulli}(0.45 + 0.05A_t - 0.4W_t + 0.4M_t + 0.1d_t).$$

Then we could compute the value total variation ($TV_{a_t, a'_t}(d_t)$), conditional average effect ($\text{CATE}_{a_t, a'_t}(d_t | m_t, w_t)$), counterfactual direct effect ($\text{Ctf-DE}_{a_t, a'_t}(d_t)$), indirect effect ($\text{Ctf-IE}_{a'_t, a_t}(d_t)$), and spurious effect ($\text{Ctf-SE}_{a'_t, a_t}(d_t)$) as following:

$$\begin{aligned} TV_{a_t, a'_t}(d_t) &= \mathbb{E}[Y_{d_t} | A_t = a'_t] - \mathbb{E}[Y_{d_t} | A_t = a_t] = 0.078. \\ \text{CATE}_{a_t, a'_t}(d_t | m_t, w_t) &= \mathbb{E}[Y_{d_t} | do(a'_t), w_t, m_t] - \mathbb{E}[Y_{d_t} | a_t, w_t, m_t] = 0.05 \\ \text{Ctf-DE}_{a_t, a'_t}(d_t) &= \sum_{w, m} [\mathbb{E}[Y_t | d_t, a'_t, m, w] - \mathbb{E}[Y_t | d_t, a_t, m, w]] \mathbb{P}(m | w, a_t) \mathbb{P}(w | a_t) = 0.05 \\ \text{Ctf-IE}_{a'_t, a_t}(d_t) &= \sum_{w, m} \mathbb{E}[Y_t | d_t, a'_t, m, w] \cdot [\mathbb{P}(m | w, a_t) - \mathbb{P}(m | w, a'_t)] \cdot \mathbb{P}(w | a_t) = -0.28 \\ \text{Ctf-SE}_{a'_t, a_t}(d_t) &= \sum_{w, m} \mathbb{E}[Y_t | d_t, a'_t, m, w] \cdot \mathbb{P}(m | w, a'_t) \cdot [\mathbb{P}(w | a_t) - \mathbb{P}(w | a'_t)] = 0.252. \end{aligned}$$

Next, we provide the identification steps for expressing the nested counterfactuals into functions of observational data.

Proposition 2 ((Plečko & Bareinboim, 2024)). *Under the extended Standard Fairness Model (SFM) in Figure 1, $\text{Ctf-DE}_{a_t, a'_t}(d_t)$, $\text{Ctf-IE}_{a_t, a'_t}(d_t)$, and $\text{Ctf-SE}_{a_t, a'_t}(d_t)$ can be estimated, respectively, from the observational distribution as*

$$\text{Ctf-DE}_{a_t, a'_t}(d_t) = \sum_{m, w} (\mathbb{E}[Y_t | d_t, a'_t, w, m] - \mathbb{E}[Y_t | d_t, a_t, w, m]) \mathbb{P}(m | a_t, w) \mathbb{P}(w | a_t),$$

$$\begin{aligned}\text{Ctf-IE}_{a_t, a'_t}(d_t) &= \sum_{m, w} \mathbb{E}[Y_t | d_t, a_t, w, m] (\mathbb{P}(m | a'_t, w) - \mathbb{P}(m | a_t, w)) \mathbb{P}(w | a_t), \\ \text{Ctf-SE}_{a_t, a'_t}(d_t) &= \sum_{m, w} \mathbb{E}[Y_t | d_t, a_t, w, m] \mathbb{P}(m | a_t, w) (\mathbb{P}(w | a'_t) - \mathbb{P}(w | a_t)).\end{aligned}$$

Proof. We firstly compute the counterfactual direct effect:

$$\text{Ctf-DE}_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{a'_t, d_t, M_{a_t}} - Y_{a_t, d_t, M_{a_t}} | a_t] \quad (19)$$

As for $\mathbb{E}[Y_{a'_t, d_t, M_{a_t}} | a_t]$, we have,

$$\mathbb{E}[Y_{a'_t, d_t, M_{a_t}} | a_t] = \sum_w \mathbb{E}[Y_{a'_t, d_t, M_{a_t}} | w, a_t] \mathbb{P}(w | a_t) \quad (20)$$

$$= \sum_{w, m} \mathbb{E}[Y_{a'_t, d_t, m} | M_{a_t} = m, w, a_t] \mathbb{P}(M_{a_t} = m | w, a_t) \mathbb{P}(w | a_t) \quad (21)$$

$$= \sum_{w, m} \mathbb{E}[Y_{a'_t, d_t, m} | w] \mathbb{P}(M_{a_t} = m | w, a_t) \mathbb{P}(w | a_t) \quad (22)$$

$$= \sum_{w, m} \mathbb{E}[Y_t | d_t, a'_t, m, w] \mathbb{P}(m | w, a_t) \mathbb{P}(w | a_t). \quad (23)$$

Where (20) follows by law of total expectation; (21) comes from the law of total expectation and consistency for nested counterfactuals, i.e., on the event $M_{a_t} = m$, $Y_{a'_t, d_t, M_{a_t}} = Y_{a'_t, d_t, m}$; (22) follows by the cross-world independence $Y_{a'_t, d_t, m} \perp\!\!\!\perp M_{a_t} | W_t$ and sequential ignorability $Y_{a'_t, d_t, m} \perp\!\!\!\perp A_t | W_t$; and the last equality (23) holds because of consistency for nested counterfactuals and no unmeasured confounding for $D_t \rightarrow Y_t$ given A_t, M_t, W_t .

Similarly, for $\mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a_t]$ we have:

$$\mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a_t] = \sum_{w, m} \mathbb{E}[Y_t | d_t, a_t, m, w] \mathbb{P}(m | w, a_t) \mathbb{P}(w | a_t). \quad (24)$$

Then by (20), (23), and (24), the counterfactual direct effect could be computed as:

$$\begin{aligned}\text{Ctf-DE}_{a_t, a'_t}(d_t) &= \mathbb{E}[Y_{a'_t, d_t, M_{a_t}} - Y_{a_t, d_t, M_{a_t}} | a_t] \\ &= \sum_{w, m} \left[\mathbb{E}[Y_t | d_t, a'_t, m, w] - \mathbb{E}[Y_t | d_t, a_t, m, w] \right] \mathbb{P}(m | w, a_t) \mathbb{P}(w | a_t)\end{aligned} \quad (25)$$

Next, we compute the counterfactual indirect effect:

$$\text{Ctf-IE}_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{a_t, d_t, M_{a'_t}} | a_t] - \mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a_t]. \quad (26)$$

As for $\mathbb{E}[Y_{a_t, d_t, M_{a'_t}} | a_t]$, we have,

$$\begin{aligned}\mathbb{E}[Y_{a_t, d_t, M_{a'_t}} | a_t] &= \sum_w \mathbb{E}[Y_{a_t, d_t, M_{a'_t}} | w, a_t] \mathbb{P}(w | a_t) \\ &= \sum_{w, m} \mathbb{E}[Y_{a_t, d_t, m} | M_{a'_t} = m, w, a_t] \mathbb{P}(M_{a'_t} = m | w, a_t) \mathbb{P}(w | a_t) \\ &= \sum_{w, m} \mathbb{E}[Y_{a_t, d_t, m} | w, a_t] \mathbb{P}(M_{a'_t} = m | w) \mathbb{P}(w | a_t) \quad (27)\end{aligned}$$

$$= \sum_{w, m} \mathbb{E}[Y_t | d_t, a_t, m, w] \mathbb{P}(m | w, a'_t) \mathbb{P}(w | a_t). \quad (28)$$

Where (27) follows by cross-world independence $Y_{a_t, d_t, m} \perp\!\!\!\perp M_{a'_t} | W_t$ and sequential ignorability $M_{a'_t} \perp\!\!\!\perp A_t | W_t$.

Then by (24), (26), and (28) the counterfactual indirect effect could be computed as:

$$\text{Ctf-IE}_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{a_t, d_t, M_{a'_t}} | a_t] - \mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a_t]$$

$$= \sum_{w,m} \mathbb{E}[Y_t | d_t, a_t, m, w] \cdot [\mathbb{P}(m|w, a'_t) - \mathbb{P}(m|w, a_t)] \cdot \mathbb{P}(w|a_t). \quad (29)$$

We finally compute the counterfactual spurious effect:

$$\text{Ctf-SE}_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a'_t] - \mathbb{E}[Y_{d_t} | a_t]. \quad (30)$$

As for $\mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a'_t]$, we have,

$$\begin{aligned} \mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a'_t] &= \sum_w \mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a'_t, w] \mathbb{P}(w|a'_t) \\ &= \sum_{w,m} \mathbb{E}[Y_{a_t, d_t, m} | a'_t, M_{a_t} = m, w] \mathbb{P}(M_{a_t} = m | a'_t, w) \mathbb{P}(w|a'_t) \\ &= \sum_{w,m} \mathbb{E}[Y_{a_t, d_t, m} | w] \mathbb{P}(M_{a_t} = m | w) \mathbb{P}(w|a'_t) \end{aligned} \quad (31)$$

$$= \sum_{w,m} \mathbb{E}[Y_t | d_t, a_t, m, w] \mathbb{P}(m|w, a_t) \mathbb{P}(w|a'_t) \quad (32)$$

Where (31) follows by cross-world independence $Y_{a_t, d_t, m} \perp\!\!\!\perp M_{a_t} | W_t$ and sequential ignorability $Y_{a_t, d_t, m} \perp\!\!\!\perp A_t | W_t, M_{a_t} \perp\!\!\!\perp A_t | W_t$.

And for $\mathbb{E}[Y_{d_t} | a_t]$ we have:

$$\mathbb{E}[Y_{d_t} | a_t] = \sum_{w,m} \mathbb{E}[Y_t | d_t, a_t, m, w] \mathbb{P}(m|w, a_t) \mathbb{P}(w|a_t). \quad (33)$$

Then by (30), (32), and (33) the indirect effect could be computed as:

$$\begin{aligned} \text{Ctf-SE}_{a_t, a'_t}(d_t) &= \mathbb{E}[Y_{a_t, d_t, M_{a_t}} | a'_t] - \mathbb{E}[Y_{d_t} | a_t] \\ &= \sum_{w,m} \mathbb{E}[Y_t | d_t, a_t, m, w] \cdot \mathbb{P}(m|w, a_t) \cdot [\mathbb{P}(w|a'_t) - \mathbb{P}(w|a_t)]. \end{aligned} \quad (34)$$

Which finishes the proof. ■

Proposition 1 ((Plečko & Bareinboim, 2024)). *For every step $t \in [T]$, under an intervention $do(D_t = d_t)$, the total variation measure can be decomposed as*

$$\text{TV}_{a_t, a'_t}(d_t) = \text{Ctf-DE}_{a_t, a'_t}(d_t) - \text{Ctf-IE}_{a'_t, a_t}(d_t) - \text{Ctf-SE}_{a'_t, a_t}(d_t). \quad (35)$$

Proof. For the left hand side, we have

$$\text{TV}_{a_t, a'_t}(d_t) = \mathbb{E}[Y_{d_t} | a'_t] - \mathbb{E}[Y_{d_t} | a_t] \quad (36)$$

$$= \sum_{w,m} \mathbb{E}[Y_t | d_t, a'_t, m, w] \mathbb{P}(m|w, a'_t) \mathbb{P}(w|a'_t) - \sum_{w,m} \mathbb{E}[Y_t | d_t, a_t, m, w] \mathbb{P}(m|w, a_t) \mathbb{P}(w|a_t). \quad (37)$$

Where (36) follows by (1); (37) comes from no unmeasured confounding for $D_t \rightarrow Y_t$ given A_t, M_t, W_t .

For the right hand side, we have (Zhang & Bareinboim, 2018),

$$\text{Ctf-DE}_{a_t, a'_t}(d_t) = \sum_{w,m} [\mathbb{E}[Y_t | d_t, a'_t, m, w] - \mathbb{E}[Y_t | d_t, a_t, m, w]] \mathbb{P}(m|w, a_t) \mathbb{P}(w|a_t) \quad (38)$$

$$\text{Ctf-IE}_{a'_t, a_t}(d_t) = \sum_{w,m} \mathbb{E}[Y_t | d_t, a'_t, m, w] \cdot [\mathbb{P}(m|w, a_t) - \mathbb{P}(m|w, a'_t)] \cdot \mathbb{P}(w|a_t) \quad (39)$$

$$\text{Ctf-SE}_{a'_t, a_t}(d_t) = \sum_{w,m} \mathbb{E}[Y_t | d_t, a'_t, m, w] \cdot \mathbb{P}(m|w, a'_t) \cdot [\mathbb{P}(w|a_t) - \mathbb{P}(w|a'_t)]. \quad (40)$$

Therefore, by Equations (38) to (40)

$$\text{Ctf-DE}_{a_t, a'_t}(d_t) - \text{Ctf-SE}_{a'_t, a_t}(d_t) - \text{Ctf-IE}_{a'_t, a_t}(d_t) = \text{TV}_{a_t, a'_t}(d_t).$$

Which finishes the proof. ■

E. Proof for Section 3

In this section, we provide the proofs of the theoretical results for non-parametric contextual bandits setting in Section 3.

E.1. Regret Guarantee

Theorem 1 (Regret Upper Bound). *Fix $\delta \in (0, 1)$, $\omega \in (0, 1)$, and let $\epsilon_t = t^{-\omega}$. Let $\mathcal{X}$ be a finite context set and the learning rate $\beta = \sqrt{|\mathcal{X}| \log(2|\mathcal{D}|)/T}$. Under Assumptions 1 and 2, Algorithm 1 guarantees that, with probability at least $1 - 3T|\mathcal{X}||\mathcal{D}|\delta$,*

$$\sum_{t=1}^T \left[\max_{\pi \in \Pi_{X_t}^f} \langle \pi, \mathbb{E}[Y_d(X_t)] \rangle - \mathbb{E}[Y_{d_t}(X_t)] \right] = \tilde{O}\left(\sqrt{T|\mathcal{X}||\mathcal{D}| \log(2/\delta) \log(2|\mathcal{D}|/\delta)} + T^{1-\omega}\right).$$

Proof. The proof proceeds as follows. Part (i) describes the notation; Part (ii) establishes a online-mirror-descent (OMD) bound for each fixed context; Part (iii) controls the estimation error of the estimated optimistic reward; Part (iv) controls the error incurred by sampling decisions from the policy; Part (v) assembles all the results.

Part (i): Notations.

Throughout, $\mu_d(X) := \mathbb{E}[Y_d(X)]$ denotes the mean reward of decision d in context X and the vector of expected rewards $\mu(X) := (\mu_d(X))_{d \in \mathcal{D}}$; for the realized pair (X_t, d_t) we write $\mu_{d_t}(X_t)$.

For $t \in \{1, \dots, T\}$, $d \in \mathcal{D}$ and a context $X = (a, m, w)$ define the count

$$N_{X,d}(t) := \sum_{\tau=1}^t \mathbf{1}\{d_\tau = d, a_\tau = a, m_\tau = m, w_\tau = w\}, \quad n_{X,d}(t) := \max\{1, N_{X,d}(t)\}, \quad (41)$$

also written $N_{d,a,m,w}(t)$ and $n_{d,a,m,w}(t)$ where $X = (a, m, w)$. Thus $N_{X,d}(t)$ records all rounds up to and including t , and is determined by the history through round t .

Define the confidence bonus⁶

$$b_{t,d}(X) := \sqrt{\frac{\log(2/\delta)}{2n_{X,d}(t)}} \in (0, B], \quad B := \sqrt{\frac{\log(2/\delta)}{2}}, \quad b_t(X) := (b_{t,d}(X))_{d \in \mathcal{D}}. \quad (42)$$

Since $\delta < 1$ we have $\log(2/\delta) > \log 2$ and hence $B > \sqrt{\frac{\log 2}{2}} > \frac{1}{2}$.

For $N_{X,d}(t) \geq 1$ let $\hat{y}_{t,d}(X)$ be the average of the $N_{X,d}(t)$ rewards observed at the pair (X, d) in rounds $\tau \leq t$. By Algorithm 1, the truncated optimistic estimate and the associated optimistic loss are

$$\tilde{y}_{t,d}(X) := \begin{cases} 1, & N_{X,d}(t) = 0, \\ \min\{1, \hat{y}_{t,d}(X) + b_{t,d}(X)\}, & N_{X,d}(t) \geq 1, \end{cases} \quad \tilde{\ell}_{t,d}(X) := 1 - \tilde{y}_{t,d}(X), \quad (43)$$

with $\tilde{y}_t(X)$, $\tilde{\ell}_t(X)$ the corresponding vectors in $\mathbb{R}^{|\mathcal{D}|}$; note $\tilde{\ell}_{t,d}(X) \in [0, 1]$.

Let s denote the previous occurrence of X before round t , as in line 4 of Algorithm 1. Because X does not occur strictly between s and t , so

$$N_{X,d}(t-1) = N_{X,d}(s), \quad N_{X,d}(t) = N_{X,d}(s) + \mathbf{1}\{d_t = d\}. \quad (44)$$

Algorithm 1 uses $\hat{y}_{s,d}(X)$: the empirical mean formed from all rounds up to and including s .

To formalize the ϵ_t -greedy step (11), let Z_t be an exploration indicator drawn at round t after X_t is revealed, with $Z_t \sim \text{Bernoulli}(\epsilon_t)$ independent of the pre-decision history. Define

$$\mathcal{F}_{t-1} := \sigma(X_1, Z_1, d_1, Y_1, \dots, X_{t-1}, Z_{t-1}, d_{t-1}, Y_{t-1}, X_t),$$

⁶Not to be confused with the fairness slack terms $b_t^{(j)}(\pi)$, $j \in \{1, 2, 3\}$, in the definition of $\hat{\Pi}_{X_t}^f$.

the history through round $t - 1$ together with the context revealed at round t ; thus Z_t is independent of $\mathcal{F}_{t-1}$.

By construction, π_{X_t} and $\tilde{\pi}_{X_t}$ are $\mathcal{F}_{t-1}$ -measurable, while $n_{X_t,d}(t)$, $b_t(X_t)$, $\tilde{y}_t(X_t)$ and $\tilde{\ell}_t(X_t)$ depend on d_t through (44) and are only $\mathcal{F}_t$ -measurable. The decision satisfies $\mathbb{P}(d_t = d \mid \mathcal{F}_{t-1}, Z_t) = Z_t \pi_0(d) + (1 - Z_t) \pi_{X_t}(d)$ by $Z_t \in \{0, 1\}$. Then we have,

$$\mathbb{P}(d_t = d \mid \mathcal{F}_{t-1}) = \mathbb{E}[Z_t \pi_0(d) + (1 - Z_t) \pi_{X_t}(d)] = (1 - \epsilon_t) \pi_{X_t}(d) + \epsilon_t \pi_0(d) = \tilde{\pi}_{X_t}(d), \quad d \in \mathcal{D}. \quad (45)$$

Because the occurrence count T_X of a context X is itself random and the Azuma–Hoeffding inequality (Lemma 2) is stated for a fixed number of increments, instead we define the two martingale events, that is, for every round $t \in [T]$,

$$\xi_t := \langle \tilde{\pi}_{X_t}, b_{t-1}(X_t) \rangle - b_{t-1,d_t}(X_t), \quad S_t := \langle \tilde{\pi}_{X_t}, \mu(X_t) \rangle - \mu_{d_t}(X_t), \quad (46)$$

(46) will be used in parts (iii) and (iv).

Both sequences in (46) are martingale differences with respect to $\{\mathcal{F}_t\}_{t=1}^T$. Instead $b_{t-1}(X_t)$, $\mu(X_t)$ and $\tilde{\pi}_{X_t}$ are $\mathcal{F}_{t-1}$ -measurable, so the only randomness in ξ_t and S_t given $\mathcal{F}_{t-1}$ is the draw of d_t , and by (45)

$$\mathbb{E}[b_{t-1,d_t}(X_t) \mid \mathcal{F}_{t-1}] = \langle \tilde{\pi}_{X_t}, b_{t-1}(X_t) \rangle, \quad \mathbb{E}[\mu_{d_t}(X_t) \mid \mathcal{F}_{t-1}] = \langle \tilde{\pi}_{X_t}, \mu(X_t) \rangle. \quad (47)$$

Then, by Equations (46) and (47), we have,

$$\mathbb{E}[\xi_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[S_t \mid \mathcal{F}_{t-1}] = 0.$$

Also, we have,

$$|\xi_t| \leq B, \quad |S_t| \leq 1, \quad t \in [T]. \quad (48)$$

Regret decomposition. Fix a context $X \in \mathcal{X}$ with occurrence times $t_1 < \dots < t_{T_X}$, and let $\pi_X \in \bigcap_{i=1}^{T_X} \hat{\Pi}_{X_{t_i}}^f$ be an arbitrary comparator. For each $i \in [T_X]$, we have,

$$\begin{aligned} \langle \pi_X, \mu(X) \rangle - \mu_{d_{t_i}}(X) &= \underbrace{\langle \pi_X, \mu(X) - \tilde{y}_{t_i}(X) \rangle}_{\text{optimism}} + \underbrace{\langle \pi_{X_{t_i}} - \pi_X, \tilde{\ell}_{t_i}(X) \rangle}_{\text{OMD}} + \\ &\quad \underbrace{\langle \pi_{X_{t_i}}, \tilde{y}_{t_i}(X) - \mu(X) \rangle}_{\text{estimation}} + \underbrace{\langle \pi_{X_{t_i}}, \mu(X) \rangle - \mu_{d_{t_i}}(X)}_{\text{sampling}}. \end{aligned} \quad (49)$$

Where (49) follows by $\langle \pi_X - \pi_{X_{t_i}}, \tilde{y}_{t_i}(X) \rangle = \langle \pi_{X_{t_i}} - \pi_X, \tilde{\ell}_{t_i}(X) \rangle$.

Part (ii): the OMD term.

We first show that the estimated fair policy set $\hat{\Pi}_{X_t}^f$ is a convex set for any fixed context $X_t = X \in \mathcal{X}$. Let $T_X = \sum_{t=1}^T \mathbb{1}\{X_t = X\}$. The simplex Π is the set of all probability distributions over decisions in $\mathcal{D}$, i.e.,

$$\Pi = \left\{ \pi \in \mathbb{R}^{|\mathcal{D}|} : \pi(d) \geq 0 \ \forall d, \sum_{d \in \mathcal{D}} \pi(d) = 1 \right\},$$

therefore Π is a convex set by definition (closed under convex combinations) and each element inside it corresponds to a probability distribution (policy). For the estimated fair policy set $\hat{\Pi}_{X_t}^f$, we have that:

$$\begin{aligned} \hat{\Pi}_{X_t}^f &\leftarrow \left\{ \pi \in \Pi : \langle \pi, \sum_{m,w} (\hat{y}_{a'_t,d,m,w} - \hat{y}_{a_t,d,m,w}) \mathbb{P}(m|a_t, w) \mathbb{P}(w|a_t) \rangle - \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) \leq c_1 \right\} \\ &\cap \left\{ \pi \in \Pi : \langle \pi, \sum_{m,w} \hat{y}_{a'_t,d,m,w} (\mathbb{P}(m|a_t, w) - \mathbb{P}(m|a'_t, w)) \mathbb{P}(w|a_t) \rangle - \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) \leq c_2 \right\} \\ &\cap \left\{ \pi \in \Pi : \langle \pi, \sum_{m,w} \hat{y}_{a'_t,d,m,w} \mathbb{P}(m|a_t, w) (\mathbb{P}(w|a'_t) - \mathbb{P}(w|a_t)) \rangle - \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) \leq c_3 \right\}, \end{aligned}$$

where $\hat{y}_{a_t, d, m, w}$ and $\hat{y}_{a'_t, d, m, w}$ denote the empirical mean rewards for decision $d \in \mathcal{D}$ evaluated for contexts $\{a_t, m, w\}$ and $\{a'_t, m, w\}$ at round t . Since $\sum_{j \in \{1, 2, 3\}} b_t^{(j)}(\pi)$ is linear in π (see Proposition 3), therefore, the estimated feasible policy space $\hat{\Pi}_{X_t}^f \subseteq \Pi$ is a convex polytope cut out by three affine inequalities over probability distribution π . $\hat{\Pi}_{X_t}^f$ is non-empty by Assumption 1 and Proposition 3. Let us define the negative entropy (“mirror map”) (Bernasconi et al., 2024) for an (unnormalized) probability distribution $\pi \in \mathbb{R}_{>0}^{|\mathcal{D}|}$ as:

$$\Psi(\pi) = \sum_{d \in \mathcal{D}} \pi(d)(\log \pi(d) - 1). \quad (50)$$

Since the above function Ψ is convex and differentiable over π , we define the Bregman divergence between two probability distributions $\pi_1, \pi_2 \in \mathbb{R}_{>0}^{|\mathcal{D}|}$

$$B_\Psi(\pi_1 || \pi_2) = \Psi(\pi_1) - \left[\Psi(\pi_2) + \nabla \Psi(\pi_2)^\top (\pi_1 - \pi_2) \right]. \quad (51)$$

Geometrically, this is the gap between the value $\Psi(\pi_1)$ and the first-order approximation of $\Psi(\pi_1)$ around the point π_2 (base point). When taking the gradient with respect to the first argument, we have

$$[\nabla_{\pi_1} B_\Psi(\pi_1 || \pi_2)]_d = \log \pi_1(d) - \log \pi_2(d) = \log \frac{\pi_1(d)}{\pi_2(d)}, \quad d \in \mathcal{D}. \quad (52)$$

By (51), for any $\pi_1 \in \mathbb{R}_{>0}^{|\mathcal{D}|}$ and any $\pi_2, \pi_3 \in \mathbb{R}_{>0}^{|\mathcal{D}|}$ (the latter two need not be normalized, since $\nabla \Psi$ is evaluated there), we have the following property (Hazan, 2016),

$$B_\Psi(\pi_1 || \pi_2) + B_\Psi(\pi_2 || \pi_3) - B_\Psi(\pi_1 || \pi_3) = (\pi_1 - \pi_2)^\top (\nabla \Psi(\pi_3) - \nabla \Psi(\pi_2)). \quad (53)$$

Moreover, for a given $\bar{\pi} \in \mathbb{R}_{>0}^{|\mathcal{D}|}$, define $\pi_{\min} = \arg \min_{\pi \in \hat{\Pi}_{X_t}^f} B_\Psi(\pi || \bar{\pi})$. Then,

$$0 \leq B_\Psi(\pi_4 || \pi_{\min}) \leq B_\Psi(\pi_4 || \pi_{\min}) + B_\Psi(\pi_{\min} || \bar{\pi}) \leq B_\Psi(\pi_4 || \bar{\pi}), \quad \forall \pi_4 \in \hat{\Pi}_{X_t}^f. \quad (54)$$

where (54) holds since $B_\Psi(\pi_{\min} || \bar{\pi}) \geq 0$ and

$$B_\Psi(\pi_4 || \pi_{\min}) + B_\Psi(\pi_{\min} || \bar{\pi}) - B_\Psi(\pi_4 || \bar{\pi}) = (\pi_4 - \pi_{\min})^\top (\nabla \Psi(\bar{\pi}) - \nabla \Psi(\pi_{\min})) \quad (55)$$

$$\leq 0. \quad (56)$$

Where (55) follows by (51); as for (56) since both $\hat{\Pi}_{X_t}^f$ and Ψ are convex over policy vector, then the first order optimality condition of the optimization problem $\pi_{\min} = \arg \min_{\pi \in \hat{\Pi}_{X_t}^f} B_\Psi(\pi || \bar{\pi})$ gives us that

$$\langle \nabla B_\Psi(\pi_{\min} || \bar{\pi}), \pi - \pi_{\min} \rangle \geq 0, \quad \forall \pi \in \hat{\Pi}_{X_t}^f.$$

and since $\nabla B_\Psi(\pi_{\min} || \bar{\pi}) = \nabla \Psi(\pi_{\min}) - \nabla \Psi(\bar{\pi})$, then we have

$$(\pi - \pi_{\min})^\top (\nabla \Psi(\bar{\pi}) - \nabla \Psi(\pi_{\min})) \leq 0, \quad \forall \pi \in \hat{\Pi}_{X_t}^f.$$

because $\pi_4 \in \hat{\Pi}_{X_t}^f$ by (54), then (56) holds.

For notational simplicity, we suppress time indices when they are clear from context. In particular, π_{X_s} denotes the policy associated with context X_s at round s , and $\tilde{\ell}_s(X)$ denotes the corresponding optimistic loss vector at round s .

For convenience, we work with losses rather than rewards. Define $\tilde{\ell}_t(X) = 1 - \tilde{y}_t(X)$ as the optimistic empirical expected loss vector for all the decisions $d \in \mathcal{D}$ and $\tilde{\ell}_{t, d_t}(X) = 1 - \tilde{y}_{t, d_t}(X)$ is the optimistic empirical expected loss for the given decision d_t . For a given time step t , let $s := \max(\{u < t : X_u = X_t\} \cup \{0\})$ be the most recent round prior to time t at which the same context X_t occurred, with the conventions $\pi_{X_0} = \pi_0$ and $\hat{\Pi}_{X_0}^f = \Pi$ at a first occurrence by Algorithm 1. Here s is a random index. We have the following online mirror update:

$$\hat{\pi}_{X_t} \leftarrow \arg \min_{\pi \in \mathbb{R}_{>0}^{|\mathcal{D}|}} \left[\beta \langle \tilde{\ell}_s(X), \pi \rangle + B_\Psi(\pi || \pi_{X_s}) \right]. \quad (57)$$

The gradient of the mirror map $\nabla_{\pi} \Psi(\pi) = \nabla_{\pi} [\sum_{d \in \mathcal{D}} \pi(d)(\log \pi(d) - 1)] = \log(\pi)$, which takes a natural log of each coordinate of the discrete distribution π , i.e., $(\log(\pi(d)))_{d \in \mathcal{D}}$. We can find the closed-form solution of (57) and obtain the following (by taking the derivative of (57)),

$$\log \hat{\pi}_{X_t} = \log \pi_{X_s} - \beta \tilde{\ell}_s(X). \quad (58)$$

Exponentiating (58), we then get the update (9) in Algorithm 1.

For each context $X \in \mathcal{X}$, we denote the round indices when $X_t = X$ as $\{t_1, \dots, t_{T_X}\}$.

Consider any $\pi_X \in \bigcap_{i=1}^{T_X} \hat{\Pi}_{X_{t_i}}^f$ ⁷. Then, by (58) and (53) with $\pi_1 = \pi_X, \pi_2 = \pi_{X_s}$ (output optimistically fair policy from (10)), $\pi_3 = \hat{\pi}_{X_t}$ (output unnormalized policy from (9)), we have the following:

$$\begin{aligned} B_{\Psi}(\pi_X || \pi_{X_s}) + B_{\Psi}(\pi_{X_s} || \hat{\pi}_{X_t}) - B_{\Psi}(\pi_X || \hat{\pi}_{X_t}) &= (\pi_X - \pi_{X_s})^{\top} (\nabla \Psi(\hat{\pi}_{X_t}) - \nabla \Psi(\pi_{X_s})) \\ &= \beta \cdot (\pi_{X_s} - \pi_X)^{\top} \tilde{\ell}_s(X). \end{aligned} \quad (59)$$

Next, we are going to establish an upper bound on $\beta(\pi_{X_s} - \pi_X)^{\top} \tilde{\ell}_s(X)$ in (59) over T_X .

For notational convenience, in the rest of the proof of part (ii), let $t_1, t_2, \dots, t_{T_X}$ be the times that context X has been observed by the learner in T rounds. T_X is the total number of occurrence of context X over T rounds. Define $\hat{\pi}_{X_{t_{T_X+1}}}$ as the unprojected OMD iterate obtained after processing the last occurrence t_{T_X} , namely,

$$\hat{\pi}_{X_{t_{T_X+1}}}(d) = \pi_{X_{t_{T_X}}}(d) \exp\left(-\beta \tilde{\ell}_{t_{T_X}, d}(X)\right), \quad d \in \mathcal{D}.$$

Here, t_{T_X+1} is a dummy index used only to label this final intermediate iterate; it is not an additional round or occurrence of context X .

$$\begin{aligned} &\beta \sum_{i=1}^{T_X} (\pi_{X_{t_i}} - \pi_X)^{\top} \tilde{\ell}_{t_i}(X) \\ &= \sum_{i=1}^{T_X} \left[B_{\Psi}(\pi_X || \pi_{X_{t_i}}) + B_{\Psi}(\pi_{X_{t_i}} || \hat{\pi}_{X_{t_{i+1}}}) - B_{\Psi}(\pi_X || \hat{\pi}_{X_{t_{i+1}}}) \right] \end{aligned} \quad (60)$$

$$= B_{\Psi}(\pi_X || \pi_{X_{t_1}}) - B_{\Psi}(\pi_X || \hat{\pi}_{X_{t_{T_X+1}}}) + \sum_{i=2}^{T_X} [B_{\Psi}(\pi_X || \pi_{X_{t_i}}) - B_{\Psi}(\pi_X || \hat{\pi}_{X_{t_i}})] + \sum_{i=1}^{T_X} B_{\Psi}(\pi_{X_{t_i}} || \hat{\pi}_{X_{t_{i+1}}}) \quad (61)$$

$$\leq B_{\Psi}(\pi_X || \pi_{X_{t_1}}) + \sum_{i=1}^{T_X} B_{\Psi}(\pi_{X_{t_i}} || \hat{\pi}_{X_{t_{i+1}}}), \quad (62)$$

where (60) follows by (59); (61) rewrites the technical terms; (62) holds since $\pi_X \in \bigcap_{i=1}^{T_X} \hat{\Pi}_{X_{t_i}}^f$ and by (54), we have, for every $i \geq 2$,

$$B_{\Psi}(\pi_X || \pi_{X_{t_i}}) \leq B_{\Psi}(\pi_X || \pi_{X_{t_i}}) + B_{\Psi}(\pi_{X_{t_i}} || \hat{\pi}_{X_{t_i}}) \leq B_{\Psi}(\pi_X || \hat{\pi}_{X_{t_i}}).$$

where $\pi_{X_{t_i}} = \arg \min_{\pi \in \hat{\Pi}_{X_{t_{i-1}}}^f} B_{\Psi}(\pi || \hat{\pi}_{X_{t_i}})$ from the Bregman projection step.

To bound $\beta(\pi_{X_s} - \pi_X)^{\top} \tilde{\ell}_s(X)$ in (59), we need to further analyze term $B_{\Psi}(\pi_{X_{t_i}} || \hat{\pi}_{X_{t_{i+1}}})$ in (62),

$$B_{\Psi}(\pi_{X_{t_i}} || \hat{\pi}_{X_{t_{i+1}}}) \leq B_{\Psi}(\pi_{X_{t_i}} || \hat{\pi}_{X_{t_{i+1}}}) + B_{\Psi}(\hat{\pi}_{X_{t_{i+1}}} || \pi_{X_{t_i}}) \quad (63)$$

$$= \langle \pi_{X_{t_i}} - \hat{\pi}_{X_{t_{i+1}}}, \nabla \Psi(\pi_{X_{t_i}}) - \nabla \Psi(\hat{\pi}_{X_{t_{i+1}}}) \rangle \quad (64)$$

$$= \beta \cdot \langle \pi_{X_{t_i}} - \hat{\pi}_{X_{t_{i+1}}}, \tilde{\ell}_{t_i}(X) \rangle \quad (65)$$

$$= \beta \sum_{d \in \mathcal{D}} \pi_{X_{t_i}}(d) \left(1 - e^{-\beta \tilde{\ell}_{t_i, d}(X)}\right) \tilde{\ell}_{t_i, d}(X) \quad (66)$$

⁷This is a nonempty set w.h.p. by Proposition 3.

$$\leq \beta^2 \sum_{d \in \mathcal{D}} \pi_{X_{t_i}}(d) \tilde{\ell}_{t_i,d}(X)^2 \leq \beta^2, \quad (67)$$

where (63) follows by $B_\Psi(\hat{\pi}_{X_{t_{i+1}}} || \pi_{X_{t_i}}) \geq 0$; (64) follows by the definition of Bregman divergence from (51); (65) is because $\nabla \Psi(\pi_{X_{t_i}}) - \nabla \Psi(\hat{\pi}_{X_{t_{i+1}}}) = \log(\pi_{X_{t_i}}) - \log(\hat{\pi}_{X_{t_{i+1}}}) = \beta \tilde{\ell}_{t_i}(X)$; (66) substitutes $\hat{\pi}_{X_{t_{i+1}}}(d) = \pi_{X_{t_i}}(d) e^{-\beta \tilde{\ell}_{t_i,d}(X)}$; and (67) uses $1 - e^{-x} \leq x$ for $x \geq 0$ (valid since $\tilde{\ell}_{t_i,d}(X) \geq 0$), followed by $\tilde{\ell}_{t_i,d}(X) \leq 1$ and $\sum_d \pi_{X_{t_i}}(d) = 1$.

For the first occurrence of context X we have $s = 0$, so the conventions $\pi_{X_0} = \pi_0$ and $\hat{\Pi}_{X_0}^f = \Pi$ apply. Since no reward has yet been observed at X , every count $N_{a_{t_1},d,m_{t_1},w_{t_1}}(0)$ is zero and the optimistic estimate returns $\tilde{y}_{0,d}(X) = 1$ for all $d \in \mathcal{D}$, hence $\tilde{\ell}_0(X) = \mathbf{0}$. The mirror step (9) therefore returns $\hat{\pi}_{X_{t_1}} = \pi_0$, and the Bregman projection (10) of π_0 returns π_0 itself; both operations are the identity at t_1 , so $\pi_{X_{t_1}} = \pi_0$. Then, we have,

$$B_\Psi(\pi_X || \pi_{X_{t_1}}) = B_\Psi(\pi_X || \pi_0) \leq \log |\mathcal{D}|. \quad (68)$$

Thus by (62), (67), and (68), we have

$$\begin{aligned} \sum_{i=1}^{T_X} (\pi_{X_{t_i}} - \pi_X)^\top \tilde{\ell}_{t_i}(X) &\leq \frac{1}{\beta} \cdot \left[B_\Psi(\pi_X || \pi_{X_{t_1}}) + \sum_{i=1}^{T_X} B_\Psi(\pi_{X_{t_i}} || \hat{\pi}_{X_{t_{i+1}}}) \right] \\ &\leq \frac{1}{\beta} \left[\log(|\mathcal{D}|) + T_X \beta^2 \right] \\ &= \frac{\log(|\mathcal{D}|)}{\beta} + \beta T_X. \end{aligned} \quad (69)$$

Then, we have,

$$\sum_{X \in \mathcal{X}} \left[\frac{\log |\mathcal{D}|}{\beta} + \beta T_X \right] \leq \frac{|\mathcal{X}| \log(2|\mathcal{D}|)}{\beta} + \beta T,$$

Select $\beta = \sqrt{|\mathcal{X}| \log(2|\mathcal{D}|)/T}$, then we have,

$$\sum_{X \in \mathcal{X}} \sum_{i=1}^{T_X} (\pi_{X_{t_i}} - \pi_X)^\top \tilde{\ell}_{t_i}(X) \leq 2\sqrt{T |\mathcal{X}| \log(2|\mathcal{D}|)}. \quad (70)$$

Part (iii): concentration and the estimation term.

Fix $X \in \mathcal{X}$, $d \in \mathcal{D}$ and a deterministic $k \in [T]$, and let $\hat{y}_{d,k}(X)$ denote the average of the first k rewards collected at the pair (X, d) ; these are i.i.d. with mean $\mu_d(X)$ and range $[0, 1]$. Applying Lemma 1, we have

$$\mathbb{P}\left(|\hat{y}_{d,k}(X) - \mu_d(X)| \geq \epsilon\right) \leq 2 \exp(-2k\epsilon^2) \quad (71)$$

$$= \delta. \quad (72)$$

where (72) follows by $\epsilon = \sqrt{\frac{\log(2/\delta)}{2k}}$.

Applying a union bound over $X \in \mathcal{X}$, $d \in \mathcal{D}$ and $k \in [T]$, therefore, yields that with probability at least $1 - T|\mathcal{X}||\mathcal{D}|\delta$,

$$|\hat{y}_{d,k}(X) - \mu_d(X)| \leq \sqrt{\frac{\log(2/\delta)}{2k}}, \quad \forall X \in \mathcal{X}, d \in \mathcal{D}, k \in [T]. \quad (73)$$

At round t the algorithm has the realized count $k = N_{X,d}(t)$, and $n_{X,d}(t) = k$ whenever $k \geq 1$; since (73) holds simultaneously for all k , then

$$\tilde{y}_{t,d}(X) = \min \{1, \hat{y}_{t,d}(X) + b_{t,d}(X)\} \quad (74)$$

$$\geq \min \{1, \mu_d(X)\} = \mu_d(X), \quad (75)$$

and

$$\tilde{y}_{t,d}(X) \leq \hat{y}_{t,d}(X) + b_{t,d}(X) \quad (76)$$

$$\leq \mu_d(X) + 2b_{t,d}(X), \quad (77)$$

while for $N_{X,d}(t) = 0$, where $\tilde{y}_{t,d}(X) = 1$ and $b_{t,d}(X) = B$,

$$\mu_d(X) \leq 1 = \tilde{y}_{t,d}(X) \leq \mu_d(X) + 2B. \quad (78)$$

In both cases,

$$\mu_d(X) \leq \tilde{y}_{t,d}(X) \leq \mu_d(X) + 2b_{t,d}(X), \quad \forall t \in [T], X \in \mathcal{X}, d \in \mathcal{D}. \quad (79)$$

Here (75) follows by (73) together with $\mu_d(X) \leq 1$; (77) follows by (73); and the last inequality of (78) holds since $2B > 1$.

Define $N_{X,d} := \sum_{\tau=1}^T \mathbf{1}\{X_\tau = X, d_\tau = d\}$, so that $\sum_{d \in \mathcal{D}} N_{X,d} = T_X$ and $\sum_{X \in \mathcal{X}} \sum_{d \in \mathcal{D}} N_{X,d} = T$. If t denotes the k -th round at which d is chosen at context X , where $1 \leq k \leq N_{X,d}$, then $N_{X,d}(t) = k$ and $n_{X,d}(t-1) = \max\{1, k-1\}$.

For every $N \geq 2$,

$$\sum_{k=1}^N \frac{1}{\sqrt{\max\{1, k-1\}}} = 1 + \sum_{j=1}^{N-1} \frac{1}{\sqrt{j}} \quad (80)$$

$$\leq 1 + (2\sqrt{N-1} - 1) \quad (81)$$

$$= 2\sqrt{N-1} \leq 2\sqrt{N},$$

where (80) isolates the $k = 1$ term and reindexes $j = k - 1$, and (81) uses the bound $\sum_{j=1}^J j^{-1/2} \leq 2\sqrt{J} - 1$, which holds for $J \geq 1$ and therefore requires $N \geq 2$.

For $N = 1$ the left-hand side equals $1 \leq 2 = 2\sqrt{N}$ directly, and for $N = 0$ it is an empty sum; hence in all cases

$$\sum_{k=1}^N \frac{1}{\sqrt{\max\{1, k-1\}}} \leq 2\sqrt{N}, \quad N \geq 0. \quad (82)$$

Therefore

$$\sum_{t=1}^T b_{t-1,d_t}(X_t) = B \sum_{X \in \mathcal{X}} \sum_{d \in \mathcal{D}} \sum_{k=1}^{N_{X,d}} \frac{1}{\sqrt{\max\{1, k-1\}}} \quad (83)$$

$$\leq 2B \sum_{X \in \mathcal{X}} \sum_{d \in \mathcal{D}} \sqrt{N_{X,d}} \quad (84)$$

$$\leq 2B \sqrt{|\mathcal{X}||\mathcal{D}| \sum_{X,d} N_{X,d}} \quad (85)$$

$$= 2B\sqrt{|\mathcal{X}||\mathcal{D}|T}, \quad (86)$$

where (83) groups the rounds by context and decision and uses $b_{t-1,d}(X) = B/\sqrt{n_{X,d}(t-1)}$; (84) follows by (82); and (85) is Cauchy-Schwarz inequality.

Now, we are going to bound policy-weighted estimation error. Set $\mathcal{R}_{\text{est}}^\pi(X) := \sum_{i=1}^{T_X} \langle \pi_{X_{t_i}}, \tilde{y}_{t_i}(X) - \mu(X) \rangle$. Since $\pi_{X_{t_i}} - \tilde{\pi}_{X_{t_i}} = \epsilon_{t_i}(\pi_{X_{t_i}} - \pi_0)$ by (11), we have,

$$\langle \pi_{X_{t_i}} - \tilde{\pi}_{X_{t_i}}, b_{t_i-1}(X) \rangle = \epsilon_{t_i} \langle \pi_{X_{t_i}} - \pi_0, b_{t_i-1}(X) \rangle \quad (87)$$

$$\leq \epsilon_{t_i} \|\pi_{X_{t_i}} - \pi_0\|_1 \|b_{t_i-1}(X)\|_\infty \quad (88)$$

$$\leq 2\epsilon_{t_i} B. \quad (89)$$

Where (88) is Hölder's inequality; (89) uses $\|\pi_{X_{t_i}} - \pi_0\|_1 \leq 2$ and $\|b_{t_i-1}(X)\|_\infty \leq B$. Hence, we have,

$$\begin{aligned} \mathcal{R}_{\text{est}}^\pi(X) &= \sum_{i=1}^{T_X} \langle \pi_{X_{t_i}}, \tilde{y}_{t_i}(X) - \mu(X) \rangle \\ &\leq 2 \sum_{i=1}^{T_X} \langle \pi_{X_{t_i}}, b_{t_i}(X) \rangle \end{aligned} \quad (90)$$

$$\leq 2 \sum_{i=1}^{T_X} \langle \pi_{X_{t_i}}, b_{t_i-1}(X) \rangle \quad (91)$$

$$\leq 2 \sum_{i=1}^{T_X} \langle \tilde{\pi}_{X_{t_i}}, b_{t_i-1}(X) \rangle + 4B \sum_{i=1}^{T_X} \epsilon_{t_i} \quad (92)$$

$$= 2 \sum_{i=1}^{T_X} b_{t_i-1, d_{t_i}}(X) + 2 \sum_{i=1}^{T_X} \xi_{t_i} + 4B \sum_{i=1}^{T_X} \epsilon_{t_i}. \quad (93)$$

Where (90) follows by (79); (91) holds because the counts are nondecreasing, $n_{X,d}(t_i-1) \leq n_{X,d}(t_i)$, hence $b_{t_i,d}(X) \leq b_{t_i-1,d}(X)$, and $b_{t_i-1}(X)$ is $\mathcal{F}_{t_i-1}$ -measurable; (92) follows by (89); and (93) adds and subtracts the realized bonus $b_{t_i-1, d_{t_i}}(X)$, and recalling the definition (46) of ξ_t .

Part (iv): the sampling term.

Define $\mathcal{R}_{\text{sam}}(X) := \sum_{i=1}^{T_X} [\langle \pi_{X_{t_i}}, \mu(X) \rangle - \mu_{d_{t_i}}(X)]$ and the definition (46) of S_t ,

$$\langle \pi_{X_{t_i}}, \mu(X) \rangle - \mu_{d_{t_i}}(X) = \langle \pi_{X_{t_i}} - \tilde{\pi}_{X_{t_i}}, \mu(X) \rangle + S_{t_i} \leq 2\epsilon_{t_i} + S_{t_i},$$

by the inequality (89) with $\|\mu(X)\|_\infty \leq 1$. Hence

$$\mathcal{R}_{\text{sam}}(X) \leq \sum_{i=1}^{T_X} S_{t_i} + 2 \sum_{i=1}^{T_X} \epsilon_{t_i}. \quad (94)$$

Part (v): assembling the bound.

We first look at one context. Summing (49) over $i \in [T_X]$ and by (79), (69), (93) and (94), we have,

$$\begin{aligned} \sum_{i=1}^{T_X} [\langle \pi_X, \mu(X) \rangle - \mu_{d_{t_i}}(X)] &\leq \left[\frac{\log |\mathcal{D}|}{\beta} + \beta T_X \right] + 2 \sum_{i=1}^{T_X} b_{t_i-1, d_{t_i}}(X) \\ &\quad + 2 \sum_{i=1}^{T_X} \xi_{t_i} + \sum_{i=1}^{T_X} S_{t_i} + (4B+2) \sum_{i=1}^{T_X} \epsilon_{t_i}, \end{aligned} \quad (95)$$

Then we sum over all the contexts. The occurrence times of distinct contexts partition $[T]$, so the per-context sums in (95) reassemble into sums over $t \in [T]$. Then by (70) and (86), we have,

$$\begin{aligned} \sum_{t=1}^T \left[\max_{\pi \in \Pi_X^f} \langle \pi, \mu(X_t) \rangle - \mu_{d_t}(X_t) \right] &\leq 2\sqrt{T|\mathcal{X}|\log(2|\mathcal{D}|)} + 4B\sqrt{|\mathcal{X}||\mathcal{D}|T} \\ &\quad + 2 \sum_{t=1}^T \xi_t + \sum_{t=1}^T S_t + (4B+2) \sum_{t=1}^T \epsilon_t. \end{aligned} \quad (96)$$

Where (96) follows by $\Pi_X^f \subseteq \hat{\Pi}_{X_t}^f$ in Proposition 3 w.p. at least $1 - T|\mathcal{X}||\mathcal{D}|\delta$.

By (46) and (48), $\{\xi_t\}_{t=1}^T$ and $\{S_t\}_{t=1}^T$ are martingale difference sequences with respect to $\{\mathcal{F}_t\}_{t=1}^T$ with $|\xi_t| \leq B$ and $|S_t| \leq 1$, over the horizon T . By Lemma 2, we have,

$$\sum_{t=1}^T \xi_t \leq B\sqrt{2T\log(1/\delta)} = \sqrt{T\log(1/\delta)\log(2/\delta)}, \quad (97)$$

$$\sum_{t=1}^T S_t \leq \sqrt{2T \log(1/\delta)}, \quad (98)$$

holds with probability at least $1 - \delta$.

And also, we have,

$$\sum_{t=1}^T t^{-\omega} \leq 1 + T^{1-\omega}/(1-\omega). \quad (99)$$

Then by Equations (96) to (99), with probability at least $1 - 3T|\mathcal{X}||\mathcal{D}|\delta$, we have,

$$\begin{aligned} \sum_{t=1}^T \left[\langle \pi_{X_t}^*, \mu(X_t) \rangle - \mu_{d_t}(X_t) \right] &\leq 10\sqrt{T|\mathcal{X}||\mathcal{D}|\log(2/\delta)\log(2|\mathcal{D}|/\delta)} + 8B\left(1 + \frac{T^{1-\omega}}{1-\omega}\right) \\ &= \tilde{\mathcal{O}}\left(\sqrt{T|\mathcal{X}||\mathcal{D}|\log(2/\delta)\log(2|\mathcal{D}|/\delta)} + T^{1-\omega}\right). \end{aligned} \quad (100)$$

Which finishes the proof. $\blacksquare$

E.2. Constraint Violations Guarantee

In this section we next prove the upper bounds for the cumulative constraint violations. We begin with the proof of Proposition 3 that the true feasible set remains within the estimated feasible set with high probability.

Proposition 3. *Given a $\delta > 0$ and define probability kernels $\mathbb{P}_{a_t}^{(1)}(m, w) = \mathbb{P}(m|a_t, w)\mathbb{P}(w|a_t)$, $\mathbb{P}_{a_t}^{(2)}(m, w) = \mathbb{P}(m|a'_t, w)\mathbb{P}(w|a_t)$, $\mathbb{P}_{a_t}^{(3)}(m, w) = \mathbb{P}(m|a'_t, w)\mathbb{P}(w|a'_t)$ on $\mathcal{M} \times \mathcal{W}$ w.r.t. the factual and counterfactual attribute a_t, a'_t . Let $b_t^{(i)}(\pi) = \mathbb{E}_{(m, w) \sim \mathbb{P}_{a_t}^{(i)}, d \sim \pi} \left(\sqrt{\frac{\log(2/\delta)}{2n_{a'_t, d, m, w}(t)}} + \sqrt{\frac{\log(2/\delta)}{2n_{a_t, d, m, w}(t)}} \right)$ such that with probability at least $1 - T|\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta$, for all $t \in [T]$, $i \in \{1, 2, 3\}$, $\pi \in \Pi$ we have:*

$$\mathbb{E}_{d \sim \pi} \left[\left| \hat{\Delta}_d^{(i)}(X_t) \right| - \left| \Delta_d^{(i)}(X_t) \right| \right] \leq \sum_{j \in \{1, 2, 3\}} b_t^{(j)}(\pi).$$

Then it holds $\Pi_{X_t}^f \subseteq \hat{\Pi}_{X_t}^f$ with probability at least $1 - T|\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta$ for all $t \in [T]$.

Proof. We first observe that the probability kernels $\mathbb{P}_{a_t}^{(2)}$ are valid distributions on $\mathcal{M} \times \mathcal{W}$ ($\mathbb{P}_{a_t}^{(1)}$ and $\mathbb{P}_{a_t}^{(3)}$ trivially is).

$$\begin{aligned} \sum_{(m, w) \in \mathcal{M} \times \mathcal{W}} \mathbb{P}_{a_t}^{(2)}(m, w) &= \sum_{(m, w) \in \mathcal{M} \times \mathcal{W}} \mathbb{P}(m|a'_t, w)\mathbb{P}(w|a_t) \\ &= \sum_{w \in \mathcal{W}} \mathbb{P}(w|a_t) \sum_{m \in \mathcal{M}} \mathbb{P}(m|a'_t, w) \\ &= \sum_{w \in \mathcal{W}} \mathbb{P}(w|a_t) = 1 \end{aligned}$$

For any $\pi \in \Pi$, we then rewrite the constraints as:

$$\begin{aligned} \mathbb{E}_{d \sim \pi} \left| \Delta_d^{(1)}(X_t) \right| - c_1 &= \mathbb{E}_{d \sim \pi} \left| \sum_{m, w} (\mathbb{E}[Y_t|a'_t, d, m, w] - \mathbb{E}[Y_t|a_t, d, m, w])\mathbb{P}(m|a_t, w)\mathbb{P}(w|a_t) \right| - c_1 \\ &= \mathbb{E}_{d \sim \pi} \left| \mathbb{E}_{(m, w) \sim \mathbb{P}_{a_t}^{(1)}} [\mathbb{E}[Y_t|a'_t, d, m, w] - \mathbb{E}[Y_t|a_t, d, m, w]] \right| - c_1 \end{aligned} \quad (101)$$

$$\begin{aligned} \mathbb{E}_{d \sim \pi} \left| \Delta_d^{(2)}(X_t) \right| - c_2 &= \mathbb{E}_{d \sim \pi} \left| \sum_{m, w} \mathbb{E}[Y_t|a'_t, d, m, w] (\mathbb{P}(m|a_t, w) - \mathbb{P}(m|a'_t, w))\mathbb{P}(w|a_t) \right| - c_2 \\ &= \mathbb{E}_{d \sim \pi} \left| \mathbb{E}_{(m, w) \sim \mathbb{P}_{a_t}^{(1)}} \mathbb{E}[Y_t|a'_t, d, m, w] - \mathbb{E}_{(m, w) \sim \mathbb{P}_{a_t}^{(2)}} \mathbb{E}[Y_t|a'_t, d, m, w] \right| - c_2 \end{aligned} \quad (102)$$

$$\begin{aligned}
 \mathbb{E}_{d \sim \pi} \left| \Delta_d^{(3)}(X_t) \right| - c_3 &= \mathbb{E}_{d \sim \pi} \left| \sum_{m,w} \mathbb{E}[Y_t | a'_t, d, m, w] \mathbb{P}(m | a'_t, w) (\mathbb{P}(w | a_t) - \mathbb{P}(w | a'_t)) \right| - c_3 \\
 &= \mathbb{E}_{d \sim \pi} \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a'_t}^3} \mathbb{E}[Y_t | a'_t, d, m, w] - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^2} \mathbb{E}[Y_t | a'_t, d, m, w] \right| - c_3.
 \end{aligned} \tag{103}$$

Here $\mathbb{E}[Y_t | a'_t, d, m, w]$ and $\mathbb{E}[Y_t | a_t, d, m, w]$ denote the true mean rewards for decision $d \in \mathcal{D}$ with contexts $\{a'_t, m, w\}$ and $\{a_t, m, w\}$ respectively; we also recall that the counterfactual fairness thresholds are bounded $c_i \in [0, 1]$, $i \in \{1, 2, 3\}$ so for each difference $\mathbb{E}_{d \sim \pi} \left| \Delta_d^{(i)}(X_t) \right| - c_i \in [-1, 1]$.

The true feasible policy set $\Pi_{X_t}^f$ is:

$$\begin{aligned}
 \Pi_{X_t}^f &= \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} \left| \Delta_d^{(1)}(X_t) \right| \leq c_1 \right\} \cap \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} \left| \Delta_d^{(2)}(X_t) \right| \leq c_2 \right\} \\
 &\quad \cap \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} \left| \Delta_d^{(3)}(X_t) \right| \leq c_3 \right\}.
 \end{aligned} \tag{104}$$

The optimistic estimated feasible policy set $\hat{\Pi}_{X_t}^f$ is:

$$\begin{aligned}
 \hat{\Pi}_{X_t}^f &= \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} \left| \hat{\Delta}_d^{(1)}(X_t) \right| - \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) \leq c_1 \right\} \cap \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} \left| \hat{\Delta}_d^{(2)}(X_t) \right| - \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) \leq c_2 \right\} \\
 &\quad \cap \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} \left| \hat{\Delta}_d^{(3)}(X_t) \right| - \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) \leq c_3 \right\}.
 \end{aligned}$$

We first show that $\mathbb{E}_{d \sim \pi} [|\hat{\Delta}_d^{(i)}(X_t)| - |\Delta_d^{(i)}(X_t)|] \leq \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi)$ for all $t \in [T]$, $i \in \{1, 2, 3\}$, $\pi \in \Pi$; then we show that all fair policies are in the optimistic estimated feasible policy set with probability at least $1 - T|\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta$ as promised.

For any $t \in [T]$ and context $X_t \in \mathcal{X}$, consider any fair policy $\pi \in \Pi_{X_t}^f$. For each $i \in \{1, 2, 3\}$, consider the estimated counterfactual fairness:

$$\mathbb{E}_{d \sim \pi} |\hat{\Delta}_d^{(i)}(X_t)| = \mathbb{E}_{d \sim \pi} |\hat{\Delta}_d^{(i)}(X_t)| - \mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(X_t)| + \mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(X_t)| \tag{105}$$

$$\leq \mathbb{E}_{d \sim \pi} |\hat{\Delta}_d^{(i)}(X_t)| - \mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(X_t)| + c_i \tag{106}$$

$$= \mathbb{E}_{d \sim \pi} [|\hat{\Delta}_d^{(i)}(X_t)| - |\Delta_d^{(i)}(X_t)|] + c_i, \tag{107}$$

where (105) adds and subtracts the expected value and (106) follows by (104), i.e., $\mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(X_t)| \leq c_i$. Now, we are going to analyze $\mathbb{E}_{d \sim \pi} [|\hat{\Delta}_d^{(i)}(X_t)| - |\Delta_d^{(i)}(X_t)|]$.

Case $i = 1$, Ctf-DE : We have

$$\begin{aligned}
 &\mathbb{E}_{d \sim \pi} [|\hat{\Delta}_d^{(1)}(X_t)| - |\Delta_d^{(1)}(X_t)|] \\
 &= \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} (\hat{y}_{a'_t, d, w, m} - \hat{y}_{a_t, d, w, m}) \mathbb{P}(m | a_t, w) \mathbb{P}(w | a_t) \right| \\
 &\quad - \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} (\mathbb{E}[Y_t | a'_t, d, w, m] - \mathbb{E}[Y_t | a_t, d, w, m]) \mathbb{P}(m | a_t, w) \mathbb{P}(w | a_t) \right|
 \end{aligned} \tag{108}$$

$$\begin{aligned}
 &\leq \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} (\mathbb{E}[Y_t | a'_t, d, w, m] - \mathbb{E}[Y_t | a_t, d, w, m]) \mathbb{P}(m | a_t, w) \mathbb{P}(w | a_t) \right| \\
 &\quad - \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} (\hat{y}_{a'_t, d, w, m} - \hat{y}_{a_t, d, w, m}) \mathbb{P}(m | a_t, w) \mathbb{P}(w | a_t)
 \end{aligned} \tag{109}$$

$$\leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \mathbb{E}_{d \sim \pi} |\mathbb{E}[Y_t | a'_t, d, w, m] - \mathbb{E}[Y_t | a_t, d, w, m] - \hat{y}_{a'_t, d, w, m} + \hat{y}_{a_t, d, w, m}| \tag{110}$$

$$\leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \mathbb{E}_{d \sim \pi} \left[|\hat{y}_{a_t, d, w, m} - \mathbb{E}[Y_t | a_t, d, w, m]| + |\mathbb{E}[Y_t | a'_t, d, w, m] - \hat{y}_{a'_t, d, w, m}| \right] \tag{111}$$

$$\leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \mathbb{E}_{d \sim \pi} \left(\sqrt{\frac{\log(2/\delta)}{2n_{a'_t, d, m, w}(t)}} + \sqrt{\frac{\log(2/\delta)}{2n_{a_t, d, m, w}(t)}} \right) \quad (112)$$

$$\leq b_t^{(1)}(\pi), \quad (113)$$

where (108) follows by Equations (6) and (101); (109) is by the triangle inequality, i.e., $|x| - |y| \leq |x - y|$, $\forall x, y \in \mathbb{R}$; (110) uses the probability kernel $\mathbb{P}_{a_t}^{(1)}(m, w) = \mathbb{P}(m|a_t, w)\mathbb{P}(w|a_t)$ and moves the expectation outside of absolute value; (111) rearranges and uses the triangle inequality; (112) is the bound on the estimation error follows by Hoeffding's inequality of Lemma 1, in which it is which holds with probability at least $1 - T|\mathcal{X}||\mathcal{D}|\delta$; and (113) holds by the definition of $b_t^{(i)}(\pi)$, $i \in \{1, 2, 3\}$.

Case $i = 2$, Ctf-IE: We have

$$\begin{aligned} & \mathbb{E}_{d \sim \pi} \left[|\hat{\Delta}_d^{(2)}(X_t)| - |\Delta_d^{(2)}(X_t)| \right] \\ &= \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} \hat{y}_{a'_t, d, w, m} (\mathbb{P}(m|a_t, w) - \mathbb{P}(m|a'_t, w)) \mathbb{P}(w|a_t) \right| - \\ & \quad \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} \mathbb{E}[Y_t|a'_t, d, w, m] (\mathbb{P}(m|a_t, w) - \mathbb{P}(m|a'_t, w)) \mathbb{P}(w|a_t) \right| \end{aligned} \quad (114)$$

$$\begin{aligned} & \leq \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} \mathbb{E}[Y_t|a'_t, d, w, m] (\mathbb{P}(m|a_t, w) - \mathbb{P}(m|a'_t, w)) \mathbb{P}(w|a_t) - \right. \\ & \quad \left. \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} \hat{y}_{a'_t, d, w, m} (\mathbb{P}(m|a_t, w) - \mathbb{P}(m|a'_t, w)) \mathbb{P}(w|a_t) \right| \end{aligned} \quad (115)$$

$$\begin{aligned} & \leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \mathbb{E}_{d \sim \pi} \left| \mathbb{E}[Y_t|a'_t, d, w, m] - \hat{y}_{a'_t, d, w, m} \right| + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \mathbb{E}_{d \sim \pi} \\ & \quad \left| \mathbb{E}[Y_t|a'_t, d, w, m] - \hat{y}_{a'_t, d, w, m} \right| \end{aligned} \quad (116)$$

$$\leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \mathbb{E}_{d \sim \pi} \sqrt{\frac{\log(2/\delta)}{2n_{a'_t, d, m, w}(t)}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \mathbb{E}_{d \sim \pi} \sqrt{\frac{\log(2/\delta)}{2n_{a'_t, d, m, w}(t)}} \quad (117)$$

$$\leq \sum_{j \in \{1, 2\}} b_t^{(j)}(\pi). \quad (118)$$

Where (114) follows by Equations (7) and (102); (115) is by the triangle inequality; (116) is to rewrite the technical terms in expectation using probability kernel $\mathbb{P}_{a_t}^{(2)}(m, w) = \mathbb{P}(m|a'_t, w)\mathbb{P}(w|a_t)$ and move the expectation outside of absolute value; (117) comes from Lemma 1, and holds with probability at least $1 - T|\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta$.

Case $i = 3$, Ctf-SE: We have

$$\begin{aligned} & \mathbb{E}_{d \sim \pi} \left[|\hat{\Delta}_d^{(3)}(X_t)| - |\Delta_d^{(3)}(X_t)| \right] \\ &= \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} \hat{y}_{a'_t, d, w, m} \mathbb{P}(m|a'_t, w) (\mathbb{P}(w|a_t) - \mathbb{P}(w|a'_t)) \right| - \\ & \quad \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} \mathbb{E}[Y_t|a'_t, d, w, m] \mathbb{P}(m|a'_t, w) (\mathbb{P}(w|a_t) - \mathbb{P}(w|a'_t)) \right| \end{aligned} \quad (119)$$

$$\begin{aligned} & \leq \mathbb{E}_{d \sim \pi} \left| \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} \mathbb{E}[Y_t|a'_t, d, w, m] \mathbb{P}(m|a'_t, w) (\mathbb{P}(w|a_t) - \mathbb{P}(w|a'_t)) - \right. \\ & \quad \left. \sum_{(m,w) \in (\mathcal{M}, \mathcal{W})} \hat{y}_{a'_t, d, w, m} \mathbb{P}(m|a'_t, w) (\mathbb{P}(w|a_t) - \mathbb{P}(w|a'_t)) \right| \end{aligned} \quad (120)$$

$$\begin{aligned} & \leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \mathbb{E}_{d \sim \pi} \left| \mathbb{E}[Y_t|a'_t, d, w, m] - \hat{y}_{a'_t, d, w, m} \right| + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \mathbb{E}_{d \sim \pi} \\ & \quad \left| \mathbb{E}[Y_t|a'_t, d, w, m] - \hat{y}_{a'_t, d, w, m} \right| \end{aligned} \quad (121)$$

$$\leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \mathbb{E}_{d \sim \pi} \sqrt{\frac{\log(2/\delta)}{2n_{a'_t, d, m, w}(t)}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \mathbb{E}_{d \sim \pi} \sqrt{\frac{\log(2/\delta)}{2n_{a'_t, d, m, w}(t)}} \quad (122)$$

$$\leq \sum_{j \in \{2,3\}} b_t^{(j)}(\pi). \quad (123)$$

Where (119) follows by Equations (8) and (103); (120) is by the triangle inequality; (121) is to rewrite the technical terms in expectation using probability kernel $\mathbb{P}_{a_t}^{(3)}(m, w) = \mathbb{P}(m|a'_t, w)\mathbb{P}(w|a'_t)$ and move the expectation outside of absolute value; (122) comes from Lemma 1, and holds with probability at least $1 - T|\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta$.

Therefore, by Equations (107), (113), (118) and (123), we have

$$\begin{aligned} \mathbb{E}_{d \sim \pi} |\hat{\Delta}_d^{(i)}(X_t)| &\leq \mathbb{E}_{d \sim \pi} \left[|\hat{\Delta}_d^{(i)}(X_t)| - |\Delta_d^{(i)}(X_t)| \right] + c_i \\ &\leq \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) + c_i, \end{aligned}$$

holds $\forall i \in \{1, 2, 3\}$, in which, by construction of the estimated feasible policy set: $\hat{\Pi}_{X_t}^f = \bigcap_{i=1}^3 \{\pi \in \Pi : \mathbb{E}_{d \sim \pi} |\hat{\Delta}_d^{(i)}(X_t)| - \sum_{j \in \{1,2,3\}} b_t^{(j)}(\pi) \leq c_i\}$, proves that $\pi \in \hat{\Pi}_{X_t}^f$, thus $\Pi_{X_t}^f \subseteq \hat{\Pi}_{X_t}^f$ holds with probability at least $1 - T|\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta$ for all $t \in [T]$. ■

Proposition 4. Select $\epsilon_t = t^{-\omega}$. Let $\mathcal{D}$ be finite domain of decisions. Consider the decisions are draw from policy $\tilde{\pi}_{X_t}(X_t = (a_t, w_t, m_t))$ output from Algorithm 1, define the number of counts:

$$N_{d,a,m,w}(t) = \sum_{s=1}^t \mathbb{1}\{(D_s, A_s, M_s, W_s) = (d, a, m, w)\}, \quad n_{d,a,m,w}(t) = \max\{1, N_{d,a,m,w}(t)\}. \quad (124)$$

Under Assumption 2, for any $a \in \mathcal{A}, d \in \mathcal{D}, w \in \mathcal{W}, m \in \mathcal{M}$, the following results hold for Algorithm 1:

$$\mathbb{E} \left[\sum_{t=1}^T \sqrt{\frac{1}{n_{a,d,w,m}(t)}} \right] \leq 2\sqrt{2} \sqrt{\frac{|\mathcal{D}|}{\gamma_1}} T^{\frac{1+\omega}{2}}.$$

Proof. By the context coverage from Assumption 2, we know that every context in the context space shows up with probability at least $\gamma_1 > 0$ for each round, that is,

$$\mathbb{P}[(A_t, W_t, M_t) = (a, w, m) | \mathcal{F}_{t-1}] \geq \gamma_1, \quad \forall (a, w, m) \in (\mathcal{A}, \mathcal{W}, \mathcal{M}), \quad \forall t \in [T]. \quad (125)$$

Next we need to figure out how probable each decision d will be selected under context (a, w, m) . By Algorithm 1, the decisions are draw from the output policy $\tilde{\pi}_{X_t}(X_t = (a, w, m))$, and we have:

$$\begin{aligned} \mathbb{P}[D_t = d | A_t = a, W_t = w, M_t = m, \mathcal{F}_{t-1}] &= \tilde{\pi}_{X_t}(d) \\ &= (1 - \epsilon_t)\pi_{X_t}(d) + \epsilon_t\pi_0(d) \end{aligned} \quad (126)$$

$$\geq \frac{\epsilon_t}{|\mathcal{D}|}, \quad (127)$$

where (126) follows by (11). Then by (125) and (127), $\forall (d, a, w, m) \in (\mathcal{D}, \mathcal{A}, \mathcal{W}, \mathcal{M}), \forall t \in [T]$, we have

$$\begin{aligned} \mathbb{P}[(D_t, A_t, W_t, M_t) = (d, a, w, m) | \mathcal{F}_{t-1}] &= \mathbb{P}[(A_t, W_t, M_t) = (a, w, m) | \mathcal{F}_{t-1}] \cdot \mathbb{P}[D_t = d | A_t, W_t, M_t, \mathcal{F}_{t-1}] \\ &\geq \frac{\gamma_1 \epsilon_t}{|\mathcal{D}|}, \end{aligned} \quad (128)$$

which implies any context-action pair (including the counterfactual one) is visited by the learner with probability at least $\frac{\gamma_1 \epsilon_t}{|\mathcal{D}|}$ for Algorithm 1.

Fix any tuple $(d, a, m, w) \in \mathcal{D} \times \mathcal{A} \times \mathcal{M} \times \mathcal{W}$. Recall that

$$n_{d,a,m,w}(t) = \max \left\{ 1, \sum_{s=1}^t \mathbb{1}\{(D_s, A_s, M_s, W_s) = (d, a, m, w)\} \right\}.$$

By (128) and $\epsilon_s = s^{-\omega}$, we have

$$\mathbb{P}((D_s, A_s, M_s, W_s) = (d, a, m, w) \mid \mathcal{F}_{s-1}) \geq \frac{\gamma_1 \epsilon_s}{|\mathcal{D}|} \geq \frac{\gamma_1}{|\mathcal{D}| T^\omega}, \quad \forall s \in [T].$$

Define

$$p := \frac{\gamma_1}{|\mathcal{D}| T^\omega}, \quad q_s := \mathbb{P}((D_s, A_s, M_s, W_s) = (d, a, m, w) \mid \mathcal{F}_{s-1}).$$

Then $q_s \geq p$ almost surely for all $s \in [T]$. We then use a standard coupling construction (see Definition 6) as a proof device. By the randomization lemma for Bernoulli variables (Lemma 8), we may work on an enlarged probability space on which there exist i.i.d. random variables

$$Z_1, \dots, Z_T \sim \text{Unif}[0, 1],$$

independent of the past filtration, such that

$$\mathbb{1}\{(D_s, A_s, M_s, W_s) = (d, a, m, w)\} = \mathbb{1}\{Z_s \leq q_s\}, \quad s \in [T], \quad \text{almost surely.}$$

Further define

$$V_s := \mathbb{1}\{Z_s \leq p\}, \quad s \in [T].$$

Then $(V_s)_{s=1}^T$ are i.i.d. Bernoulli variables with probability p . Moreover, since $q_s \geq p$, we have

$$\{Z_s \leq p\} \subseteq \{Z_s \leq q_s\},$$

and hence distribution $\mathbb{1}\{(D_s, A_s, M_s, W_s) = (d, a, m, w)\}$ stochastically dominates distribution V_s , i.e.,

$$V_s \leq \mathbb{1}\{(D_s, A_s, M_s, W_s) = (d, a, m, w)\}, \quad \forall s \in [T], \quad \text{almost surely.}$$

Let $S_t := \sum_{s=1}^t V_s$, then $S_t \sim \text{Bin}(t, p)$ and

$$S_t = \sum_{s=1}^t V_s \leq \sum_{s=1}^t \mathbb{1}\{(D_s, A_s, M_s, W_s) = (d, a, m, w)\}, \quad \forall t \in [T], \quad \text{almost surely.}$$

Therefore,

$$n_{d,a,m,w}(t) \geq \max\{1, S_t\} \geq \frac{1 + S_t}{2}, \quad \forall t \in [T], \quad \text{almost surely,} \quad (129)$$

where the first inequality uses $\sum_{s \leq t} \mathbb{1}\{\cdot\} \geq S_t$, and the second holds because $\max\{1, x\} \geq (1 + x)/2$ for every integer $x \geq 0$ (with equality at $x = 1$).

Now, we can bound our results:

$$\mathbb{E} \left[\sum_{t=1}^T \sqrt{\frac{1}{n_{d,a,m,w}(t)}} \right] \leq \sqrt{2} \sum_{t=1}^T \mathbb{E} \left[\sqrt{\frac{1}{1 + S_t}} \right] \quad (130)$$

$$\leq \sqrt{2} \sum_{t=1}^T \sqrt{\mathbb{E} \left[\frac{1}{1 + S_t} \right]} \quad (131)$$

$$= \sqrt{2} \sum_{t=1}^T \sqrt{\frac{1 - (1 - p)^{t+1}}{p(t+1)}} \quad (132)$$

$$\begin{aligned} &\leq \sqrt{2} \sum_{t=1}^T \sqrt{\frac{1}{p(t+1)}} \\ &\leq \frac{2\sqrt{2T}}{\sqrt{p}} \end{aligned} \quad (133)$$

$$= 2\sqrt{2} \sqrt{\frac{|\mathcal{D}|}{\gamma_1}} T^{\frac{1+\omega}{2}}. \quad (134)$$

Where (130) follows by (129); (131) holds because Jensen's inequality and the square-root function is concave; (132) follows since $S_t \sim \text{Bin}(t, p)$ and

$$\begin{aligned} \mathbb{E} \left[\frac{1}{1+S_t} \right] &= \sum_{k=0}^t \frac{1}{1+k} \binom{t}{k} p^k (1-p)^{t-k} \\ &= \sum_{k=0}^t \frac{1}{p(t+1)} \binom{t+1}{k+1} p^{k+1} (1-p)^{t-k} \\ &= \frac{1}{p(t+1)} \sum_{k=0}^t \binom{t+1}{k+1} p^{k+1} (1-p)^{t-k} \\ &= \frac{1}{p(t+1)} \sum_{j=1}^{t+1} \binom{t+1}{j} p^j (1-p)^{t+1-j} \\ &= \frac{1}{p(t+1)} \left[\sum_{j=0}^{t+1} \binom{t+1}{j} p^j (1-p)^{t+1-j} - (1-p)^{t+1} \right] \\ &= \frac{1 - (1-p)^{t+1}}{p(t+1)}; \end{aligned}$$

(133) holds by $\sum_{t=1}^T \frac{1}{\sqrt{t+1}} \leq \sum_{t=1}^T \frac{1}{\sqrt{t}} \leq 2\sqrt{T}$; and (134) is because $p := \frac{\gamma_1}{|\mathcal{D}|T^\omega}$. $\blacksquare$

Theorem 1 (Constraint Violations Upper Bound). *Select $\epsilon_t = t^{-\omega}$. Let $\mathcal{A}, \mathcal{M}, \mathcal{W}, \mathcal{D}$ be finite domain of protected attribute, intermediate features, confounded features, and decisions. Under Assumptions 1 and 2, Algorithm 1 holds that:*

$$V_1(T) = \tilde{\mathcal{O}} \left(|\mathcal{D}| \sqrt{\frac{|\mathcal{A}| |\mathcal{M}| |\mathcal{W}|}{\gamma_1}} T^{\frac{1+\omega}{2}} + T^{1-\omega} \right). \quad (135)$$

Proof. We recall the cumulative constraints violations $V_1(T)$:

$$V_1(T) = \max_{i \in \{1,2,3\}} \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{d \sim \tilde{\pi}_{X_t}} |\Delta_d^{(i)}(X_t)| - c_i \right]. \quad (136)$$

On the high-probability event in Proposition 3, for all $t \in [T]$, $i \in \{1, 2, 3\}$, and all policies $\pi \in \Pi$, we could show the following results as we did in Appendix E.2:

$$\mathbb{E}_{d \sim \pi} \left[|\Delta_d^{(i)}(X_t)| - |\hat{\Delta}_d^{(i)}(X_t)| \right] \leq \sum_{j=1}^3 b_t^{(j)}(\pi). \quad (137)$$

For any $i \in \{1, 2, 3\}$,

$$\begin{aligned} V_1^{(i)}(T) &= \mathbb{E} \left[\sum_{t=1}^T \left[\mathbb{E}_{d \sim \tilde{\pi}_{X_t}} |\Delta_d^{(i)}(X_t)| - c_i \right] \right] \\ &= \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{d \sim \tilde{\pi}_{X_t}} \left[|\Delta_d^{(i)}(X_t)| - |\hat{\Delta}_d^{(i)}(X_t)| \right] + \sum_{t=1}^T \left[\mathbb{E}_{d \sim \tilde{\pi}_{X_t}} |\hat{\Delta}_d^{(i)}(X_t)| - c_i \right] \right] \end{aligned} \quad (138)$$

Define the good event:

$$\mathcal{E} = \left\{ \forall t \in [T], \forall a \in \mathcal{A}, d \in \mathcal{D}, m \in \mathcal{M}, w \in \mathcal{W} : |\hat{g}_{a,d,m,w}(t) - \mathbb{E}[Y_t | A_t = a, D_t = d, M_t = m, W_t = w]| \right\}$$

$$\leq \sqrt{\frac{\log(2/\delta)}{2n_{a,d,m,w}(t)}} \Big\}.$$

Then, by a union bound over all t, a, d, m, w (see Proposition 3), this event holds with probability at least

$$\mathbb{P}(\mathcal{E}) \geq 1 - T|\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta.$$

And the bad event is denoted as $\bar{\mathcal{E}}$. Then

$$\begin{aligned} & \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{d \sim \tilde{\pi}_{X_t}} \left[\left| \Delta_d^{(i)}(X_t) \right| - \left| \hat{\Delta}_d^{(i)}(X_t) \right| \right] \right] \\ &= \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{d \sim \tilde{\pi}_{X_t}} \left[\left| \Delta_d^{(i)}(X_t) \right| - \left| \hat{\Delta}_d^{(i)}(X_t) \right| \right] \middle| \mathcal{E} \right] \cdot \mathbb{P}(\mathcal{E}) + \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{d \sim \tilde{\pi}_{X_t}} \left[\left| \Delta_d^{(i)}(X_t) \right| - \left| \hat{\Delta}_d^{(i)}(X_t) \right| \right] \middle| \bar{\mathcal{E}} \right] \cdot \mathbb{P}(\bar{\mathcal{E}}) \end{aligned} \quad (139)$$

$$\leq \mathbb{E} \left[\sum_{t=1}^T \sum_{j=1}^3 b_t^{(j)}(\tilde{\pi}_{X_t}) \middle| \mathcal{E} \right] \cdot \mathbb{P}(\mathcal{E}) + \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{d \sim \tilde{\pi}_{X_t}} \left[\left| \Delta_d^{(i)}(X_t) \right| - \left| \hat{\Delta}_d^{(i)}(X_t) \right| \right] \middle| \bar{\mathcal{E}} \right] \cdot \mathbb{P}(\bar{\mathcal{E}}) \quad (140)$$

$$\leq \mathbb{E} \left[\sum_{t=1}^T \sum_{j=1}^3 b_t^{(j)}(\tilde{\pi}_{X_t}) \right] + \sum_{t=1}^T \mathbb{P}(\bar{\mathcal{E}}) \quad (141)$$

$$\leq \mathbb{E} \left[\sum_{t=1}^T \sum_{j=1}^3 b_t^{(j)}(\tilde{\pi}_{X_t}) \right] + T^2 |\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta \quad (142)$$

$$= \mathbb{E} \left[\sum_{t=1}^T \sum_{j=1}^3 b_t^{(j)}(\tilde{\pi}_{X_t}) \right] + 1, \quad (143)$$

where (139) is from law of total expectation; (140) follows by (137) when the good event happens; (141) is because $\mathbb{P}(\mathcal{E}) \leq 1$, $|\Delta_d^{(i)}(X_t)| \leq 1$, $|\hat{\Delta}_d^{(i)}(X_t)| \leq 1$, $\forall i \in \{1, 2, 3\}$; (142) follows by $\mathbb{P}(\bar{\mathcal{E}}) \leq T|\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|\delta$; (143) comes from $\delta = \frac{1}{T^2 |\mathcal{A}||\mathcal{D}||\mathcal{M}||\mathcal{W}|}$.

Since $\tilde{\pi}_{X_t}$ is a mixture distribution, i.e., $\tilde{\pi}_{X_t} = (1 - \epsilon_t)\pi_{X_t} + \epsilon_t\pi_0$. Then,

$$\begin{aligned} \mathbb{E}_{d \sim \tilde{\pi}_{X_t}} \left[\left| \hat{\Delta}_d^{(i)}(X_t) \right| - c_i \right] &= (1 - \epsilon_t) \left[\mathbb{E}_{d \sim \pi_{X_t}} \left[\left| \hat{\Delta}_d^{(i)}(X_t) \right| - c_i \right] \right] + \epsilon_t \left[\mathbb{E}_{d \sim \pi_0} \left[\left| \hat{\Delta}_d^{(i)}(X_t) \right| - c_i \right] \right] \\ &\leq \sum_{j=1}^3 b_t^{(j)}(\tilde{\pi}_{X_t}) + \epsilon_t \left[\mathbb{E}_{d \sim \pi_0} \left[\left| \hat{\Delta}_d^{(i)}(X_t) \right| - c_i \right] \right] \end{aligned} \quad (144)$$

$$\leq \sum_{j=1}^3 b_t^{(j)}(\tilde{\pi}_{X_t}) + \epsilon_t, \quad (145)$$

where (144) is by the definition of $\hat{\Pi}_{X_t}^f$.

Combining Equations (138), (143) and (145), we obtain

$$\begin{aligned} V_1^{(i)}(T) &\leq \mathbb{E} \left[2 \sum_{t=1}^T \sum_{j=1}^3 b_t^{(j)}(\tilde{\pi}_{X_t}) + \sum_{t=1}^T \epsilon_t \right] + 1 \\ &= \mathbb{E} \left[2 \sum_{t=1}^T \sum_{j=1}^3 \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(j)}, d \sim \tilde{\pi}_{X_t}} \left[\sqrt{\frac{\log(2/\delta)}{2n_{a'_t,d,m,w}(t)}} + \sqrt{\frac{\log(2/\delta)}{2n_{a_t,d,m,w}(t)}} \right] + \sum_{t=1}^T \epsilon_t \right] + 1 \end{aligned} \quad (146)$$

To further bound the constraint violations, we need to analyze $\sum_{t=1}^T \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(j)}, d \sim \tilde{\pi}_{X_t}} \left[\sqrt{\frac{\log(2/\delta)}{2n_{a'_t,d,m,w}(t)}} + \sqrt{\frac{\log(2/\delta)}{2n_{a_t,d,m,w}(t)}} \right]$.

Taking $\sqrt{\frac{\log(2/\delta)}{2n_{a'_t,d,m,w}(t)}}$ as a start, for each $j \in \{1, 2, 3\}$,

$$\mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(j)}, d \sim \tilde{\pi}_{X_t}} \sqrt{\frac{1}{n_{a'_t,d,m,w}(t)}} \right]$$

$$\begin{aligned}
 &= \mathbb{E} \left[\sum_{t=1}^T \sum_{d \in \mathcal{D}} \sum_{m \in \mathcal{M}} \sum_{w \in \mathcal{W}} \tilde{\pi}_{X_t}(d) \mathbb{P}_{a_t}^{(j)}(m, w) \sqrt{\frac{1}{n_{a'_t, d, m, w}(t)}} \right] \\
 &\leq \mathbb{E} \left[\sum_{t=1}^T \sqrt{\sum_{d \in \mathcal{D}} \sum_{m \in \mathcal{M}} \sum_{w \in \mathcal{W}} \tilde{\pi}_{X_t}(d) \mathbb{P}_{a_t}^{(j)}(m, w)} \sqrt{\sum_{d \in \mathcal{D}} \sum_{m \in \mathcal{M}} \sum_{w \in \mathcal{W}} \tilde{\pi}_{X_t}(d) \mathbb{P}_{a_t}^{(j)}(m, w) \frac{1}{n_{a'_t, d, m, w}(t)}} \right] \quad (147)
 \end{aligned}$$

$$= \mathbb{E} \left[\sum_{t=1}^T \sqrt{\sum_{d \in \mathcal{D}} \sum_{m \in \mathcal{M}} \sum_{w \in \mathcal{W}} \tilde{\pi}_{X_t}(d) \mathbb{P}_{a_t}^{(j)}(m, w) \frac{1}{n_{a'_t, d, m, w}(t)}} \right] \quad (148)$$

$$\begin{aligned}
 &= \mathbb{E} \left[\sum_{t=1}^T \sqrt{\mathbb{E}_{(m, w) \sim \mathbb{P}_{a_t}^{(j)}, d \sim \tilde{\pi}_{X_t}} \frac{1}{n_{a'_t, d, m, w}(t)}} \right] \\
 &\leq \mathbb{E} \left[\sqrt{T \sum_{t=1}^T \mathbb{E}_{(m, w) \sim \mathbb{P}_{a_t}^{(j)}, d \sim \tilde{\pi}_{X_t}} \frac{1}{n_{a'_t, d, m, w}(t)}} \right] \quad (149)
 \end{aligned}$$

$$\leq \sqrt{T \mathbb{E} \left[\sum_{t=1}^T \mathbb{E}_{(m, w) \sim \mathbb{P}_{a_t}^{(j)}, d \sim \tilde{\pi}_{X_t}} \frac{1}{n_{a'_t, d, m, w}(t)} \right]} \quad (150)$$

$$\leq \sqrt{T \sum_{a, d, w, m} \mathbb{E} \left[\sum_{t=1}^T \frac{1}{n_{a, d, m, w}(t)} \right]} \quad (151)$$

$$\leq \sqrt{T |\mathcal{A}| |\mathcal{D}| |\mathcal{W}| |\mathcal{M}| \sum_{t=1}^T \frac{|\mathcal{D}| T^\omega}{\gamma_1 (t+1)}} \quad (152)$$

$$= \tilde{\mathcal{O}} \left(|\mathcal{D}| \sqrt{\frac{|\mathcal{A}| |\mathcal{M}| |\mathcal{W}|}{\gamma_1}} T^{\frac{1+\omega}{2}} \right), \quad (153)$$

where (147) is by Cauchy-Schwarz inequality; (148) holds by $\sum_{d \in \mathcal{D}} \sum_{m \in \mathcal{M}} \sum_{w \in \mathcal{W}} \tilde{\pi}_{X_t}(d) \mathbb{P}_{a_t}^{(j)}(m, w) = 1$; (149) comes from Cauchy-Schwarz inequality across time; (150) follows by Jensen's inequality and square root is concave; (151) is by $\mathbb{E}_{(m, w) \sim \mathbb{P}_{a_t}^{(j)}, d \sim \tilde{\pi}_{X_t}} \left[\frac{1}{n_{a'_t, d, m, w}(t)} \right] = \sum_{d \in \mathcal{D}} \sum_{m \in \mathcal{M}} \sum_{w \in \mathcal{W}} \tilde{\pi}_{X_t}(d) \mathbb{P}_{a_t}^{(j)}(m, w) \frac{1}{n_{a'_t, d, m, w}(t)}$; (152) is from the coupling in Proposition 4 (see (129)). The same steps apply for term $\sqrt{\frac{\log(2/\delta)}{2n_{a'_t, d, m, w}(t)}}$.

Since $\sum_{t=1}^T \epsilon_t = \mathcal{O}(T^{1-\omega})$, therefore, by Equations (136), (146) and (153), we have

$$V_1(T) = \tilde{\mathcal{O}} \left(|\mathcal{D}| \sqrt{\frac{|\mathcal{A}| |\mathcal{M}| |\mathcal{W}|}{\gamma_1}} T^{\frac{1+\omega}{2}} + T^{1-\omega} \right),$$

which finishes the proof. ■

F. Proof for Section 4

In this section, we provide the proofs of the theoretical results for logistic bandits setting in Section 4. We prove the confidence set for logistic bandits for completeness.

F.1. Regret Guarantee

Proposition 5 ((Zhang et al., 2025) Theorem 1). *Let the learning rate $\eta = 1 + S$ and the regularization parameter $\lambda = 2 \max\{7n\eta, 3\eta S, 1\}$. For each $t \in [T]$ and any $\alpha \in (0, 1]$, define the confidence set as*

$$\mathcal{C}_t(\alpha) = \left\{ \theta \in \Theta : \|\hat{\theta}_t - \theta\|_{H_t} \leq \gamma_t(\alpha) \right\}.$$

Where the radius is given by $\gamma_t(\alpha) = \sqrt{4\lambda S^2 + 2\eta \ln(\frac{1}{\alpha}) + 6n\eta^2 \ln(2 + \frac{2t}{\lambda})}$. Then, we have $\mathbb{P}(\forall t \geq 1, \theta_* \in \mathcal{C}_t(\alpha)) \geq 1 - \alpha$.

Proof. We include the proof for completeness and to illustrate how the confidence set is constructed and later used in our analysis.

To prove the above results, it suffices to show that the estimation error of OMD estimator $\|\hat{\theta}_{t+1} - \theta_*\|_{H_t}$ is bounded. First, by Lemma 4 and Lemma 5, when setting $\eta = 1 + S$, we can upper bound the estimation error of the online estimator:

$$\begin{aligned} \|\hat{\theta}_{t+1} - \theta_*\|_{H_t}^2 &\leq 2\eta \left(\sum_{s=1}^t \mathcal{L}_s(\theta_*) - \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1}) \right) + 4\lambda S^2 + 2\eta S \sum_{s=1}^t \|\hat{\theta}_s - \hat{\theta}_{s+1}\|_2^2 \\ &\quad - \sum_{s=1}^t \|\hat{\theta}_s - \hat{\theta}_{s+1}\|_{H_s}^2, \end{aligned} \quad (154)$$

where $\mathcal{L}_s$ is logistic loss function. The above inequality has been shown in the logistic bandits literature (Zhang & Sugiyama, 2023).

Next, we are going to upper bound $\sum_{s=1}^t \mathcal{L}_s(\theta_*) - \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1})$. It is inspired and shares close connections with the work of (Jézéquel et al., 2020; Faury et al., 2022; Zhang & Sugiyama, 2023). To bound (154), we can decompose it into

$$\sum_{s=1}^t \mathcal{L}_s(\theta_*) - \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1}) = \sum_{s=1}^t \mathcal{L}_s(\theta_*) - \sum_{s=1}^t m_s(P_s) + \sum_{s=1}^t m_s(P_s) - \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1}), \quad (155)$$

where in (155), we define $P_s = \mathcal{N}(\hat{\theta}_s, hH_s^{-1})$ as an n -variate Gaussian distribution with mean $\hat{\theta}_s \in \mathbb{R}^n$ and covariance matrix $hH_s^{-1} \in \mathbb{R}^{n \times n}$, where $h > 0$ is a constant to be specified, and we define $m_s : \mathcal{P}(\mathbb{R}^n) \rightarrow \mathbb{R}$ to be

$$m_s(P_s) = -\log(\mathbb{E}_{\theta \sim P_s}[\exp\{-\mathcal{L}_s(\theta)\}]).$$

Then we are going to bound $\sum_{s=1}^t \mathcal{L}_s(\theta_*) - \sum_{s=1}^t m_s(P_s)$. Since P_s is a Gaussian distribution with mean $\hat{\theta}_s$ and covariance hH_s^{-1} , it is $\mathcal{F}_s$ -measurable (σ -algebra generated by everything that has happened so far $\{X_\tau, d_\tau, y_\tau\}_{\tau=1}^{s-1}$). By Lemma 6, we have that for any $\alpha \in (0, 1]$, with probability at least $1 - \alpha$,

$$\sum_{s=1}^t \mathcal{L}_s(\theta_*) - \sum_{s=1}^t m_s(P_s) = \sum_{s=1}^t \left[\mathcal{L}_s(\theta_*) + \log(\mathbb{E}_{\theta \sim P_s}[\exp\{-\mathcal{L}_s(\theta)\}]) \right] \leq \log\left(\frac{1}{\alpha}\right). \quad (156)$$

Next, we proceed to analyze $\sum_{s=1}^t m_s(P_s) - \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1})$. Under the condition that

$$\lambda \geq \max\{14n\eta, 2\},$$

by Lemma 7 with $h = \frac{3}{2}\eta$,

$$\sum_{s=1}^t m_s(P_s) - \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1}) \leq \frac{1}{3\eta} \sum_{s=1}^t \|\hat{\theta}_{s+1} - \hat{\theta}_s\|_{H_s}^2 + 3n\eta \ln\left(2 + \frac{2t}{\lambda}\right). \quad (157)$$

Finally, by (154), (155), (156), and (157), we obtain

$$\begin{aligned} \|\hat{\theta}_{t+1} - \theta_*\|_{H_t}^2 &\leq 4\lambda S^2 + 2\eta \log\left(\frac{1}{\alpha}\right) + 6n\eta^2 \ln\left(2 + \frac{2t}{\lambda}\right) + 2\eta S \sum_{s=1}^t \|\hat{\theta}_s - \hat{\theta}_{s+1}\|_2^2 \\ &\quad - \frac{1}{3} \sum_{s=1}^t \|\hat{\theta}_s - \hat{\theta}_{s+1}\|_{H_s}^2 \\ &\leq 4\lambda S^2 + 2\eta \log\left(\frac{1}{\alpha}\right) + 6n\eta^2 \ln\left(2 + \frac{2t}{\lambda}\right). \end{aligned} \quad (158)$$

Where (158) follows by $\lambda \geq 6\eta S$. ■

Theorem 2 (Regret Upper Bound). *Let $\eta = 1 + S$ and $\lambda = 2 \max\{7n\eta, 3\eta S, 1\}$. Under Assumptions 1 and 5, Algorithm 2 achieves the following bound with probability at least $1 - \alpha - \delta_1$ for any $\alpha \in (0, 1]$ and $\delta_1 \in (0, 1]$,*

$$R_2(T) = \mathcal{O}\left(nS\sqrt{S + \log(T)}\sqrt{(T - T_0)\log(T)} + \kappa n^2 S^2 \log^2(T) + \sqrt{(T - T_0)\log(1/\delta_1)} + T_0\right).$$

Proof. In this section, we prove the sublinear upper bound for regret, we first show the proofs for upper bounding errors in mean rewards between θ_* and $\tilde{\theta}_t$ for logistic bandits. Here, we denote $\tilde{\theta}_t = \arg \max_{\theta \in \mathcal{C}_t(\alpha)} \sigma(f(Z_{d_t})^\top \theta)$ as the optimistic parameter for the decision d_t for context X_t .

$$R'_{\log} := \sum_{t=1}^T \sigma(f(Z_{d_t})^\top \tilde{\theta}_t) - \sigma(f(Z_{d_t})^\top \theta_*) \quad (159)$$

$$= \sum_{t=1}^T \dot{\sigma}(f(Z_{d_t})^\top \theta_*) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*) + \sum_{t=1}^T \int_{f(Z_{d_t})^\top \theta_*}^{f(Z_{d_t})^\top \tilde{\theta}_t} \ddot{\sigma}(u) (f(Z_{d_t})^\top \tilde{\theta}_t - u) du \quad (160)$$

$$= \underbrace{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*)}_{\text{term (1)}} + \underbrace{\sum_{t \in [T]/I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*)}_{\text{term (2)}} \\ + \underbrace{\sum_{t=1}^T \int_{f(Z_{d_t})^\top \theta_*}^{f(Z_{d_t})^\top \tilde{\theta}_t} \ddot{\sigma}(u) (f(Z_{d_t})^\top \tilde{\theta}_t - u) du}_{\text{term (3)}}, \quad (161)$$

where (160) follows by the Taylor series expansion of $\sigma(f(Z_{d_t})^\top \tilde{\theta}_t)$ on $\sigma(f(Z_{d_t})^\top \theta_*)$ with a first order integral remainder. As for (161), we divide the horizon into two parts $I = \{t \in [T] : \dot{\sigma}(f(Z_{d_t})^\top \theta_*) \geq \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1})\}$ and $[T]/I$.

Analysis for term (1). For the first term, we have:

$$\text{term (1)} = \sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*) \\ = \sum_{t \in I} \left[\dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1}) + \ddot{\sigma}(\zeta_t) f(Z_{d_t})^\top (\theta_* - \hat{\theta}_{t+1}) \right] f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*) \quad (162)$$

$$\leq \underbrace{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1}) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*)}_{\text{term (1-1)}} + \underbrace{\sum_{t \in I} |\dot{\sigma}(f(Z_{d_t})^\top \theta_* - \hat{\theta}_{t+1}) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*)|}_{\text{term (1-2)}}, \quad (163)$$

where (162) follows by mean value theorem and ζ_t is the (unknown) point between $f(Z_{d_t})^\top \theta_*$ and $f(Z_{d_t})^\top \hat{\theta}_{t+1}$ for which the mean value theorem holds; and (163) is because $|\ddot{\sigma}(\zeta_t)| \leq 1$.

We then prove the upper bound for term (1-1):

$$\text{term (1-1)} = \sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1}) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*) \\ \leq \sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1}) \|f(Z_{d_t})\|_{H_t^{-1}} \|\tilde{\theta}_t - \theta_*\|_{H_t} \quad (164)$$

$$\leq \sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1}) \|f(Z_{d_t})\|_{H_t^{-1}} (\|\tilde{\theta}_t - \hat{\theta}_t\|_{H_t} + \|\hat{\theta}_t - \theta_*\|_{H_t}) \quad (165)$$

$$\leq 2\gamma_T(\alpha) \sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1}) \|f(Z_{d_t})\|_{H_t^{-1}} \quad (166)$$

$$\leq 2\gamma_T(\alpha) \sqrt{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1})} \sqrt{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1}) \|f(Z_{d_t})\|_{H_t^{-1}}^2} \quad (167)$$

$$= 2\gamma_T(\alpha) \sqrt{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1})} \sqrt{\sum_{t \in I} \|\sqrt{\dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1})} f(Z_{d_t})\|_{H_t^{-1}}^2}$$

$$\leq 4\gamma_T(\alpha)\sqrt{n\log\left(\lambda + \frac{T}{n}\right)}\sqrt{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1})} \quad (168)$$

$$\leq 4\gamma_T(\alpha)\sqrt{n\log\left(\lambda + \frac{T}{n}\right)}\sqrt{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*)}, \quad (169)$$

where (164) is by the Cauchy–Schwarz inequality; (165) comes from the triangle inequality; (166) follows by the confidence set in Proposition 5 where $\gamma_T(\alpha) = \max_{t \in [T]} \gamma_t(\alpha)$; (167) is by the Cauchy–Schwarz inequality; (168) holds because of Lemma 3; and (169) follows by $\dot{\sigma}(f(Z_{d_t})^\top \theta_*) \geq \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1})$.

In order to bound term (1-1), we need to further upper bound $\sqrt{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*)}$.

$$\begin{aligned} & \sum_{t=1}^T \dot{\sigma}(f(Z_{d_t})^\top \theta_*) \\ &= \sum_{t=1}^T \dot{\sigma}(f(Z_{d_t})^\top \tilde{\theta}_t) + \sum_{t=1}^T \int_{f(Z_{d_t})^\top \tilde{\theta}_t}^{f(Z_{d_t})^\top \theta_*} \ddot{\sigma}(u) du \end{aligned} \quad (170)$$

$$= \sum_{t=1}^T \dot{\sigma}(f(Z_{d_t})^\top \tilde{\theta}_t) + \sum_{t=1}^T \left[\int_{v=0}^1 \ddot{\sigma}\left(f(Z_{d_t})^\top \tilde{\theta}_t + v f(Z_{d_t})^\top (\theta_* - \tilde{\theta}_t)\right) dv \right] f(Z_{d_t})^\top (\theta_* - \tilde{\theta}_t) \quad (171)$$

$$\leq \sum_{t=1}^T \dot{\sigma}(f(Z_{d_t})^\top \tilde{\theta}_t) + \sum_{t=1}^T \left[\int_{v=0}^1 \ddot{\sigma}\left(f(Z_{d_t})^\top \tilde{\theta}_t + v f(Z_{d_t})^\top (\theta_* - \tilde{\theta}_t)\right) dv \right] f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*) \quad (172)$$

$$\leq \sum_{t=1}^T \dot{\sigma}(f(Z_{d_t})^\top \tilde{\theta}_t) + \sum_{t=1}^T \left[\sigma\left(f(Z_{d_t})^\top \tilde{\theta}_t\right) - \sigma\left(f(Z_{d_t})^\top \theta_*\right) \right] \quad (173)$$

$$= \sum_{t=1}^T \dot{\sigma}(f(Z_{d_t})^\top \tilde{\theta}_t) + R'_{\log} \quad (174)$$

$$\leq T + R'_{\log}, \quad (175)$$

where (170) comes from Taylor expansion with an integral remainder; (171) follows by changing variables; and (172) holds since $\dot{\sigma} \geq -\ddot{\sigma}$ and $\sigma(f(Z_{d_t})^\top \tilde{\theta}_t) \geq \sigma(f(Z_{d_t})^\top \theta_*)$ w.h.p.; (173) follows by the mean-value theorem on (172); (174) is from (159); and (175) is because $\dot{\sigma}(f(Z_{d_t})^\top \tilde{\theta}_t) \leq 1$.

Therefore, we could bound term (1-1) as follows:

$$\begin{aligned} \text{term (1-1)} &\leq 4\gamma_T(\alpha)\sqrt{n\log\left(\lambda + \frac{T}{n}\right)}\sqrt{\sum_{t \in I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*)} \\ &\leq 4\gamma_T(\alpha)\sqrt{n\log\left(\lambda + \frac{T}{n}\right)}\sqrt{\sum_{t \in [T]} \dot{\sigma}(f(Z_{d_t})^\top \theta_*)} \\ &\leq 4\gamma_T(\alpha)\sqrt{n\log\left(\lambda + \frac{T}{n}\right)} \cdot \sqrt{R'_{\log} + T}, \end{aligned} \quad (176)$$

where (176) follows by (169) and (175).

Next, we are going to bound term (1-2):

$$\begin{aligned} \text{term (1-2)} &= \sum_{t \in I} |f(Z_{d_t})^\top (\theta_* - \hat{\theta}_{t+1}) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*)| \\ &\leq \sum_{t \in I} \|f(Z_{d_t})\|_{H_{t+1}^{-1}} \|\theta_* - \hat{\theta}_{t+1}\|_{H_{t+1}} \|f(Z_{d_t})\|_{H_t^{-1}} \|\tilde{\theta}_t - \theta_*\|_{H_t} \end{aligned} \quad (177)$$

$$\leq \sum_{t \in I} \|f(Z_{d_t})\|_{H_t^{-1}}^2 \|\theta_* - \hat{\theta}_{t+1}\|_{H_{t+1}} \|\tilde{\theta}_t - \theta_*\|_{H_t} \quad (178)$$

$$\leq 2\gamma_T^2(\alpha) \sum_{t \in I} \|f(Z_{d_t})\|_{H_t^{-1}}^2 \quad (179)$$

$$\leq 4\gamma_T^2(\alpha) \kappa n \log\left(\lambda + \frac{T}{n}\right), \quad (180)$$

where (177) is by Cauchy–Schwarz inequality; (178) holds since $H_{t+1} \geq H_t$ then $H_{t+1}^{-1} \leq H_t^{-1}$; (179) follows by the confidence set in Proposition 5 and the triangle inequality; and (180) comes from Lemma 3 and the following relationship:

$$H_t = \sum_{s=1}^{t-1} \dot{\sigma}\left(f(Z_s)^\top \hat{\theta}_{s+1}\right) f(Z_s) f(Z_s)^\top + \lambda I_n \quad (181)$$

$$\geq \frac{1}{\kappa} \left[\sum_{s=1}^{t-1} f(Z_s) f(Z_s)^\top + \lambda I_n \right], \quad (182)$$

where (181) follows by the definition of cumulative Hessian matrix H_t and (182) is by the definition of $\kappa = \max_{Z \in \mathcal{Z}} \max_{\theta \in \Theta} \frac{1}{\dot{\sigma}(f(Z)^\top \theta)}$, thus we have $H_t^{-1} \leq \kappa \left[\sum_{s=1}^{t-1} f(Z_s) f(Z_s)^\top + \lambda I_n \right]^{-1}$.

Therefore, by (176) and (180), we have:

$$\text{term (1)} \leq 4\gamma_T^2(\alpha) \kappa n \log\left(\lambda + \frac{T}{n}\right) + 4\gamma_T(\alpha) \sqrt{n \log\left(\lambda + \frac{T}{n}\right)} \cdot \sqrt{R'_{\log} + T}. \quad (183)$$

Analysis for term (2). For the second term, we have:

$$\begin{aligned} \text{term (2)} &= \sum_{t \in [T]/I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*) f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*) \\ &\leq \sum_{t \in [T]/I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*) \|f(Z_{d_t})\|_{H_t^{-1}} \|\tilde{\theta}_t - \theta_*\|_{H_t} \end{aligned} \quad (184)$$

$$\leq 2\gamma_T(\alpha) \sqrt{\sum_{t \in [T]/I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*)} \sqrt{\sum_{t \in [T]/I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*) \|f(Z_{d_t})\|_{H_t^{-1}}^2} \quad (185)$$

$$\leq 2\gamma_T(\alpha) \sqrt{\sum_{t \in [T]/I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*)} \sqrt{\sum_{t \in [T]/I} \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1}) \|f(Z_{d_t})\|_{H_t^{-1}}^2} \quad (186)$$

$$\begin{aligned} &= 2\gamma_T(\alpha) \sqrt{\sum_{t \in [T]/I} \dot{\sigma}(f(Z_{d_t})^\top \theta_*)} \sqrt{\sum_{t \in [T]/I} \|\sqrt{\dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1})} f(Z_{d_t})\|_{H_t^{-1}}^2} \\ &\leq 4\gamma_T(\alpha) \sqrt{n \log\left(\lambda + \frac{T}{n}\right)} \cdot \sqrt{R'_{\log} + T}. \end{aligned} \quad (187)$$

Where (184) is by Cauchy–Schwarz inequality; (185) follows by Proposition 5 and the triangle inequality; (186) holds since $\dot{\sigma}(f(Z_{d_t})^\top \theta_*) \leq \dot{\sigma}(f(Z_{d_t})^\top \hat{\theta}_{t+1})$; and (187) comes from (176) and Lemma 3.

Analysis for term (3). For the third term, we have:

$$\begin{aligned} \text{term (3)} &= \sum_{t=1}^T \int_{f(Z_{d_t})^\top \theta_*}^{f(Z_{d_t})^\top \tilde{\theta}_t} \ddot{\sigma}(u) (f(Z_{d_t})^\top \tilde{\theta}_t - u) du \\ &= \sum_{t=1}^T \left[\int_{v=0}^1 (1-v) \ddot{\sigma}(f(Z_{d_t})^\top \theta_* + v f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*)) dv \right] (f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*))^2 \end{aligned} \quad (188)$$

$$\leq \sum_{t=1}^T (f(Z_{d_t})^\top (\tilde{\theta}_t - \theta_*))^2 \quad (189)$$

$$\leq \sum_{t=1}^T \|f(Z_{d_t})\|_{H_t^{-1}}^2 \|\tilde{\theta}_t - \theta_*\|_{H_t}^2 \quad (190)$$

$$\leq 2\gamma_T^2(\alpha) \sum_{t=1}^T \|f(Z_{d_t})\|_{H_t^{-1}}^2 \quad (191)$$

$$\leq 4\gamma_T^2(\alpha) \kappa n \log\left(\lambda + \frac{T}{n}\right), \quad (192)$$

where (188) follows by changing variables; (189) holds because of $\bar{\sigma} \leq 1$; (190) follows by Cauchy-Schwarz inequality; (191) is from Proposition 5 and the triangle inequality; and (192) is by (180).

Analysis for $R'_{\log}$. Together with all the results above, we could bound the errors in mean reward.

$$\begin{aligned} R'_{\log} &\leq 8\gamma_T(\alpha) \sqrt{n \log\left(\lambda + \frac{T}{n}\right)} \cdot \sqrt{R'_{\log} + T} + 8\gamma_T^2(\alpha) \kappa n \log\left(\lambda + \frac{T}{n}\right) \\ &\leq 8\gamma_T(\alpha) \sqrt{n \log\left(\lambda + \frac{T}{n}\right)} \cdot (\sqrt{R'_{\log}} + \sqrt{T}) + 8\gamma_T^2(\alpha) \kappa n \log\left(\lambda + \frac{T}{n}\right). \end{aligned} \quad (193)$$

Where (193) follows by (161), (183), (187), and (192). And by (Abeille et al., 2021, Proposition 7), we have,

$$\begin{aligned} R'_{\log} &\leq \left(8\gamma_T(\alpha) \sqrt{n \log\left(\lambda + \frac{T}{n}\right)} + \sqrt{8\gamma_T(\alpha) \sqrt{nT \log\left(\lambda + \frac{T}{n}\right)} + 8\gamma_T^2(\alpha) \kappa n \log\left(\lambda + \frac{T}{n}\right)}\right)^2 \\ &\leq 128n\gamma_T^2(\alpha) \log\left(\lambda + \frac{T}{n}\right) + 16\gamma_T(\alpha) \sqrt{nT \log\left(\lambda + \frac{T}{n}\right)} + 16\gamma_T^2(\alpha) \kappa n \log\left(\lambda + \frac{T}{n}\right). \end{aligned} \quad (194)$$

Where (194) holds because of $(a+b)^2 \leq 2a^2 + 2b^2$ for $a > 0, b > 0$.

By the confidence set in Proposition 5, we have $\gamma_t(\alpha) = \sqrt{4\lambda S^2 + 2\eta \ln(\frac{1}{\alpha}) + 6n\eta^2 \ln\left(2 + \frac{2t}{\lambda}\right)} = \mathcal{O}(S\sqrt{n(S + \log(t))})$, thus

$$R'_{\log} = \mathcal{O}\left(nS\sqrt{S + \log(T)}\sqrt{T \log(T)} + \kappa n^2 S^2 \log^2(T)\right). \quad (195)$$

Bound for logistic regret. Above all, we could upper bound the logistic regret as follows:

$$\begin{aligned} R_2(T) &= \sum_{t=1}^T \left[\max_{\pi \in \Pi_t^f} \langle \pi, \sigma(f(Z_d)^\top \theta_*) \rangle - \sigma(f(Z_{d_t})^\top \theta_*) \right] \\ &= \sum_{t=1}^{T_0} \left[\max_{\pi \in \Pi_t^f} \langle \pi, \sigma(f(Z_d)^\top \theta_*) \rangle - \sigma(f(Z_{d_t})^\top \theta_*) \right] + \sum_{t=T_0+1}^T \left[\max_{\pi \in \Pi_t^f} \langle \pi, \sigma(f(Z_d)^\top \theta_*) \rangle - \sigma(f(Z_{d_t})^\top \theta_*) \right] \end{aligned} \quad (196)$$

$$\leq T_0 + \sum_{t=T_0+1}^T \left[\max_{\pi \in \Pi_t^f} \langle \pi, \sigma(f(Z_d)^\top \theta_*) \rangle - \sigma(f(Z_{d_t})^\top \theta_*) \right] \quad (197)$$

$$\leq T_0 + \sum_{t=T_0+1}^T \left[\max_{\pi \in \Pi_t^f} \langle \pi, \tilde{y}_t \rangle - \sigma(f(Z_{d_t})^\top \theta_*) \right] \quad (198)$$

$$\leq T_0 + \sum_{t=T_0+1}^T \left[\max_{\pi \in \hat{\Pi}_t^f} \langle \pi, \tilde{y}_t \rangle - \sigma(f(Z_{d_t})^\top \theta_*) \right] \quad (199)$$

$$= T_0 + \sum_{t=T_0+1}^T \left[\langle \pi_t, \tilde{y}_t \rangle - \sigma(f(Z_{d_t})^\top \theta_*) \right] \quad (200)$$

$$= T_0 + \sum_{t=T_0+1}^T \left[\langle \pi_t, \tilde{y}_t \rangle - \tilde{y}_t(d_t) + \tilde{y}_t(d_t) - \sigma(f(Z_{d_t})^\top \theta_*) \right] \quad (201)$$

$$\leq T_0 + \sum_{t=T_0+1}^T [\langle \pi_t, \tilde{y}_t \rangle - \tilde{y}_t(d_t)] + R'_{\log} \quad (202)$$

$$\begin{aligned}
 &\leq T_0 + R'_{\log} + \sqrt{2T \ln \frac{2}{\delta_1}} \\
 &= \mathcal{O}\left(nS\sqrt{S + \log(T)}\sqrt{(T - T_0)\log(T)} + \kappa n^2 S^2 \log^2(T) + \sqrt{(T - T_0)\log(1/\delta_1)} + T_0\right),
 \end{aligned} \tag{203}$$

where (196) is by splitting the sum; (197) holds since the rewards is bounded in $[0, 1]$; (198) comes from the optimistic reward, i.e., $\tilde{y}_t(d) = \max_{\theta \in \mathcal{C}_t(\alpha)} \sigma(f(Z_d)^\top \theta)$, $\forall d \in \mathcal{D}$; (199) is from Proposition 6; (200) follows by the OFU principle in (14); (201) is by adding technical terms; (202) follows by (159) with $\tilde{y}_t(d_t) = \sigma(f(Z_{d_t})^\top \tilde{\theta}_t)$; and as for (203), let $\mathcal{G}_{t-1} = \sigma(\mathcal{F}_{t-1}, X_t)$ be the sigma-field before sampling d_t . Since π_t and $\tilde{y}_t$ are $\mathcal{G}_{t-1}$ -measurable and $d_t \sim \pi_t$ conditional on $\mathcal{G}_{t-1}$, the sequence

$$M_t = \langle \pi_t, \tilde{y}_t \rangle - \tilde{y}_t(d_t)$$

is a martingale difference sequence with $|M_t| \leq 1$. By the Azuma–Hoeffding inequality (Lemma 2), with probability at least $1 - \delta_1$,

$$\sum_{t=1}^T M_t \leq \sqrt{2T \ln \left(\frac{2}{\delta_1} \right)}.$$

Combining this with the confidence-set event (which holds with probability at least $1 - \alpha$ and taking a union bound yields overall probability at least $1 - \alpha - \delta_1$. $\blacksquare$

F.2. Constraint Violations Guarantee

Proposition 6. *In the logistic bandits setting, with probability at least $1 - \alpha$, it holds that $\Pi_t^f \subseteq \hat{\Pi}_t^f$ for all $t \in [T]$.*

Proof. Firstly, we recall the structural counterfactual measures as following:

$$\mathbb{E}_{d \sim \pi} \left| \Delta_d^{(1)}(\theta_*) \right| = \mathbb{E}_{d \sim \pi} \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right] \right| \tag{204}$$

$$\mathbb{E}_{d \sim \pi} \left| \Delta_d^{(2)}(\theta_*) \right| = \mathbb{E}_{d \sim \pi} \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right| \tag{205}$$

$$\mathbb{E}_{d \sim \pi} \left| \Delta_d^{(3)}(\theta_*) \right| = \mathbb{E}_{d \sim \pi} \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right|. \tag{206}$$

Where $\mathbb{P}_{a_t}^{(1)}(m, w) = \mathbb{P}(m|a_t, w)\mathbb{P}(w|a_t)$, $\mathbb{P}_{a_t}^{(2)}(m, w) = \mathbb{P}(m|a'_t, w)\mathbb{P}(w|a_t)$, $\mathbb{P}_{a_t}^{(3)}(m, w) = \mathbb{P}(m|a'_t, w)\mathbb{P}(w|a'_t)$.

By definition of the true feasible region Π_t^f and the optimistic estimated feasible region $\hat{\Pi}_t^f$, we have

$$\Pi_t^f = \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(\theta_*)| \leq c_i, \forall i \in \{1, 2, 3\} \right\} \tag{207}$$

$$\hat{\Pi}_t^f = \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} [|\Delta_d^{(i)}(\hat{\theta}_t)| - \xi_t(d)] \leq c_i, \forall i \in \{1, 2, 3\} \right\}. \tag{208}$$

We first look at $i = 1$:

$$\begin{aligned}
 \left| \Delta_d^{(1)}(\hat{\theta}_t) - \Delta_d^{(1)}(\theta_*) \right| &= \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) \right] - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\right. \right. \\
 &\quad \left. \left. \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right] \right| \\
 &\leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left| \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) \right. \\
 &\quad \left. + \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right|
 \end{aligned} \tag{209}$$

$$\begin{aligned}
 &\leq \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\left| \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right| + \left| \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) \right. \right. \\
 &\quad \left. \left. - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right| \right]
 \end{aligned} \tag{210}$$

$$\leq \frac{1}{4} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\left| f(Z'_{a'_t, d, m, w})^\top (\hat{\theta}_t - \theta_*) \right| + \left| f(Z_{a_t, d, m, w})^\top (\hat{\theta}_t - \theta_*) \right| \right] \tag{211}$$

$$\leq \frac{1}{4} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\|f(Z'_{a'_t, d, m, w})\|_{H_t^{-1}} \|\hat{\theta}_t - \theta_*\|_{H_t} + \|f(Z_{a_t, d, m, w})\|_{H_t^{-1}} \|\hat{\theta}_t - \theta_*\|_{H_t} \right] \quad (212)$$

$$\leq \frac{1}{4} \gamma_t(\alpha) \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\|f(Z'_{a'_t, d, m, w})\|_{H_t^{-1}} + \|f(Z_{a_t, d, m, w})\|_{H_t^{-1}} \right]. \quad (213)$$

Where (209) follows by Jensen's inequality; (210) comes from the triangle inequality; (211) holds since $|\sigma(u) - \sigma(v)| \leq \frac{1}{4}|u - v|$, $\forall u, v \in \mathbb{R}$; (212) follows by Cauchy-Schwarz inequality; and (213) comes from Proposition 5.

Similarly, for $i = 2$ and $i = 3$, we have:

$$\begin{aligned} \left| \Delta_d^{(2)}(\hat{\theta}_t) - \Delta_d^{(2)}(\theta_*) \right| &= \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \right. \\ &\quad \left. \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right| \\ &\leq \frac{1}{4} \gamma_t(\alpha) \left(\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \|f(Z'_{a'_t, d, m, w})\|_{H_t^{-1}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \|f(Z'_{a'_t, d, m, w})\|_{H_t^{-1}} \right) \\ \\ \left| \Delta_d^{(3)}(\hat{\theta}_t) - \Delta_d^{(3)}(\theta_*) \right| &= \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \right. \\ &\quad \left. \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right| \\ &\leq \frac{1}{4} \gamma_t(\alpha) \left(\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \|f(Z'_{a'_t, d, m, w})\|_{H_t^{-1}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \|f(Z'_{a'_t, d, m, w})\|_{H_t^{-1}} \right). \end{aligned}$$

Since

$$\begin{aligned} \xi_t(d) &= \frac{1}{4} \gamma_t(\alpha) \left(\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \|f(Z'_{a'_t, d, M, W})\|_{H_t^{-1}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \|f(Z'_{a'_t, d, M, W})\|_{H_t^{-1}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \|f(Z'_{a'_t, d, M, W})\|_{H_t^{-1}} \right. \\ &\quad \left. + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \|f(Z_{a_t, d, M, W})\|_{H_t^{-1}} \right), \end{aligned}$$

for any $\pi \in \Pi_t^f$, we obtain for every $i \in \{1, 2, 3\}$:

$$\mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(\hat{\theta}_t)| - \mathbb{E}_{d \sim \pi} |\Delta_d^{(i)}(\theta_*)| \leq \mathbb{E}_{d \sim \pi} [|\Delta_d^{(i)}(\hat{\theta}_t) - \Delta_d^{(i)}(\theta_*)|] \leq \mathbb{E}_{d \sim \pi} \xi_t(d).$$

Therefore,

$$\mathbb{E}_{d \sim \pi} [|\Delta_d^{(i)}(\hat{\theta}_t)| - \xi_t(d)] \leq c_i,$$

which implies $\pi \in \hat{\Pi}_t^f$. Hence $\Pi_t^f \subseteq \hat{\Pi}_t^f$. ■

Proposition 7. With probability at least $1 - \delta_2$, simultaneously for every $a \in \mathcal{A}$, $d \in \mathcal{D}$, $m \in \mathcal{M}$, and $w \in \mathcal{W}$,

$$\sum_{t=1}^T \|f(Z_{a, d, m, w})\|_{H_t^{-1}} = \mathcal{O} \left(\sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}} \right) \right).$$

Proof. For any positive definite matrix $H_t = \lambda I_n + \sum_{s=1}^{t-1} \dot{\sigma}(f(Z_{a_s, d_s, m_s, w_s})^\top \hat{\theta}_{s+1}) f(Z_{a_s, d_s, m_s, w_s}) f(Z_{a_s, d_s, m_s, w_s})^\top$ and any vector $f(Z_{a, d, m, w})$, by standard eigenvalue inequality, we have

$$\|f(Z_{a, d, m, w})\|_{H_t^{-1}} = \sqrt{f(Z_{a, d, m, w})^\top H_t^{-1} f(Z_{a, d, m, w})} \leq \|f(Z_{a, d, m, w})\|_2 \sqrt{\lambda_{\max}(H_t^{-1})}.$$

Since $\lambda_{\max}(H_t^{-1}) = \frac{1}{\lambda_{\min}(H_t)}$ and $\|f(Z_{a, d, m, w})\|_2 \leq 1$ (Assumption 5), the following holds

$$\|f(Z_{a, d, m, w})\|_{H_t^{-1}} \leq \frac{1}{\sqrt{\lambda_{\min}(H_t)}}. \quad (214)$$

The next is to bound $\lambda_{\min}(H_t)$, which is a property of H_t alone. Hence the final bound is uniform over $a \in \mathcal{A}$, $d \in \mathcal{D}$, $m \in \mathcal{M}$, and $w \in \mathcal{W}$.

For $s \leq T_0$, the played decision is $d_s \sim \text{Unif}(\mathcal{D})$ and $X_s = (a_s, m_s, w_s) \sim \mathbb{P}_X$, both d_s and X_s are i.i.d.. Define the following matrix

$$V_s = f(Z_{a_s, d_s, m_s, w_s}) f(Z_{a_s, d_s, m_s, w_s})^\top.$$

By the definition of $\kappa = \max_{Z \in \mathcal{Z}} \max_{\theta \in \Theta} \frac{1}{\dot{\sigma}(f(Z)^\top \theta)}$, we have $\dot{\sigma}(f(Z)) V_s \geq \frac{1}{\kappa} V_s$, $\forall Z \in \mathcal{Z}$. Therefore,

$$H_t = \lambda I_n + \sum_{s=1}^{t-1} \dot{\sigma}(f(Z_{a_s, d_s, m_s, w_s})^\top \hat{\theta}_{s+1}) f(Z_{a_s, d_s, m_s, w_s}) f(Z_{a_s, d_s, m_s, w_s})^\top \geq \lambda I_n + \frac{1}{\kappa} \sum_{s=1}^{t-1} V_s, \quad t \leq T_0.$$

By Assumption 5, we have $0 \leq V_s \leq I_n$, then $H_t \geq \lambda I_n$. And for V_s , we have

$$\begin{aligned} \mathbb{E}[V_s] &= \mathbb{E}[f(Z_{X_s, d_s}) f(Z_{X_s, d_s})^\top] \\ &= \mathbb{E}_{X_s} \left[\frac{1}{|\mathcal{D}|} \sum_{d \in \mathcal{D}} f(Z_{X_s, d}) f(Z_{X_s, d})^\top \right] \end{aligned} \quad (215)$$

$$\geq \gamma_2 I_n. \quad (216)$$

Where (215) holds since d_s is sampled uniformly from $\mathcal{D}$ when $s \leq T_0$; and (216) comes from Assumption 4. Therefore,

$$\lambda_{\min} \left(\sum_{s=1}^{t-1} \mathbb{E}[V_s] \right) \geq (t-1) \gamma_2. \quad (217)$$

By the matrix Chernoff lower-tail bound (Lemma 9), we have, for $\epsilon \in [0, 1)$,

$$\mathbb{P} \left\{ \lambda_{\min} \left(\sum_{s=1}^{t-1} V_s \right) \leq (1-\epsilon)(t-1) \gamma_2 \right\} \leq n \left[\frac{e^{-\epsilon}}{(1-\epsilon)^{1-\epsilon}} \right]^{(t-1) \gamma_2} \quad (218)$$

$$\leq n \exp \left(- \frac{\epsilon^2 (t-1) \gamma_2}{2} \right), \quad (219)$$

where (218) follows by (259); and (219) is because $\frac{e^{-\epsilon}}{(1-\epsilon)^{1-\epsilon}} \leq \exp \left(- \frac{\epsilon^2}{2} \right)$.

Select $\epsilon = 1/2$, we have,

$$\mathbb{P} \left[\lambda_{\min} \left(\sum_{s=1}^{t-1} V_s \right) \leq \frac{1}{2} (t-1) \gamma_2 \right] \leq n \exp \left(- \frac{(t-1) \gamma_2}{8} \right).$$

Define $t_{\min} := 1 + \frac{8}{\gamma_2} \log \frac{n}{\delta_2}$. For every $t \geq t_{\min}$ we have $(t-1) \gamma_2 / 8 \geq \log(n/\delta_2)$, hence $n \exp \left(- (t-1) \gamma_2 / 8 \right) \leq \delta_2$, so that $\lambda_{\min} \left(\sum_{s=1}^{t-1} V_s \right) \geq \frac{1}{2} (t-1) \gamma_2$ fails with probability at most δ_2 .

Define a good event as follows:

$$\mathcal{E} := \left\{ \forall t \in [t_{\min}, T_0] : \lambda_{\min} \left(\sum_{s=1}^{t-1} V_s \right) \geq \frac{1}{2} (t-1) \gamma_2 \right\}.$$

Thus, for $t_{\min} \leq t \leq T_0$, on the event $\mathcal{E}$ which occurs with probability at least $1 - \delta_2$, we have,

$$\lambda_{\min}(H_t) \geq \lambda_{\min} \left(\lambda I_n + \frac{1}{\kappa} \sum_{s=1}^{t-1} V_s \right) \geq \lambda + \frac{(t-1) \gamma_2}{2\kappa} \geq \frac{(t-1) \gamma_2}{2\kappa}. \quad (220)$$

Then for $t \geq T_0 + 1$, on the event $\mathcal{E}$ we have,

$$\lambda_{\min}(H_t) \geq \lambda_{\min}(H_{T_0+1}) \geq \lambda_{\min}(H_{T_0}) \geq \frac{(T_0-1) \gamma_2}{2\kappa}. \quad (221)$$

Where (221) holds since $H_t \geq H_{T_0+1} \geq H_{T_0}$ when $t \geq T_0 + 1$.

Then, for the pure exploration phase $t \in [1, T_0]$,

$$\sum_{t=1}^{T_0} \frac{1}{\sqrt{\lambda_{\min}(H_t)}} = \sum_{t=1}^{t_{\min}} \frac{1}{\sqrt{\lambda_{\min}(H_t)}} + \sum_{t=t_{\min}+1}^{T_0} \frac{1}{\sqrt{\lambda_{\min}(H_t)}} \quad (222)$$

$$\leq \frac{t_{\min}}{\sqrt{\lambda}} + \sqrt{\frac{2\kappa}{\gamma_2}} \sum_{t=t_{\min}+1}^{T_0} \frac{1}{\sqrt{t-1}} \quad (223)$$

$$\leq \frac{t_{\min}}{\sqrt{\lambda}} + 4\sqrt{\frac{\kappa T_0}{\gamma_2}} \quad (224)$$

$$= \mathcal{O}\left(\frac{\log(n/\delta_2)}{\gamma_2 \sqrt{\lambda}} + \sqrt{\frac{\kappa T_0}{\gamma_2}}\right), \quad (225)$$

where (222) follows by splitting the sum; (223) follows by (220) and $\lambda_{\min}(H_t) \geq \lambda$ for every t ; (224) comes from $\sum_{t=t_{\min}+1}^{T_0} (t-1)^{-1/2} \leq \sum_{j=1}^{T_0-1} j^{-1/2} \leq 2\sqrt{T_0}$; and (225) is because $t_{\min} := 1 + \frac{8}{\gamma_2} \log \frac{n}{\delta_2}$.

And for the pure exploitation phase $t \in [T_0 + 1, T]$,

$$\sum_{t=T_0+1}^T \frac{1}{\sqrt{\lambda_{\min}(H_t)}} \leq (T - T_0) \sqrt{\frac{2\kappa}{\gamma_2(T_0 - 1)}} \quad (226)$$

$$= \mathcal{O}\left(\sqrt{\frac{\kappa}{\gamma_2}} \frac{T}{\sqrt{T_0}}\right), \quad (227)$$

where (226) follows by (221); and (227) is because $T_0 \leq T$.

Combining the above, we have

$$\sum_{t=1}^T \|f(Z_{a,d,m,w})\|_{H_t^{-1}} \leq \sum_{t=1}^T \frac{1}{\sqrt{\lambda_{\min}(H_t)}} \quad (228)$$

$$= \sum_{t=1}^{T_0} \frac{1}{\sqrt{\lambda_{\min}(H_t)}} + \sum_{t=T_0+1}^T \frac{1}{\sqrt{\lambda_{\min}(H_t)}} \quad (229)$$

$$\begin{aligned} &= \mathcal{O}\left(\frac{\log(n/\delta_2)}{\gamma_2 \sqrt{\lambda}} + \sqrt{\frac{\kappa T_0}{\gamma_2}} + \sqrt{\frac{\kappa}{\gamma_2}} \frac{T}{\sqrt{T_0}}\right) \\ &= \mathcal{O}\left(\sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}}\right)\right). \end{aligned} \quad (230)$$

Where (228) follows by (214); (229) is by splitting the sum; and (230) comes from Equations (225) and (227). $\blacksquare$

Theorem 2 (Constraint Violations Upper Bound). *Under Assumptions 1, 4, and 5, Algorithm 2 achieves the following bound with probability at least $1 - \alpha - \delta_2$ for any $\alpha \in (0, 1]$, $\delta_2 \in (0, 1]$,*

$$V_2(T) = \tilde{\mathcal{O}}\left(S \sqrt{\frac{n\kappa}{\gamma_2}} \frac{T}{\sqrt{T_0}} + T_0\right).$$

Proof. We prove the cumulative constraint violations for logistic bandits in this section. To bound the constraint violations, we look at the Lipschitz form of the three counterfactual fairness constraints.

By construction, the estimated fair policy set at round t is $\hat{\Pi}_t^f = \bigcap_{i \in [3]} \left\{ \pi \in \Pi : \mathbb{E}_{d \sim \pi} [|\Delta_d^{(i)}(\hat{\theta}_t)| - \xi_t(d)] \leq c_i \right\}$, since $\pi_t \in \hat{\Pi}_t^f$ when $t > T_0$, we then have the following when $t > T_0$,

$$\mathbb{E}_{d \sim \pi_t} [|\Delta_d^{(i)}(\hat{\theta}_t)| - \xi_t(d)] \leq c_i, \quad \forall i \in [3]. \quad (231)$$

Throughout the proof we compare θ_* against $\hat{\theta}_t$.

Case 1: when $i = 1$, we have

$$\begin{aligned} & \mathbb{E}_{d \sim \pi} |\Delta_d^{(1)}(\theta_*)| - \mathbb{E}_{d \sim \pi} |\Delta_d^{(1)}(\hat{\theta}_t)| \\ &= \mathbb{E}_{d \sim \pi} \left[|\Delta_d^{(1)}(\theta_*)| - |\Delta_d^{(1)}(\hat{\theta}_t)| \right] \\ &= \mathbb{E}_{d \sim \pi} \left[\left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right] \right| - \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) \right] \right| \right] \end{aligned} \quad (232)$$

$$\leq \mathbb{E}_{d \sim \pi} \left[\left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right] \right| - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) \right] \right] \quad (233)$$

$$\leq \mathbb{E}_{d \sim \pi} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left| \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) - \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) + \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) \right| \quad (234)$$

$$\begin{aligned} &= \mathbb{E}_{d \sim \pi} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left| \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) + \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| \\ &\leq \mathbb{E}_{d \sim \pi} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\left| \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right| + \left| \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| \right] \end{aligned} \quad (235)$$

Where (232) comes from (204); (233) and (235) holds from the triangle inequality, i.e., $|a| - |b| \leq |a - b|$ and $|a + b| \leq |a| + |b|$, $\forall a, b \in \mathbb{R}$; (234) follows by the Jensen's inequality where $|\cdot|$ is a convex function;

Case 2: when $i = 2$, we have

$$\begin{aligned} & \mathbb{E}_{d \sim \pi} |\Delta_d^{(2)}(\theta_*)| - \mathbb{E}_{d \sim \pi} |\Delta_d^{(2)}(\hat{\theta}_t)| \\ &= \mathbb{E}_{d \sim \pi} \left[|\Delta_d^{(2)}(\theta_*)| - |\Delta_d^{(2)}(\hat{\theta}_t)| \right] \\ &= \mathbb{E}_{d \sim \pi} \left[\left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right| - \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| \right] \end{aligned} \quad (236)$$

$$\leq \mathbb{E}_{d \sim \pi} \left[\left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| \right] \quad (237)$$

$$\begin{aligned} &= \mathbb{E}_{d \sim \pi} \left[\left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right| \right] \\ &\leq \mathbb{E}_{d \sim \pi} \left[\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \left| \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left| \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right| \right] \end{aligned} \quad (238)$$

Where (236) come from (205); (237) follows by the triangle inequality; (238) holds because of Jensen's inequality and triangle inequality.

Case 3: similarly, when $i = 3$ one could show that:

$$\begin{aligned} & \mathbb{E}_{d \sim \pi} |\Delta_d^{(3)}(\theta_*)| - \mathbb{E}_{d \sim \pi} |\Delta_d^{(3)}(\hat{\theta}_t)| \\ &= \mathbb{E}_{d \sim \pi} \left[|\Delta_d^{(3)}(\theta_*)| - |\Delta_d^{(3)}(\hat{\theta}_t)| \right] \end{aligned}$$

$$\begin{aligned}
 &= \mathbb{E}_{d \sim \pi} \left[\left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) \right| - \right. \\
 &\quad \left. \left| \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) - \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| \right] \quad (239)
 \end{aligned}$$

$$\begin{aligned}
 &\leq \mathbb{E}_{d \sim \pi} \left[\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \left| \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \right. \\
 &\quad \left. \left| \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| \right]. \quad (240)
 \end{aligned}$$

Where (239) come from (206).

Bound for logistic cumulative constraint violations. The cumulative constraint violation $V_2(T)$ over the horizon T is defined as

$$\begin{aligned}
 V_2(T) &= \max_{i \in \{1, 2, 3\}} \left\{ \sum_{t=1}^T \left[\mathbb{E}_{d \sim \pi_t} |\Delta_d^{(i)}(\theta_*)| - c_i \right] \right\}. \\
 &= \max_{i \in \{1, 2, 3\}} \left\{ \sum_{t=1}^{T_0} \left[\mathbb{E}_{d \sim \pi_t} |\Delta_d^{(i)}(\theta_*)| - c_i \right] + \sum_{t=T_0+1}^T \left[\mathbb{E}_{d \sim \pi_t} |\Delta_d^{(i)}(\theta_*)| - c_i \right] \right\} \quad (241)
 \end{aligned}$$

$$\leq \max_{i \in \{1, 2, 3\}} \left\{ \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} |\Delta_d^{(i)}(\theta_*)| - c_i \right\} + T_0. \quad (242)$$

Where (241) follows by splitting the sum; (242) is because $\mathbb{E}_{d \sim \pi_t} |\Delta_d^{(i)}(\theta_*)| - c_i \leq 1$. Since $\pi_t \in \hat{\Pi}_t^f$, we have $\mathbb{E}_{d \sim \pi_t} [|\Delta_d^{(i)}(\hat{\theta}_t)| - \xi_t(d)] \leq c_i$, and hence

$$\sum_{t=T_0+1}^T \left[\mathbb{E}_{d \sim \pi_t} |\Delta_d^{(i)}(\theta_*)| - c_i \right] \leq \sum_{t=T_0+1}^T \left[\mathbb{E}_{d \sim \pi_t} |\Delta_d^{(i)}(\theta_*)| - \mathbb{E}_{d \sim \pi_t} (|\Delta_d^{(i)}(\hat{\theta}_t)| - \xi_t(d)) \right] \quad (243)$$

$$= \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \left[|\Delta_d^{(i)}(\theta_*)| - |\Delta_d^{(i)}(\hat{\theta}_t)| \right] + \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \xi_t(d). \quad (244)$$

Here (243) uses $\pi_t \in \hat{\Pi}_t^f$ and the LCB form of the feasibility constraint (231). We bound the two terms in (244) separately.

Estimation-error term $\sum_{t=1}^T \mathbb{E}_{d \sim \pi_t} [|\Delta_d^{(i)}(\theta_*)| - |\Delta_d^{(i)}(\hat{\theta}_t)|]$.

(1). When $i = 1$, by the reverse triangle inequality and (235),

$$\begin{aligned}
 &\sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \left[|\Delta_d^{(1)}(\theta_*)| - |\Delta_d^{(1)}(\hat{\theta}_t)| \right] \\
 &\leq \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\left| \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right| \right. \\
 &\quad \left. + \left| \sigma(f(Z'_{a'_t, d, m, w})^\top \theta_*) - \sigma(f(Z'_{a'_t, d, m, w})^\top \hat{\theta}_t) \right| \right]. \quad (245)
 \end{aligned}$$

First, we look at the factual case (with attribute value a_t), that is

$$\begin{aligned}
 &\sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left| \sigma(f(Z_{a_t, d, m, w})^\top \hat{\theta}_t) - \sigma(f(Z_{a_t, d, m, w})^\top \theta_*) \right| \\
 &\leq \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\left| f(Z_{a_t, d, m, w})^\top (\theta_* - \hat{\theta}_t) \right| \right] \quad (246)
 \end{aligned}$$

$$\leq \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\|f(Z_{a_t, d, m, w})\|_{H_t^{-1}} \|\theta_* - \hat{\theta}_t\|_{H_t} \right] \quad (247)$$

$$\leq \sum_{t=T_0+1}^T \gamma_t(\alpha) \cdot \mathbb{E}_{d \sim \pi_t} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\|f(Z_{a_t,d,m,w})\|_{H_t^{-1}} \right] \quad (248)$$

$$\leq \gamma_T(\alpha) \cdot \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left[\|f(Z_{a_t,d,m,w})\|_{H_t^{-1}} \right]. \quad (249)$$

Where (246) is because of the mean value theorem, i.e., for any scalars z, z' ,

$$|\sigma(z) - \sigma(z')| \leq \sup_{\zeta} |\dot{\sigma}(\zeta)| |z - z'|,$$

where ζ is between z and z' and $|\dot{\sigma}(\zeta)| \leq 1$; (247) come from the Cauchy-Schwarz inequality; and the last inequality (248) follows by Proposition 5 and the triangle inequality, i.e., $\|\theta_* - \hat{\theta}_t\|_{H_t} \leq \gamma_t(\alpha)$.

Then, by Proposition 7 with probability at least $1 - \delta_2$, we have,

$$\sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left| \sigma(f(Z_{a_t,d,m,w})^\top \hat{\theta}_t) - \sigma(f(Z_{a_t,d,m,w})^\top \theta_*) \right| = \mathcal{O} \left(\gamma_T \sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}} \right) \right). \quad (250)$$

Similarly, as for the counterfactual case, we have

$$\sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \left| \sigma(f(Z'_{a_t,d,m,w})^\top \hat{\theta}_t) - \sigma(f(Z'_{a_t,d,m,w})^\top \theta_*) \right| = \mathcal{O} \left(\gamma_T \sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}} \right) \right). \quad (251)$$

Then by Equations (245), (250) and (251), we have,

$$\sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \left[|\Delta_d^{(1)}(\theta_*)| - |\Delta_d^{(1)}(\hat{\theta}_t)| \right] = \mathcal{O} \left(\gamma_T \sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}} \right) \right). \quad (252)$$

(2). When $i = 2$, we have,

$$\begin{aligned} & \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \left[|\Delta_d^{(2)}(\theta_*)| - |\Delta_d^{(2)}(\hat{\theta}_t)| \right] \\ & \leq \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \left[\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \left| \sigma(f(Z'_{a_t,d,w,m})^\top \theta_*) - \sigma(f(Z'_{a_t,d,w,m})^\top \hat{\theta}_t) \right| + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \right. \\ & \quad \left. \left| \sigma(f(Z'_{a_t,d,w,m})^\top \theta_*) - \sigma(f(Z'_{a_t,d,w,m})^\top \hat{\theta}_t) \right| \right]. \end{aligned} \quad (253)$$

$$= \mathcal{O} \left(\gamma_T \sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}} \right) \right). \quad (254)$$

Where (253) comes from (238); (254) follows by (251).

(3). When $i = 3$, similarly, we have,

$$\begin{aligned} & \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \left[|\Delta_d^{(3)}(\theta_*)| - |\Delta_d^{(3)}(\hat{\theta}_t)| \right] \\ & \leq \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \left[\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \left| \sigma(f(Z'_{a_t,d,w,m})^\top \theta_*) - \sigma(f(Z'_{a_t,d,w,m})^\top \hat{\theta}_t) \right| + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \right. \\ & \quad \left. \left| \sigma(f(Z'_{a_t,d,w,m})^\top \theta_*) - \sigma(f(Z'_{a_t,d,w,m})^\top \hat{\theta}_t) \right| \right] \end{aligned} \quad (255)$$

$$= \mathcal{O} \left(\gamma_T \sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}} \right) \right). \quad (256)$$

Where (255) comes from (240); similarly, (256) follows by (251).

As for $\sum_{t=1}^T \mathbb{E}_{d \sim \pi_t} \xi_t(d)$, we have

$$\begin{aligned} \sum_{t=T_0+1}^T \mathbb{E}_{d \sim \pi_t} \xi_t(d) &= \frac{1}{4} \sum_{t=T_0+1}^T \gamma_t(\alpha) \mathbb{E}_{d \sim \pi_t} \left(\mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(3)}} \|f(Z'_{a_t', d, w, m})\|_{H_t^{-1}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(2)}} \|f(Z'_{a_t', d, w, m})\|_{H_t^{-1}} + \right. \\ &\quad \left. + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \|f(Z'_{a_t', d, w, m})\|_{H_t^{-1}} + \mathbb{E}_{(m,w) \sim \mathbb{P}_{a_t}^{(1)}} \|f(Z_{a_t, d, m, w})\|_{H_t^{-1}} \right) \\ &= \mathcal{O} \left(\gamma_T \sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}} \right) \right). \end{aligned} \quad (257)$$

Therefore, by Equations (242), (244), (252), (254), (256) and (257), we obtain the following bound on the cumulative constraint violations:

$$\begin{aligned} V_2(T) &= \mathcal{O} \left(\gamma_T \sqrt{\frac{\kappa}{\gamma_2}} \left(\sqrt{T_0} + \frac{T}{\sqrt{T_0}} \right) + T_0 \right) \\ &= \tilde{\mathcal{O}} \left(S \sqrt{\frac{n\kappa}{\gamma_2}} \frac{T}{\sqrt{T_0}} + T_0 \right). \end{aligned} \quad (258)$$

Where (258) follows by $\gamma_t(\alpha) = \mathcal{O}(S\sqrt{n(S + \log(t))})$. ■

G. Technical Lemmas

Lemma 1 (Hoeffding's Inequality). *Let $X_1, \dots, X_n$ be independent random variables on $\mathbb{R}$ such that X_i is bounded in the interval $[a_i, b_i]$. Let $S_n = \sum_{i=1}^n X_i$. Then for all $t > 0$,*

$$\begin{aligned} \mathbb{P}(S_n - \mathbb{E}[S_n] \geq t) &\leq \exp \left(\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right) \\ \mathbb{P}(S_n - \mathbb{E}[S_n] \leq -t) &\leq \exp \left(\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2} \right). \end{aligned}$$

Lemma 2 (Azuma-Hoeffding Inequality). *Let $X_0, X_1, \dots, X_n$ be a martingale sequence of random variables such that for all i , there exists a constant c_i such that $|X_i - X_{i-1}| < c_i$, then*

$$\begin{aligned} \mathbb{P}(X_n - X_0 \geq t) &\leq \exp \left(\frac{-t^2}{2 \sum_{i=1}^n c_i^2} \right) \\ \mathbb{P}(X_n - X_0 \leq -t) &\leq \exp \left(\frac{-t^2}{2 \sum_{i=1}^n c_i^2} \right). \end{aligned}$$

Lemma 3 (Elliptical Potential Lemma). *Let $\{u_s\}_{s=1}^\infty$ be a sequence in $\mathbb{R}^n$ such that $\|u_s\| \leq B$ for all $s \in \mathbb{N}$. Further let $\{\lambda_s\}_{s=1}^\infty$ be a non-decreasing sequence in $\mathbb{R}^+$ s.t. $\lambda_1 = 1$. For $t \geq 1$ define $\mathbf{V}_t = \sum_{s=1}^{t-1} u_s u_s^\top + \lambda_t \mathbf{I}_n$. Then:*

$$\sum_{t=1}^T \|u_t\|_{\mathbf{V}_t^{-1}}^2 \leq 2n(1 + B^2) \log \left(\lambda_T + \frac{TB^2}{n} \right).$$

Lemma 4 ((Zhang et al., 2025) Lemma 1). *Let $f : \Theta \rightarrow \mathbb{R}$ be a convex function on a convex set Θ and $A \in \mathbb{R}^{n \times n}$ be a symmetric positive definite matrix. Then, the update*

$$\hat{\theta}_{t+1} = \arg \min_{\theta \in \Theta} f(\theta) + \frac{1}{2\eta} \|\theta - \hat{\theta}_t\|_A^2.$$

satisfies

$$\|\hat{\theta}_{t+1} - u\|_A^2 \leq 2\eta \langle \nabla f(\hat{\theta}_{t+1}), u - \hat{\theta}_{t+1} \rangle + \|\hat{\theta}_t - u\|_A^2 - \|\hat{\theta}_t - \hat{\theta}_{t+1}\|_A^2, \quad \forall u \in \Theta.$$

Lemma 5 ((Zhang et al., 2025) Lemma 4). *Under Assumption 5, $\|\theta_*\|_2 \leq S$, and setting $\eta = 1 + S$, then for any $\lambda > 0$, the online estimator returned by (15) satisfies*

$$\begin{aligned} \|\hat{\theta}_{t+1} - \theta_*\|_{H_{t+1}}^2 &\leq (2 + 2S) \left(\sum_{s=1}^t \mathcal{L}_s(\theta_*) - \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1}) \right) + 4\lambda S^2 \\ &\quad + (2 + 2S) \sum_{s=1}^t \|\hat{\theta}_s - \hat{\theta}_{s+1}\|_2^2 - \sum_{s=1}^t \|\hat{\theta}_s - \hat{\theta}_{s+1}\|_{H_s}^2. \end{aligned}$$

Furthermore, if $\lambda \geq 6\eta S$, we can further have

$$\|\hat{\theta}_{t+1} - \theta_*\|_{H_{t+1}}^2 \leq (2 + 2S) \left(\sum_{s=1}^t \mathcal{L}_s(\theta_*) - \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1}) \right) - \frac{2}{3} \sum_{s=1}^t \|\hat{\theta}_s - \hat{\theta}_{s+1}\|_{H_s}^2 + 4\lambda S^2.$$

Lemma 6 ((Zhang et al., 2025) Lemma 5). *Let $\{\mathcal{F}_t\}_{t=1}^\infty$ be a filtration defined by $\mathcal{F}_t = \sigma(\{(f(Z_{d_s}), y_s)\}_{s=1}^{t-1})$. Let $\{P_t\}_{t=1}^\infty$ be a stochastic process such that the random variable P_t is a distribution over $\mathbb{R}^n$ and is $\mathcal{F}_t$ -measurable. Moreover, assume that the loss function*

$$\mathcal{L}_s(\theta) = -[(1 - y_s) \log(1 - \sigma(f(Z_s)^\top \theta)) + y_s \log \sigma(f(Z_s)^\top \theta)].$$

is $\mathcal{F}_{t+1}$ -measurable. For any $t \geq 1$, define

$$L_t(\theta_*) = \sum_{s=1}^t \mathcal{L}_s(\theta_*) \quad \text{and} \quad F_t = - \sum_{s=1}^t \ln \left(\mathbb{E}_{\theta \sim P_s} [e^{-\mathcal{L}_s(\theta)}] \right).$$

Then, for any $\alpha \in (0, 1]$, we have

$$\Pr \left[\forall t \geq 1, L_t(\theta_*) \leq F_t + \log \left(\frac{1}{\alpha} \right) \right] \geq 1 - \alpha.$$

Lemma 7 ((Zhang et al., 2025) Lemma 6). *Under Assumption 5 and $\|\theta_*\|_2 \leq S$, let $P_s = \mathcal{N}(\hat{\theta}_s, \alpha H_s^{-1})$ be a Gaussian distribution with mean $\hat{\theta}_s$ and covariance αH_s^{-1} , where*

$$H_s = \lambda I_n + \sum_{\tau=1}^{s-1} \nabla^2 \mathcal{L}_\tau(\hat{\theta}_{\tau+1}) \quad \text{and} \quad \alpha > 0 \text{ is any constant.}$$

We denote by $\hat{\theta}_s$ the model returned by (15). Then, setting

$$\lambda \geq \max \left\{ \frac{64n\alpha}{7}, 2 \right\},$$

we have

$$\sum_{s=1}^t m_s(P_s) \leq \sum_{s=1}^t \mathcal{L}_s(\hat{\theta}_{s+1}) + \frac{1}{2\alpha} \sum_{s=1}^t \|\hat{\theta}_{s+1} - \hat{\theta}_s\|_{H_s}^2 + 2n\alpha \ln \left(2 + \frac{2t}{\lambda} \right),$$

where the mix loss is defined as

$$m_s(P_s) = - \ln \left(\mathbb{E}_{\theta \sim P_s} [\exp\{-\mathcal{L}_s(\theta)\}] \right).$$

Lemma 8 (Randomization lemma for Bernoulli variables). *Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $\mathcal{H} \subseteq \mathcal{F}$ a sub- σ -field. Let $q : \Omega \rightarrow [0, 1]$ be $\mathcal{H}$ -measurable. Then there exists an extension of $(\Omega, \mathcal{F}, \mathbb{P})$ and random variables $Z \sim \text{Unif}[0, 1]$ (independent of $\mathcal{H}$) and U such that*

$$U = \mathbb{1}\{Z \leq q\} \quad \text{and} \quad \mathbb{P}(U = 1 \mid \mathcal{H}) = q \quad \text{a.s.}$$

The above result specializes Kallenberg (2021, Lemma 4.22) to a Bernoulli kernel.

Lemma 9 (Matrix Chernoff (Tropp, 2015, Theorem 5.1.1)). *Consider a finite sequence $\{Z_k\}$ of independent, random, Hermitian matrices with common dimension n . Assume that*

$$0 \leq \lambda_{\min}(Z_k) \quad \text{and} \quad \lambda_{\max}(Z_k) \leq L$$

for each index k .

Introduce the random matrix $Y = \sum_k Z_k$. Define the minimum eigenvalue $\lambda_{\min}$ and maximum eigenvalue $\lambda_{\max}$ of the expectation $\mathbb{E}[Y]$:

$$\lambda_{\min} = \lambda_{\min}(\mathbb{E}[Y]) = \lambda_{\min}\left(\sum_k \mathbb{E}[Z_k]\right), \quad \lambda_{\max} = \lambda_{\max}(\mathbb{E}[Y]) = \lambda_{\max}\left(\sum_k \mathbb{E}[Z_k]\right).$$

Then, for $r > 0$,

$$\mathbb{E}[\lambda_{\min}(Y)] \geq \frac{1 - e^{-r}}{r} \lambda_{\min} - \frac{1}{r} L \log n, \quad \mathbb{E}[\lambda_{\max}(Y)] \leq \frac{e^r - 1}{r} \lambda_{\max} + \frac{1}{r} L \log n.$$

Furthermore,

$$\mathbb{P}\{\lambda_{\min}(Y) \leq (1 - \varepsilon) \lambda_{\min}\} \leq n \left[\frac{e^{-\varepsilon}}{(1 - \varepsilon)^{1 - \varepsilon}} \right]^{\lambda_{\min}/L}, \quad \text{for } \varepsilon \in [0, 1), \quad (259)$$

$$\mathbb{P}\{\lambda_{\max}(Y) \geq (1 + \varepsilon) \lambda_{\max}\} \leq n \left[\frac{e^{\varepsilon}}{(1 + \varepsilon)^{1 + \varepsilon}} \right]^{\lambda_{\max}/L}, \quad \text{for } \varepsilon \geq 0. \quad (260)$$

H. Additional Discussions

In this section, we list below several discussion points that may be of interest.

Discussion 1 (Aggregate Causal Effects). An “indirect effect” is itself an aggregate over the mediated pathways included in its definition; in multiple-mediator settings, specific indirect effects may have opposing signs and partially cancel, yielding a small aggregate indirect effect despite strong pathway-level effects (MacKinnon et al., 2000; 2007).

Discussion 2 (Extension to other GLMs). In the parametric bandits setting, we consider logistic bandits, which naturally model binary outcomes via a logistic link applied to an inner product. While our approach extends directly to the linear case, extending to other Generalized Linear Models (GLMs) (e.g., Poisson) may require additional assumptions to control link curvature and tail behavior of rewards.

Discussion 3. Some readers maybe wonder if the total variation is small (or even zero) but Ctf-DE, Ctf-IE, and Ctf-SE are non-zero, why isn’t it enough to declare fairness if the unfairness along different paths (some positive, some negative) cancel out? This ultimately depends on the notion of fairness of interest. If one only cares about outcome parity measured by total variation (TV), then TV is small would be fair. However, even when TV=0, direct, indirect, and spurious discrimination may still persist: pathway-specific effects can be nonzero with opposite signs (e.g., a penalty along one path offset by a boost along another). Notice that for Example 1, although $|TV_{a_t, a'_t}(Y_{d_t})|$ and $|CATE_{a_t, a'_t}(Y_{d_t} | m_t, w_t)|$ are relatively small, the pathway-specific terms $|Ctf-IE_{a'_t, a_t}(Y_{d_t})|$ and $|Ctf-SE_{a'_t, a_t}(Y_{d_t})|$ are large in magnitude.

It can be meaningful to constrain the spurious effect under some fairness goals since the spurious effect is coming from the sensitive attribute through the (unobserved) confounding paths. TV is an aggregate that collapses all mechanisms into a single number but our fairness notion is defined in mechanism-level. Therefore, to achieve the goal of fairness for our problem, we should constrain the mechanism-specific effects, i.e., Ctf-DE, Ctf-IE, and Ctf-SE. The spurious effect are those correlations that are not supported by causal links, typically, coming from the environment, for instance, background socioeconomic factors. However, the discrimination carried over spurious effect could be amplified by the policy for some applications, therefore, it would be meaningful to bound it as well.

I. Experimental Details

In this section, we present the experimental details for parametric logistic bandits setting. For the non-parametric contextual bandit setting, we report only theoretical results for our method, since implementations of two closely related baselines

(Huang et al., 2022b; Bernasconi et al., 2024) (see Table 1) were not publicly available at the time of writing, moreover, Bernasconi et al. (2024) is primarily theoretical and does not report empirical experiments. We note that (Bernasconi et al., 2024) is designed for general constraints and subsequently requires constraint violation feedback, which is not available for this setting.

All experiments are run on a high-performance computing cluster. For each horizon T , we launch 16 independent replications in parallel; each job is allocated one compute node with 8 CPU cores (Intel Xeon Platinum 8358), 64 GB of memory. Across the 16 replications we report the empirical mean together with standard deviation.

The decision set $\mathcal{D}$ is fixed across rounds. We take the feature map f to be the identity, so that the factual feature vector is

$$f(Z_{A_t, W_t, M_t, D_t}) = [A_t, W_t, M_t, D_t]^\top,$$

and its counterfactual counterpart under an intervention on the sensitive attribute $A_t \mapsto A'_t$ is

$$f(Z_{A'_t, W_t, M_t, D_t}) = [A'_t, W_t, M_t, D_t]^\top.$$

The fairness thresholds are set to $c_1 = c_2 = c_3 = 0.157$; the true parameter is $\theta_* = (\sqrt{\frac{1}{2}}, 0, 0, \sqrt{\frac{1}{2}})^\top$, with norm bound $S = 1$ and link-function constant $\kappa \approx e^3$, the exploration period is $T_0 = \sqrt{T}$.

As for evaluation in our experiments, the penalized cumulative regret, cumulative regret, cumulative constraint violation, and cumulative reward are all computed at the level of individual actions rather than policies.

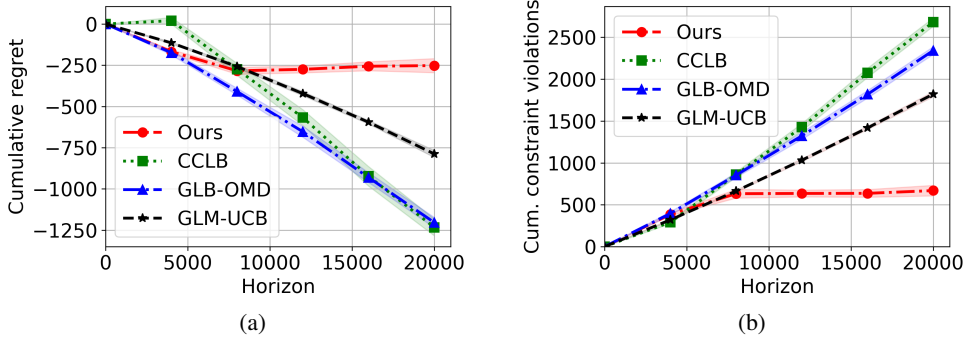


Figure 4. Plots for different algorithms: : Algorithm 2 (ours), GLB-OMD, CCLB, GLM-UCB on (a) cumulative regret; (b) cumulative constraint violations.

In Figure 4a, the instantaneous regret at round t is computed against the best fair action, i.e. the action attaining the highest expected reward within the fair action set. In Figure 2c, the running time is reported for $T = 10,000$ steps.